

CNS

By Dr.Riffat Ibrahim Riffat



Table of contents

01

**Alzheimer's
disease**

02

Psychosis

03

**Parkinson's
disease**

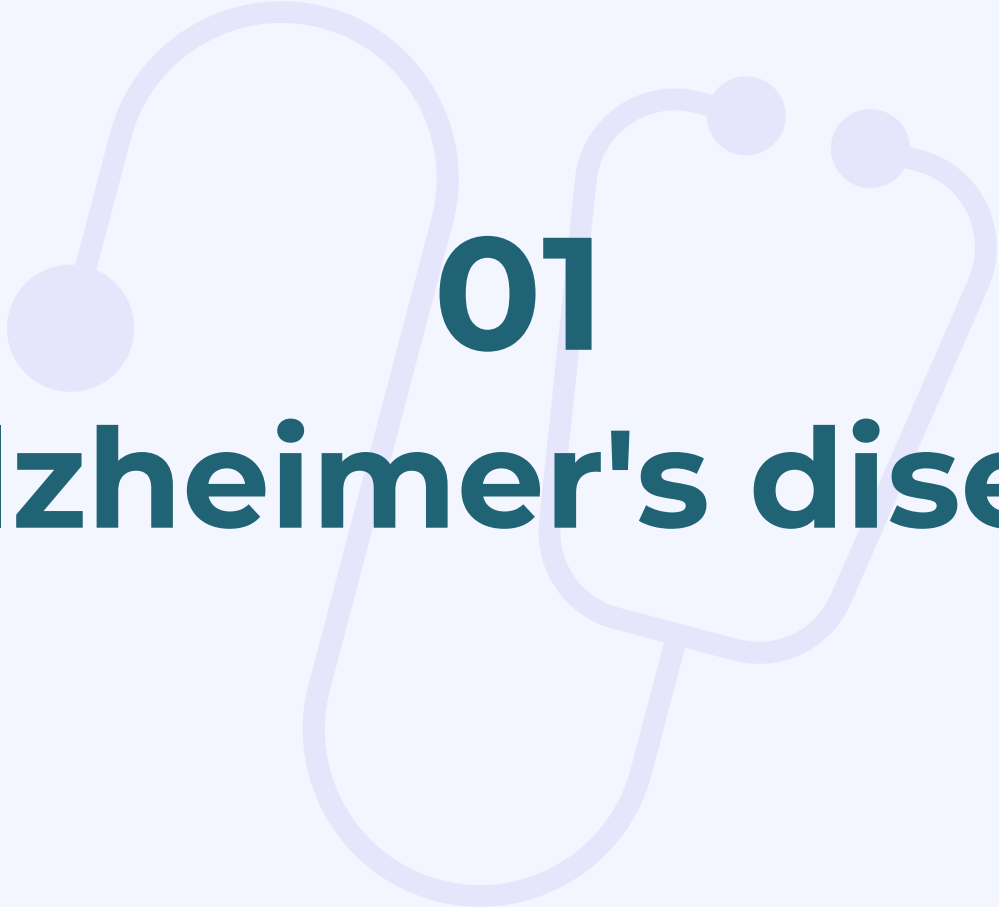
04

Epilepsy

05

Depression



A large, light purple outline of a human brain is centered in the background. A stethoscope is draped over the brain, with its chest piece resting on the right side and its ear pieces extending towards the top. The stethoscope's tubing is a slightly darker shade of purple.

01

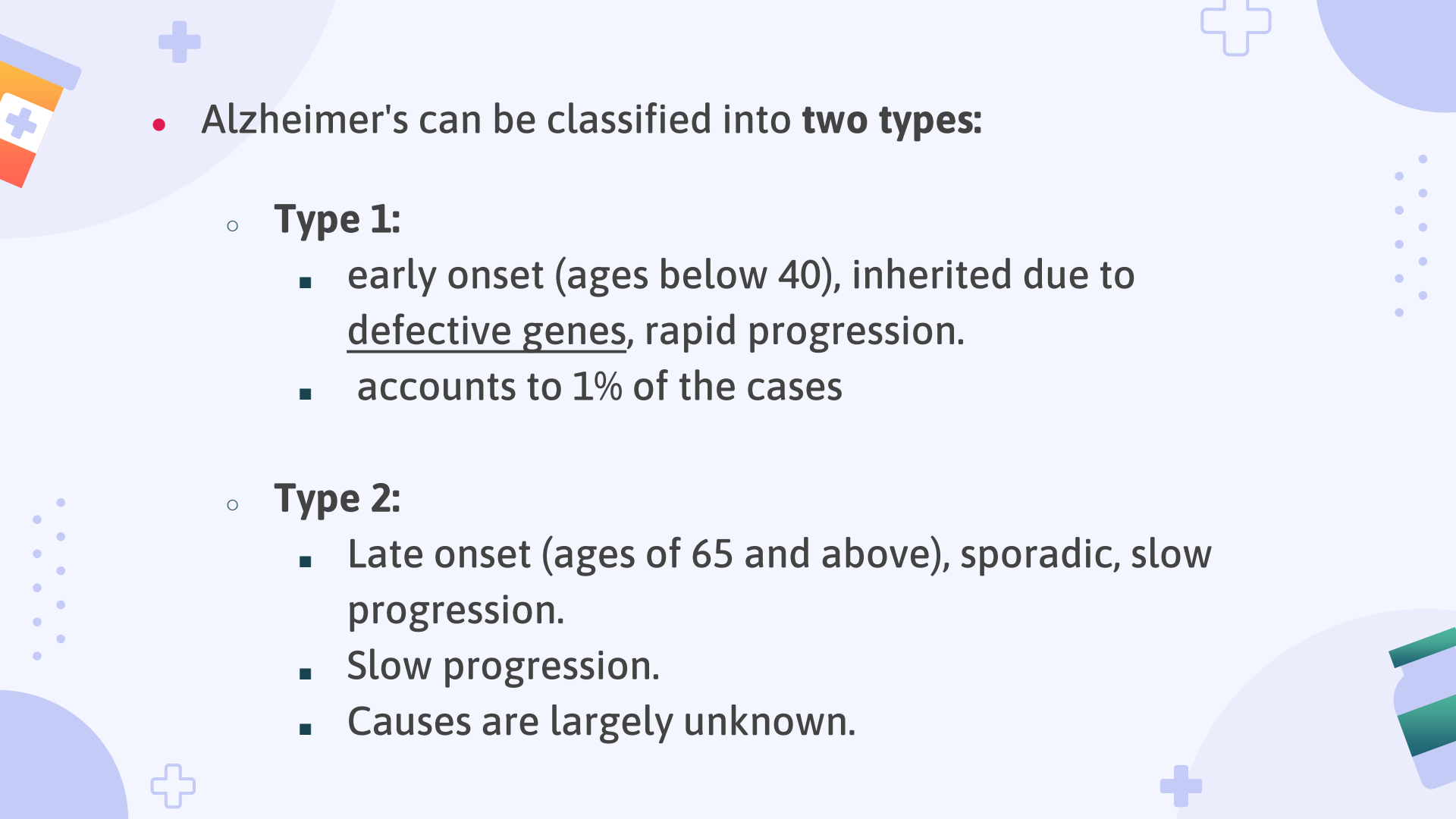
Alzheimer's disease



Introduction

- **Alzheimer's** disease is a neurodegenerative disease (irreversible damage as affected neurons can not regenerate) characterized by:
 - Progressive **M**emory loss.
 - **B**ehavioral abnormalities (Dementia) : depression ,aggression, motor hyperactivity.
 - Progressive **C**ognitive impairment: unable to recognize people and in later stages concepts.



- 
- Alzheimer's can be classified into **two types**:

- **Type 1:**

- early onset (ages below 40), inherited due to defective genes, rapid progression.
- accounts to 1% of the cases

- **Type 2:**

- Late onset (ages of 65 and above), sporadic, slow progression.
- Slow progression.
- Causes are largely unknown.

pathology

On the brain:

- **Gross brain atrophy, with significant reduction in hippocampus** (plays a role in memory function)

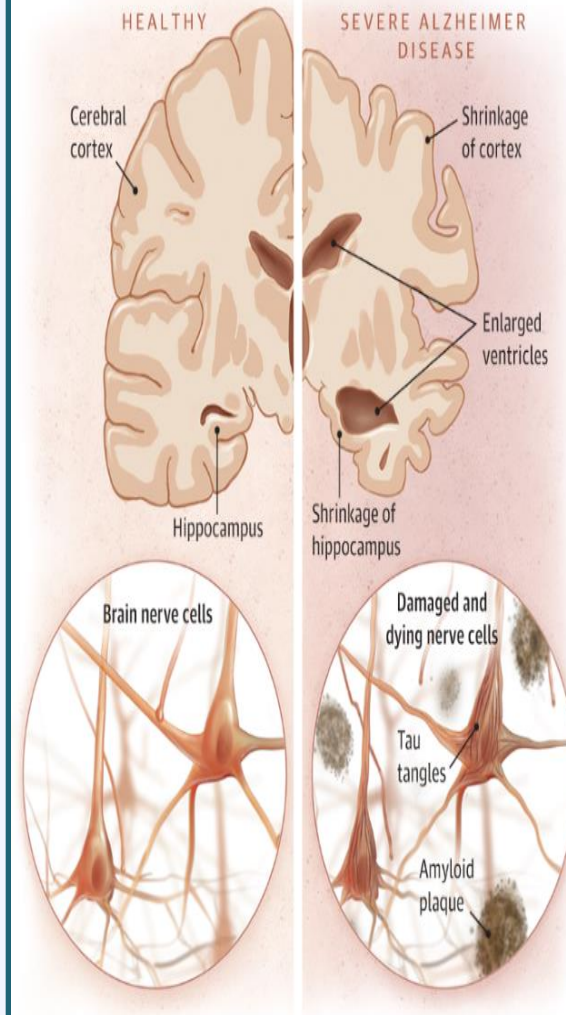
On neurons:

- Formation of **β - amyloid plaques** which reduces neuron connection.
- Formation of neurofibrillary tangles (tau tangles) which cause further degeneration of neurons.

Biochemistry:

- Decreased acetylcholine (an important neurotransmitter in memory function).
- Increase of glutamate (an excitatory neurotransmitter which causes neuron damage in Alzheimer's disease).

Brain changes in Alzheimer disease



Treatment

Alzheimer's is not curable, medication only serves to slow the progression of disease and enhance cognitive functions, however, total memory loss and worsening of patient cognitive functioning is inevitable, failing to recognize objects, concepts, and even loved ones.

- **Alzheimer's is a disease of the family** and impacts not only the patient but his family and caregiver as mental state deteriorates.

Treatment available are:

- Cholinesterase inhibitors (increase Ach)
- NMDA blockers (block the action of glutamate)





Cholinesterase Inhibitors (First line) eg: Donepezil, Rivastigmine, Galantamine.

- **M.O.A:**

- Selective Inhibition of Acetyl cholinesterase enzyme (AChE) in the brain and improvement of cholinergic transmission & Memory.



- Patients with Alzheimer's disease who treated with anti-cholinesterase drugs show improvement in memory and cognition symptoms by 10%.



- **Adverse effects include:**

- Selectivity on cerebral AChE is not absolute.
- Bradycardia.
- Salivation and Urination.
- Diarrhea and abdominal spasms.

- **Drug interactions:**

- All these drugs are substrates to CYP-450 except rivastigmine and so clinically significant drug interactions are present with drugs that may inhibit CYP-450.
- The use of drugs that block Ach receptor centrally (TCA anti-depressants) will negatively impact any improvements of AChE.I



Brands



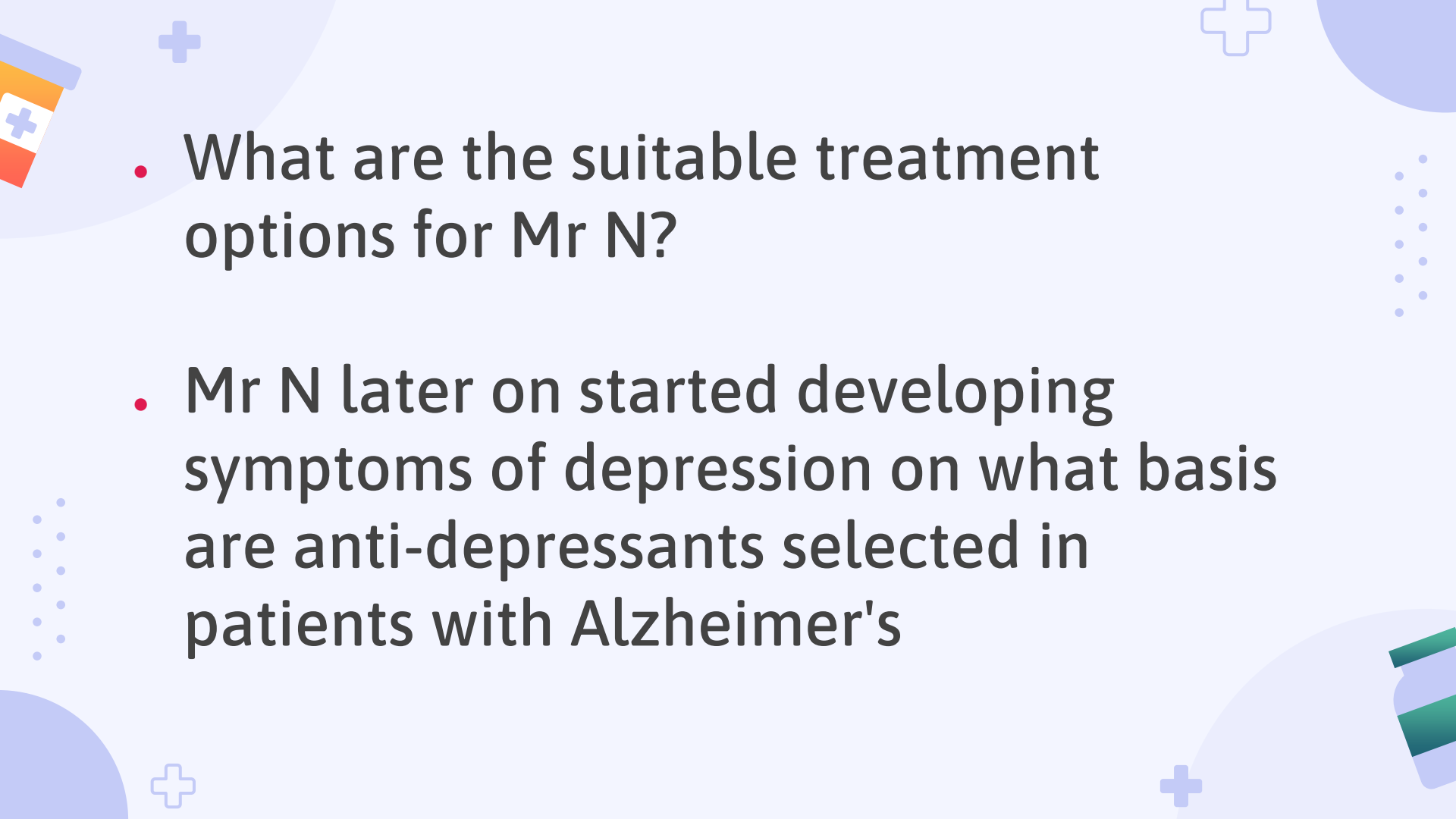
- 
- 
- **NMDA(N-Methyl-D-Aspartate) Receptor antagonists (Second line) eg: Memantine**
 - NMDA-receptor antagonists are usually used as Add-on drug in case of Severe and rapid progression of Alzheimer's Disease.
 - **M.O.A**
 - Blocking of NMDA-receptors (glutamate receptors) thus reducing its excitatory effect.
 - **Adverse effects include:** Insomnia, agitation, confusion and headache.
- 
- 

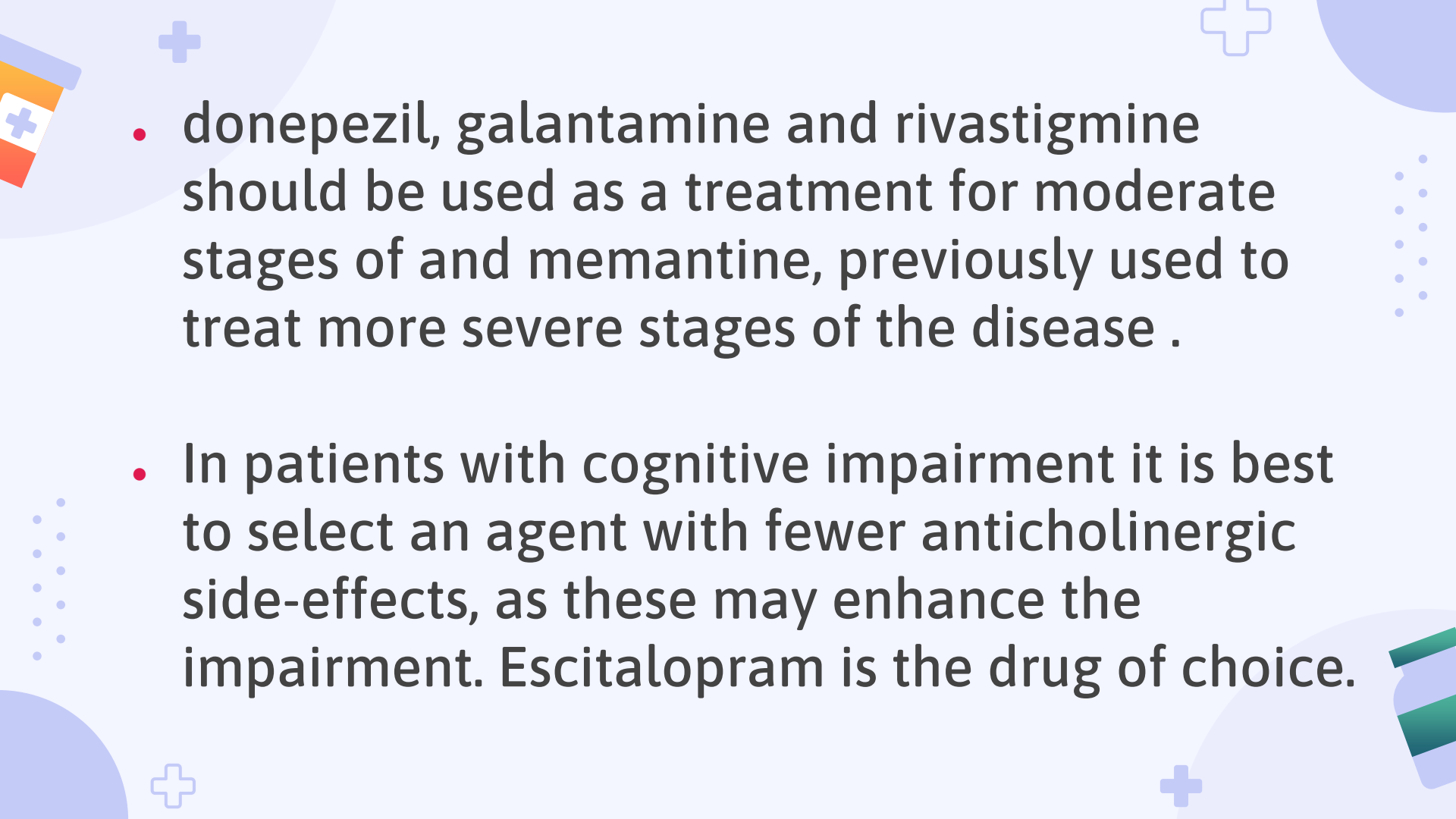
Brands



Cases

- The wife of Mr N , a 68-year-old black Caribbean male, is asking for some advice from the pharmacist because she is concerned about the changes in behavior her husband has been showing over the last few months. In contrast with Mr N's previous personality, his mood is now very volatile and somewhat unpredictable, his memory and concentration seem to be very poor and, at times, he may be very impulsive indeed. For the first time in his life he is now using words and language expressions that are both rude and vulgar .

- 
- What are the suitable treatment options for Mr N?
 - Mr N later on started developing symptoms of depression on what basis are anti-depressants selected in patients with Alzheimer's

- 
- donepezil, galantamine and rivastigmine should be used as a treatment for moderate stages of and memantine, previously used to treat more severe stages of the disease .
 - In patients with cognitive impairment it is best to select an agent with fewer anticholinergic side-effects, as these may enhance the impairment. Escitalopram is the drug of choice.

Cases

- Modest improvement in the memory of patients with Alzheimer's disease may occur with drugs that increase transmission at which of the following receptors?
 - A. Adrenergic.
 - B. Cholinergic.
 - C. Dopaminergic.
 - D. GABAergic.
 - E. Serotonergic.

Answers

- Correct answer = B. AChE inhibitors, such as rivastigmine, increase cholinergic transmission in the CNS and may cause a modest delay in the progression of Alzheimer's disease. Increased transmission at the other types of receptors listed does not result in improved memory.

Cases

- Which medication is a glutamate receptor antagonist that can be used in combination with an acetylcholinesterase inhibitor to manage the symptoms of Alzheimer's disease?
 - A. Rivastigmine.
 - B. Ropinirole.
 - C. Fluoxetine.
 - D. Memantine.
 - E. Donepezil.

Answers

- Correct answer = D. When combined with an acetylcholinesterase inhibitor, memantine has modest efficacy in keeping patients with Alzheimer's disease at or above baseline for at least 6 months and may delay disease progression.

CNS

By Dr.Riffat Ibrahim Riffat

02 Psychosis



Dopaminergic pathways in brain

Mesolimbic and mesocortical pathway:

- They are responsible for Cognition, memory, attention, behavior, and learning.
- Dopamine stimulates these functions.

Basal ganglia:

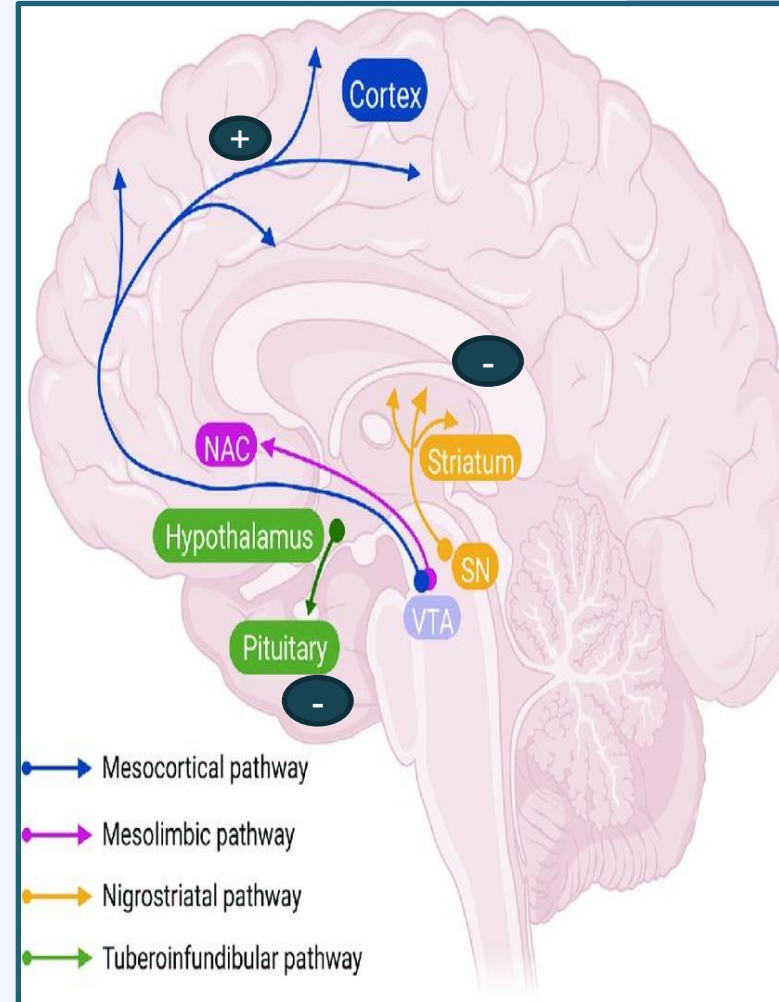
- Acts as inhibitory N.T to balance the stimulatory acetylcholine (regulation of fine movements)

On pituitary gland

- Suppresses prolactin secretion

On V.C:

- induces vomiting.



- **Psychosis or schizophrenia** is a mental illness that affects 1% of the population. psychosis has a strong genetic predisposition and is characterized by:

- **Positive symptoms:**

- Dopamine increase in mesolimbic and mesocortical track is responsible for these abnormalities.
- Series of delusions, as a set of ideas that the patient compulsively believes in and can not be convinced of their fallacy.
- Auditory and visual hallucinations.



- **Negative symptoms:**

- Serotonin is responsible for these manifestations.
- Depression, apathy, blunt emotions.
- Social isolation often resulting in lack of self-care And hygiene.



Overview

- The use of antipsychotic medications involves a difficult trade-off between the benefit of alleviating psychotic symptoms and the risk of a wide variety of troubling adverse effects.
- Antipsychotic drugs are not curative and do not eliminate chronic thought disorders, but they often decrease the intensity of hallucinations and delusions and permit the person with schizophrenia to function in a supportive environment.



First generation (Typical or Classic drugs)

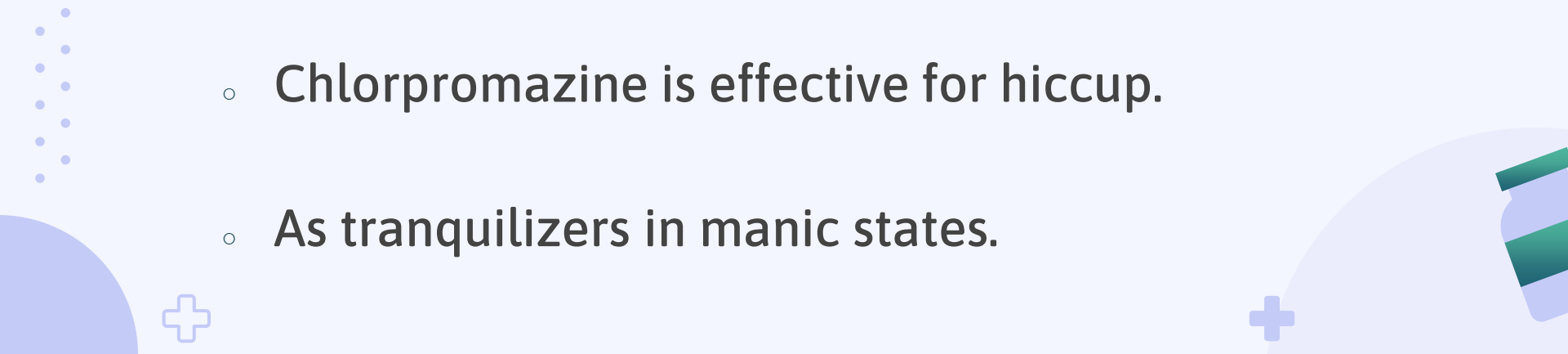
- Phenothiazines class :Chlorpromazine (CPZ)
 - Butyrophenones class : Haloperidol and Droperidol .
- 

- **MOA:**

- Competitive inhibition of D2 receptors, however, they are not selective to mesolimbic and mesocortical D2.
- 
- 

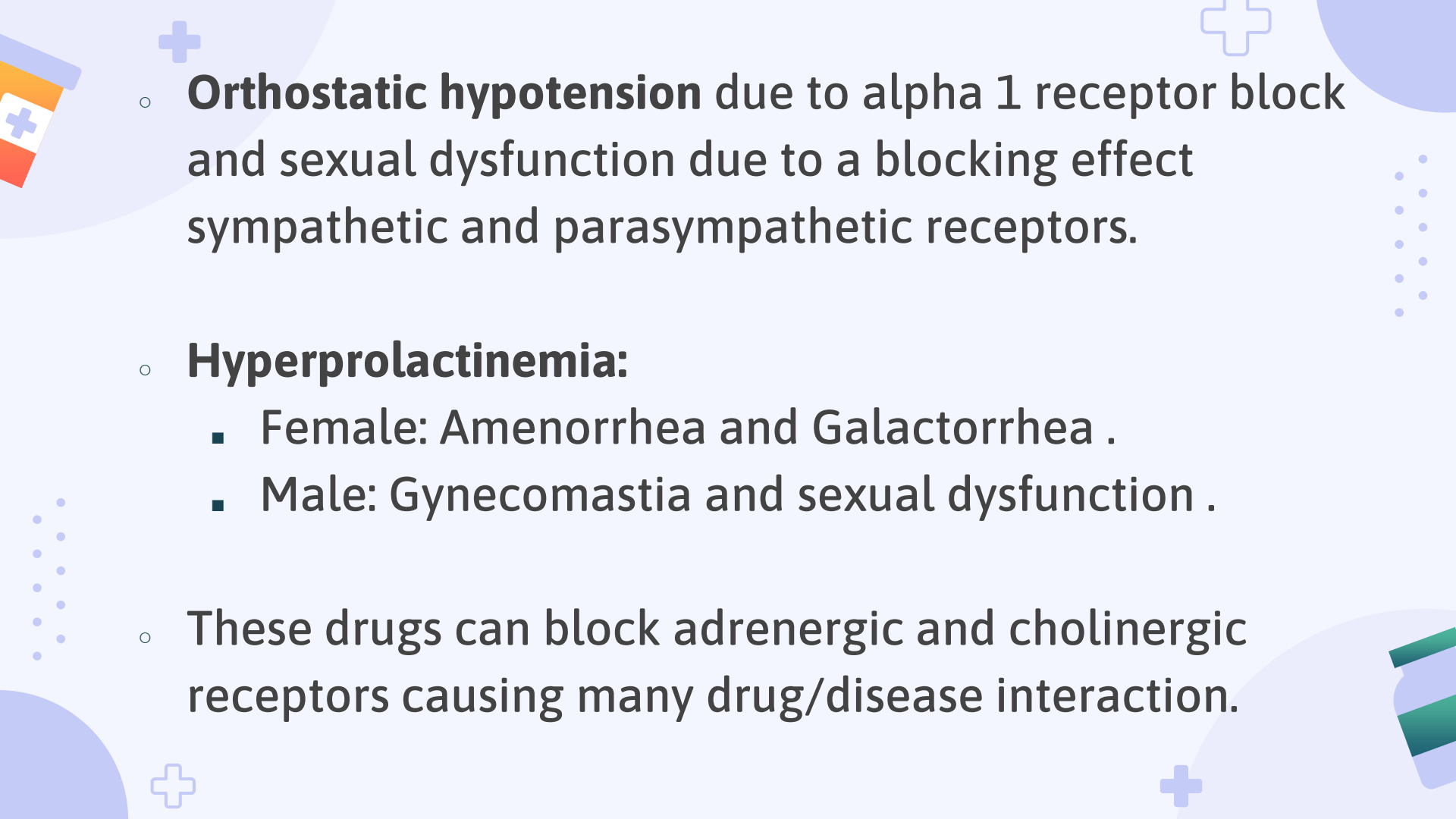


• **Therapeutic uses:**

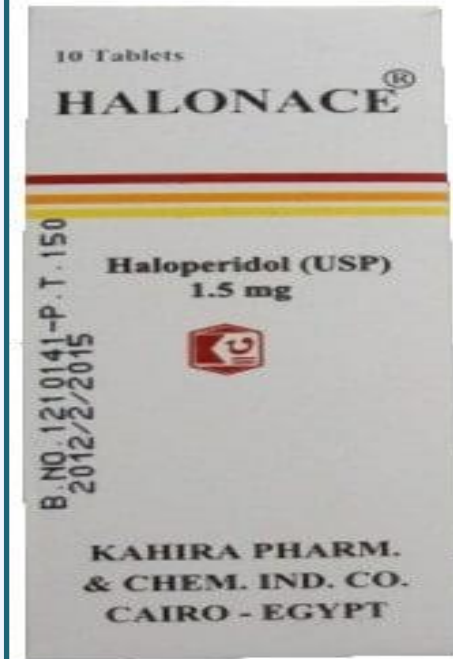
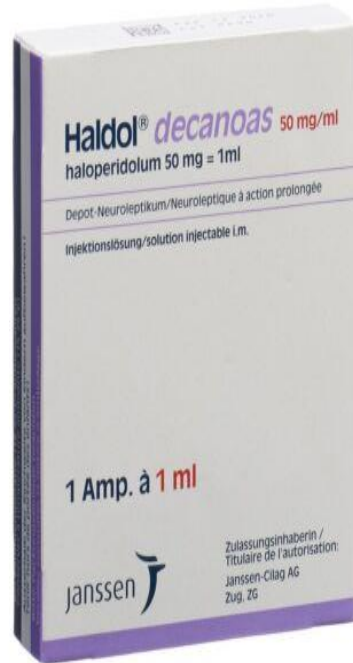
- Treatment of psychosis and Schizophrenia (+ve symptoms).
 - In severe vomiting.
 - Chlorpromazine is effective for hiccup.
 - As tranquilizers in manic states.
- 

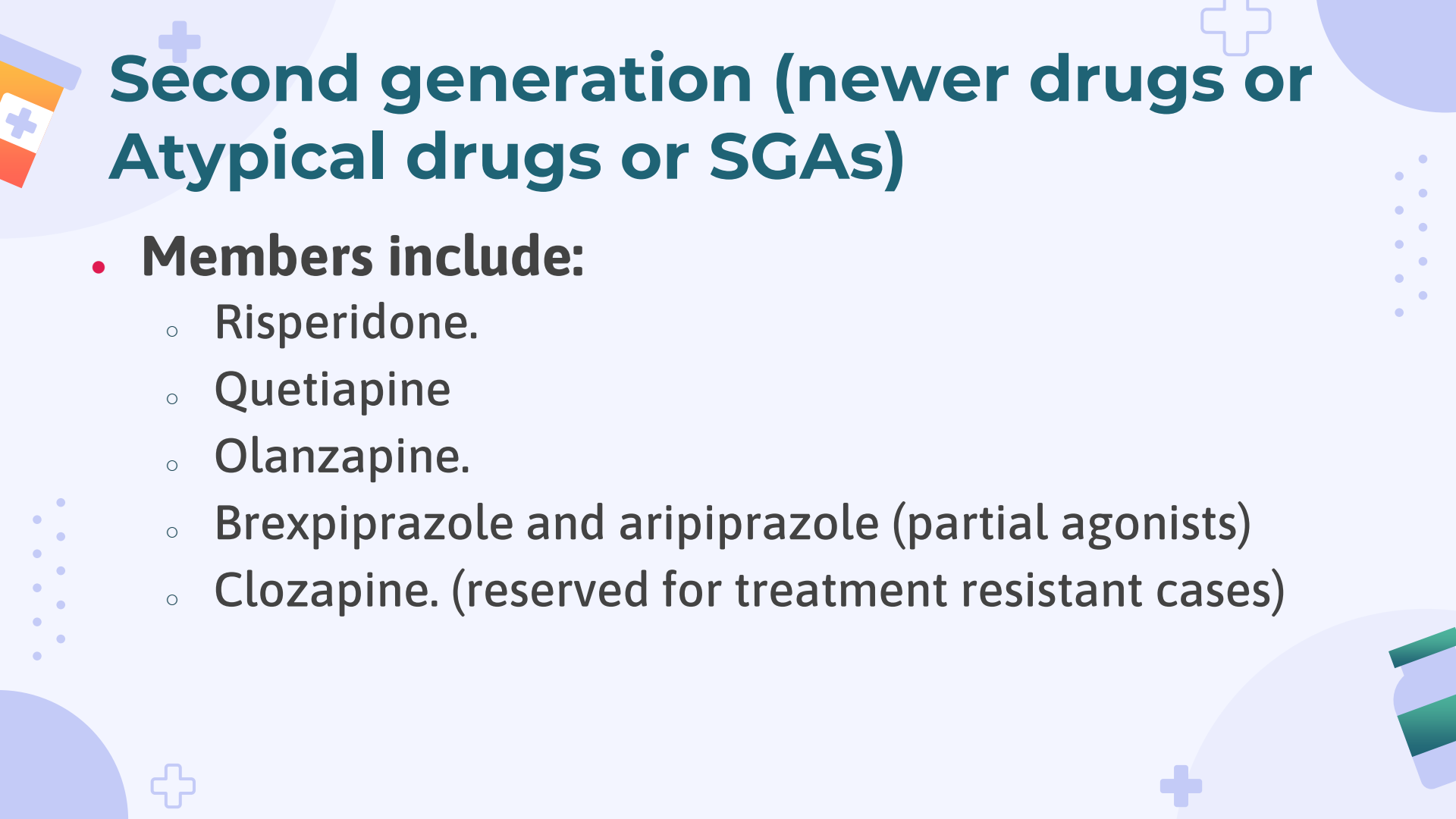
Adverse effects

- **Extrapyramidal side effects (EPS)** is a collection of CNS related side effects due to D2 receptor block in basal ganglia.
 - Akathisia which is a feeling of restlessness and anxiety (reversible)
 - Dyskinesia defined by abnormal body movements specially facial.
 - Dystonia abnormal uncontrolled body motions (irreversible)
- **Neuroleptic Malignant Syndrome (NMS):**
 - They are a set of autonomic dysfunctions characterized by:
 - Hyperthermia ,Convulsion ,Muscle rigidity ,hypertension and sweating.

- 
- **Orthostatic hypotension** due to alpha 1 receptor block and sexual dysfunction due to a blocking effect sympathetic and parasympathetic receptors.
 - **Hyperprolactinemia:**
 - Female: Amenorrhea and Galactorrhea .
 - Male: Gynecomastia and sexual dysfunction .
 - These drugs can block adrenergic and cholinergic receptors causing many drug/disease interaction.

Brands





Second generation (newer drugs or Atypical drugs or SGAs)

- **Members include:**

- Risperidone.
- Quetiapine
- Olanzapine.
- Brexpiprazole and aripiprazole (partial agonists)
- Clozapine. (reserved for treatment resistant cases)



• MOA:

- These drugs are antagonist of D2 receptors and 5-HT2A receptors.
 - Not all members show the same affinity to D2 and 5-HT2A receptor. Some have an equal affinity to both receptors while others prefer one target over the other.
 - The distributed affinity explains two effects of SGAs:
 - Reduced incidences of E.P.S since the drug acts on 2 targets rather than D2 receptors only.
 - SGAs improve treatment of –ve and +ve symptoms when compared to FGAs that treat only –ve symptoms. 
- 



Core Pharmacological Difference:

- Full Antagonists: These drugs (e.g., Haloperidol, Risperidone) block the D2 receptor. They sit on the receptor without activating it, preventing dopamine from binding. This directly reduces dopamine signaling in the mesolimbic pathway, which is thought to be overactive in psychosis.
- Partial Agonists: These drugs (e.g., Aripiprazole, Brexpiprazole, Cariprazine) act as a "dimmer switch." They have lower intrinsic activity than dopamine itself. In areas of high dopamine (like the mesolimbic pathway in psychosis), they act as functional antagonists by displacing the more potent dopamine and producing a weaker signal. In areas of low dopamine (like the mesocortical pathway, linked to negative/cognitive symptoms), they can provide a gentle boost to dopamine signaling.





• **Therapeutic uses:**

- Treatment of psychosis and schizophrenia (1st line).
 - As tranquilizers in manic states.
 - Risperidone is used in short-term treatment of persistent aggression in patients with moderate to severe dementia of the Alzheimer's
- 
- 
- 
- 



• Adverse effects:

- Weight gain, diabetes, and hypercholesterolemia due to 5-HT_{2A} block. This effect is milder with partial agonists but this also makes them weaker in severe cases or poor responders
- NMS
- Leukopenia: often fatal and is common with Clozapine.
- EPS can still occur with the atypical drugs although the risk is less than that of the classic drugs.
- Both first- and second-generation drugs can prolong QT interval making them torsadogenic drugs.



Brands





Maintenance treatment



- Patients who have had two or more psychotic episodes secondary to schizophrenia should receive maintenance therapy for at least 5 years, and some experts prefer indefinite therapy.
- Low doses of antipsychotic drugs are not as effective as higher-dose maintenance therapy in preventing relapse.
- The rate of relapse may be lower with second-generation drugs





+ Challenges with antipsychotic therapy

- Adherence to antipsychotic treatment can be challenging to patients for many reasons including:
 - Poor insight into the illness
 - Denial of the illness
 - Cognitive impairments
 - Adverse effect profile (80% clinically significant adverse effects)
 - The use of long-acting formulation (haloperidol retard inj) will not prevent all forms of non-adherence, however, it can give the clinician more insight in the patient's state.
 - When using FGAs or multiple SGAs a central acting anti-muscarinic agent maybe added to reduce tremors and Parkinson-like symptoms as long as it doesn't progress into dyskinesia. +
- 

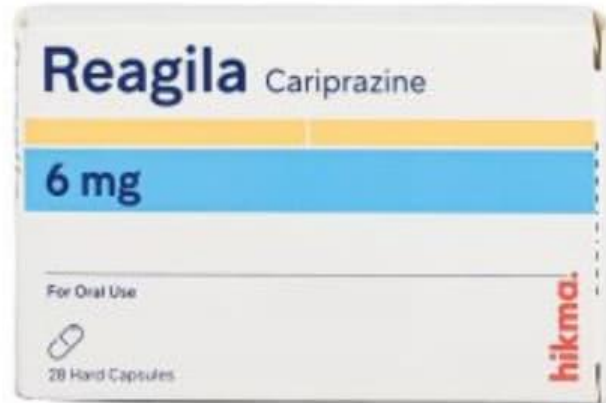
Further developments

- **Cariprazine** is a dopamine D2/D3 receptor partial agonist, with a distinctively high affinity for D3.
- This preferential D3 activity is unique among its class and may contribute to improvements in negative symptoms and cognitive function Psychiatry.
- It also modestly acts as a 5-HT1A partial agonist, and antagonizes 5-HT2A/2B and H1 receptors, while having minimal binding to 5-HT2C and α -adrenergic receptors, and negligible affinity for muscarinic cholinergic receptors
- Cariprazine is considered a third-generation anti-psychotic.

Further developments

- **Cariprazine is FDA-approved for:**
 - Schizophrenia (maintenance and symptom reduction)
 - Bipolar I disorder (manic or mixed episodes, and bipolar depression)
 - Major depressive disorder (MDD) as adjunctive therapy in unipolar depression (approved December 2022)
- Olanzapine and others are widely used in schizophrenia and bipolar mania, but not typically for depression.
- It has a low liability for metabolic effects—minimal weight gain, changes in glucose, lipids, prolactin, or QTc prolongation—all notably lower than with many SGAs like olanzapine or risperidone

Brands



 UnitedPharmacy0

 United.Pharmacy

Cases

- Mr S, a 25-year-old white pharmacy student, comes to your pharmacy in quite a distressed state. He is totally unkempt and smells of body odors; his mood is volatile and he reports to you that he has sensations of insects crawling under his skin. Over the last few days he has been feeling sick, and he is absolutely convinced that this is because his neighbor somehow put some toxic gases into his flat. He brings you some of his vomit to be analyzed, so that his concerns will now be 'demonstrated' and he will be able to take appropriate action against his neighbor .

Questions

- What is your diagnosis ?
- If Mr S start his treatment with Chlorpromazine then his mother noticed ha had developed some repeated involuntary movements, he told you that he had never suffer from it before .
- Write your comment about this and your recommendation .

Answers

- Schizophrenia .
- He had developed Extrapiramidal side effects , Which can be avoided by shifting to Atypical anti psychotic drugs (Risperidone)

Cases

An adolescent male is newly diagnosed with schizophrenia. Which of the following antipsychotic agents may have the best chance to improve his apathy and blunted affect?

- A. Chlorpromazine.
- B. Fluphenazine.
- C. Haloperidol.
- D. Risperidone.
- E. Thioridazine.

Answers

- Correct answer = D. Risperidone is the only antipsychotic on the list that has some reported benefit in improving the negative symptoms of schizophrenia. It is a second-generation antipsychotic, and the other drugs listed are first generation antipsychotic agents. All of the agents have the potential to diminish the hallucinations and delusional thought processes (positive symptoms).

Cases

- Which one of the following antipsychotics has been shown to be a partial agonist at the dopamine D2 receptor?
 - A. Aripiprazole.
 - B. Clozapine.
 - C. Haloperidol.
 - D. Risperidone.
 - E. Thioridazine.

Answers

- Correct answer = A. Aripiprazole is the agent that acts as a partial agonist at D2 receptors. Theoretically, the drug would enhance action at these receptors when there is a low concentration of dopamine and would block the actions of high concentrations of dopamine. All of the other drugs are only antagonistic at D2 receptors, with haloperidol being particularly potent.

Discussions

- **The Dopamine Dilemma: Mechanism vs. Reality**
- **Key Question:** "We know that blocking D2 receptors in the mesolimbic pathway treats positive symptoms, but why does this same action in the nigrostriatal pathway cause EPS? How does this fundamental 'trade-off' shape our entire approach to antipsychotic therapy?"
- **Follow-up/Prompt:** "If you had to explain to a patient why their medication might cause muscle stiffness or restlessness, how would you frame it in simple terms?"

Discussions

- **Case Study Deep Dive: Mr. S (4 minutes)**
- **Key Question:** "Let's revisit Mr. S. Beyond the EPS, what other adverse effects of Chlorpromazine should we be monitoring for, based on its receptor profile (e.g., alpha, muscarinic, histaminic blockade)?"
- **Follow-up/Prompt:** "His mother is now worried about switching to Risperidone after reading online about 'metabolic side effects.' How would you, as the pharmacist, counsel her on balancing the risk of EPS with the risk of weight gain/diabetes?"

Discussions

- **The Pharmacologist's Puzzle:** Partial Agonism vs. Full Antagonism
- **Key Concept:** Moving from the simple "blockade" model to the nuanced "stabilization" model of partial agonism.
- **Key Question:** "We have Aripiprazole and Cariprazine as partial agonists. The lecture says they 'block' high dopamine and 'stimulate' low dopamine. How does this theoretical mechanism translate into practical clinical advantages and disadvantages compared to a pure antagonist like Risperidone?"
- **Follow-up/Prompt:** "Imagine a patient on Risperidone is stable but experiencing significant emotional blunting (a negative symptom). Why might switching to a partial agonist like Aripiprazole be a considered strategy? What new side effect might you warn them about initially?" (This leads them to think about the initial activation or akathisia sometimes seen with partial agonists).

Discussions

- **The Ultimate Clinical Challenge:** Treatment Resistance and Clozapine
- **Key Concept:** Applying knowledge of drug mechanisms and side effect profiles to a high-stakes clinical scenario.
- **Key Question:** "Let's say Mr. S from our case was switched to Risperidone, but after an adequate trial, his positive symptoms (the delusions about his neighbor) persist. The team is now considering Clozapine. Why is Clozapine the gold standard for treatment-resistant schizophrenia, and what monumental responsibility does this place on the pharmacist?"
- **Follow-up/Prompt:** "The slide mentions 'leukopenia: often fatal.' What specific, concrete monitoring protocol must the pharmacist ensure is in place before and during Clozapine therapy? What would you tell the patient and their family is the most important reason for these weekly blood tests?"

CNS

By Dr.Riffat Ibrahim Riffat



03

Parkinson's disease

- Parkinson's disease is a chronic neurodegenerative disease with an unknown cause.

- It is characterized by:

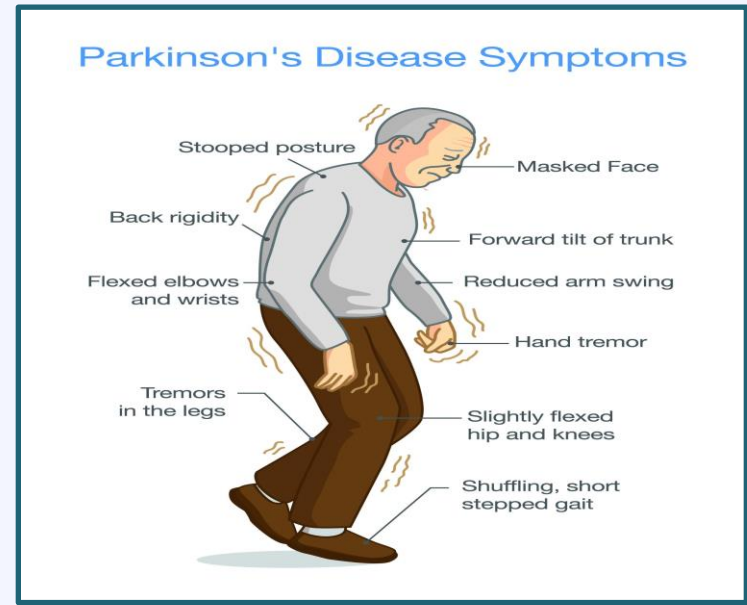
- Neurodegeneration of dopaminergic neurons (inhibitory) in the nigrostriatal pathway which causes acetylcholine (stimulatory) to take the upper hand.

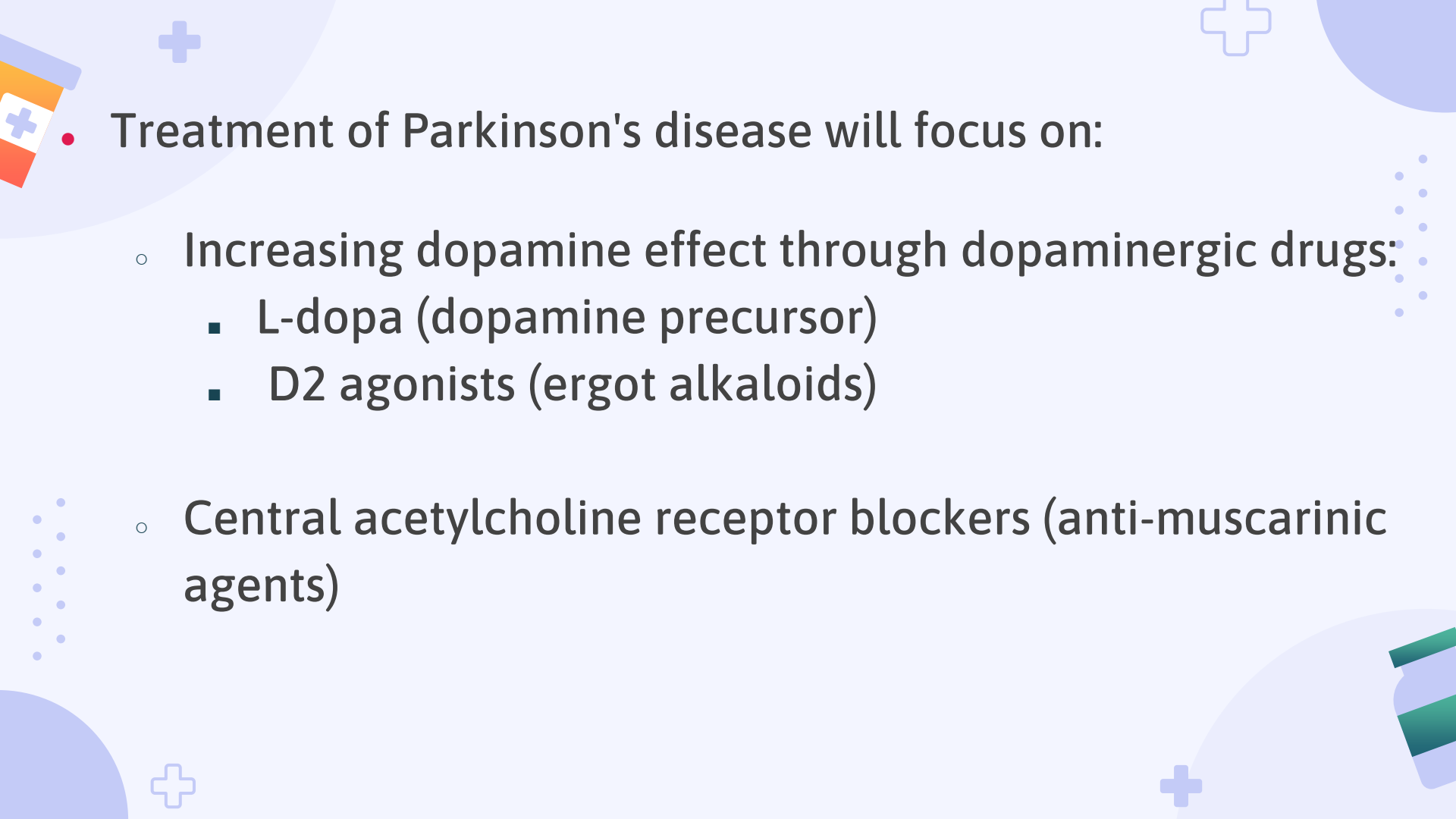
- Motor manifestations:

- Tremors.
- Muscle rigidity.
- Bradykinesia (shuffling gait).

- Non motor manifestations:

- Depression.
- Disturbance of sleep.
- Dementia

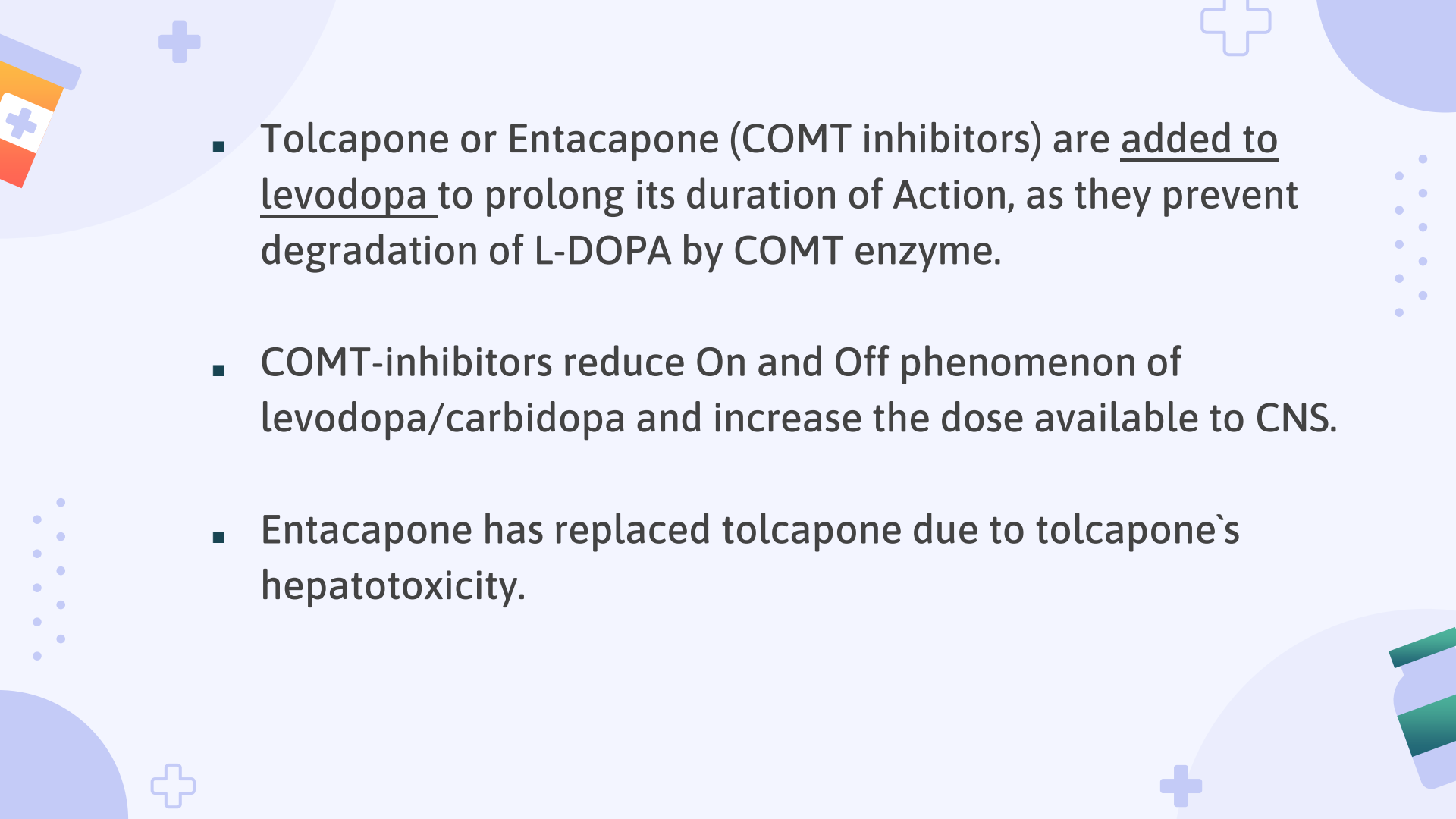


- 
- Treatment of Parkinson's disease will focus on:
 - Increasing dopamine effect through dopaminergic drugs:
 - L-dopa (dopamine precursor)
 - D2 agonists (ergot alkaloids)
 - Central acetylcholine receptor blockers (anti-muscarinic agents)

L-DOPA

Kinetics:

- It is a dopamine precursor as dopamine itself can not cross B.B.B.
- Well absorbed orally, but absorption is greatly affected by food.
- Very short half-life (1-2 hours) due to peripheral decarboxylation by hepatic decarboxylases and COMT metabolism resulting in:
 - the on/off phenomena, as most of the dose is metabolized peripherally not centrally which significantly reduces the amount that reaches CNS (actual target site).
 - Various peripheral side effects.
- Carbidopa is added to reduce peripheral decarboxylation of L-DOPA increasing the amount available to CNS from 5% of the dose to 60%

- 
- Tolcapone or Entacapone (COMT inhibitors) are added to levodopa to prolong its duration of Action, as they prevent degradation of L-DOPA by COMT enzyme.
 - COMT-inhibitors reduce On and Off phenomenon of levodopa/carbidopa and increase the dose available to CNS.
 - Entacapone has replaced tolcapone due to tolcapone's hepatotoxicity.

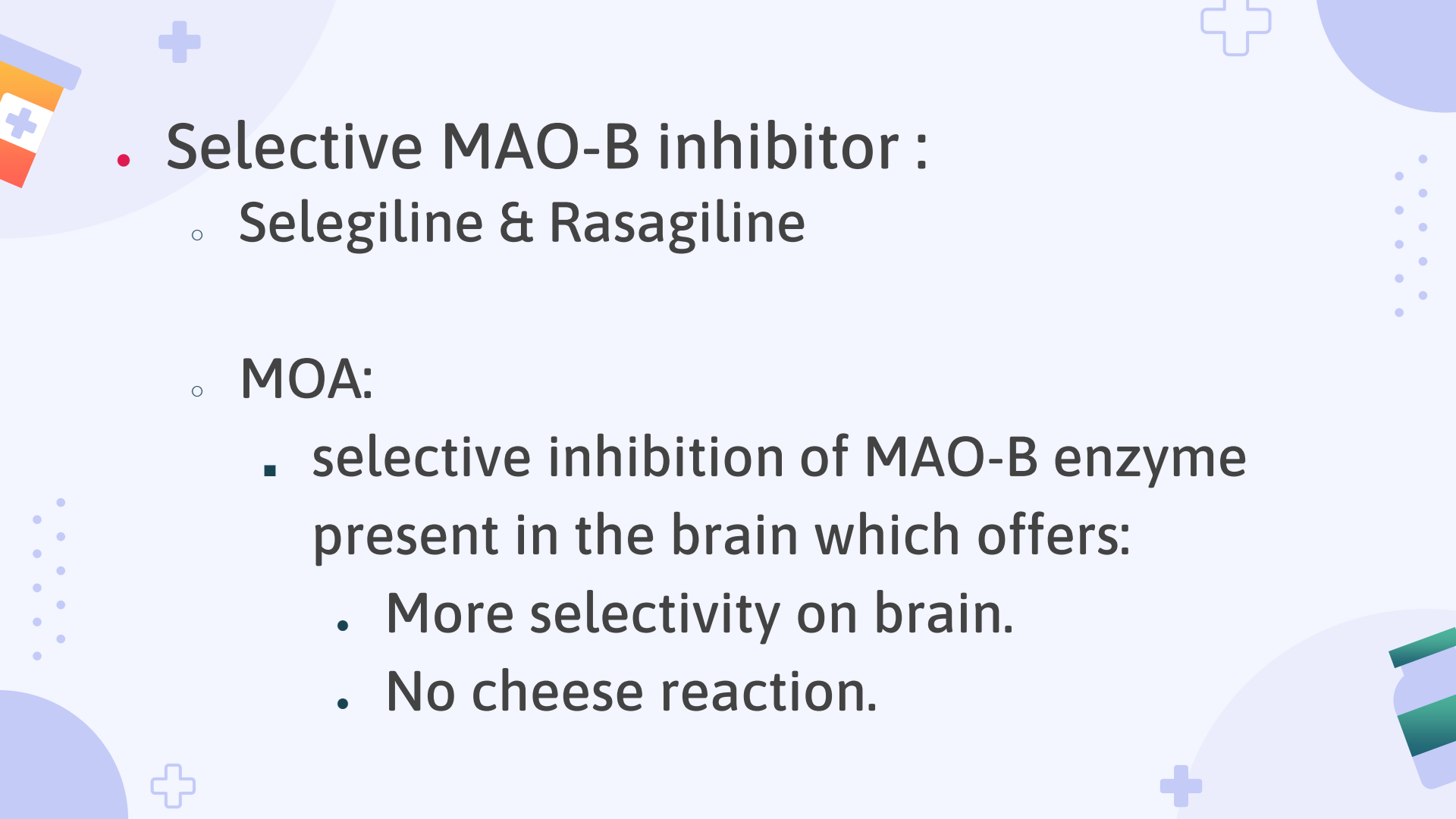
- Adverse effects:
 - CNS related side effects:
 - Nausea and Vomiting due to increased dopamine level in the vomiting center.
 - To solve this problem , use a Dopaminergic Blocker that can not pass BBB eg : (Domperidone).
 - Mood Changes (hallucination and delusion) due to increased dopamine level in mesocortical and mesolimbic pathways.
 - On and Off phenomenon (Sudden improvement followed by sudden immobility).

- **Peripheral side effects:**

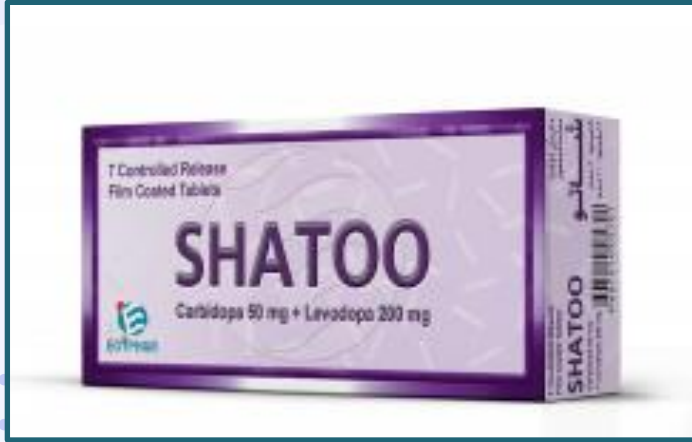
- Orthostatic hypotension and Arrhythmia due to:
 - Dopamine receptor activation in the heart (+ve Chrono and inotropic effect)
 - Dopamine receptor activation in the blood vessels (Vasodilatation)
- The most specific side effect of levodopa is brownish urine and sweat due to secretion of homovanillic acid (the metabolite of l-dopa) in urine and sweat.

- Important drug interactions:

- Levodopa should be taken on empty stomach to avoid leucine and Isoleucine interaction with levodopa absorption.
 - Pyridoxine(Vitamin B6) is contraindicated with Levodopa because of increased peripheral decarboxylation of levodopa.
 - Antipsychotic drugs (eg: Olanzapine) will prevent therapeutic effect of levodopa.
- 
- 

- 
- Selective MAO-B inhibitor :
 - Selegiline & Rasagiline
 - MOA:
 - selective inhibition of MAO-B enzyme present in the brain which offers:
 - More selectivity on brain.
 - No cheese reaction.

Brands



- Dopaminergic agonist:

- Ergot alkaloid : bromocriptin.

- MOA:

- Dopaminergic agonist.

- Bromocriptine has long Duration of action, It is not necessary to combine with COMT or MAO inhibitors. This explains why the drug exhibits no on and off phenomenon.

- Side effects:

- Orthostatic hypotension , Hallucination ,delusion , Nausea And vomiting (very similar to L-dopa due to it's agonistic action of dopamine receptors centrally and peripherally.
- Pulmonary fibrosis very specific to bromocriptin.



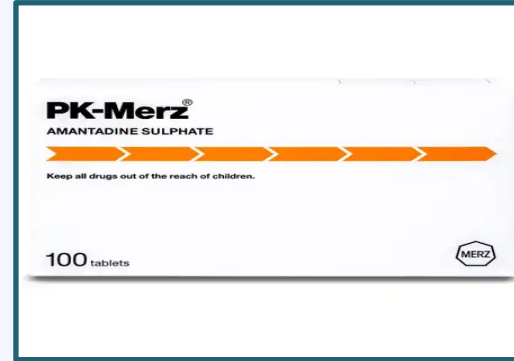
Amantadine: Antiviral drug that treats influenza and was by chance found to have anti parkinsonian effect.

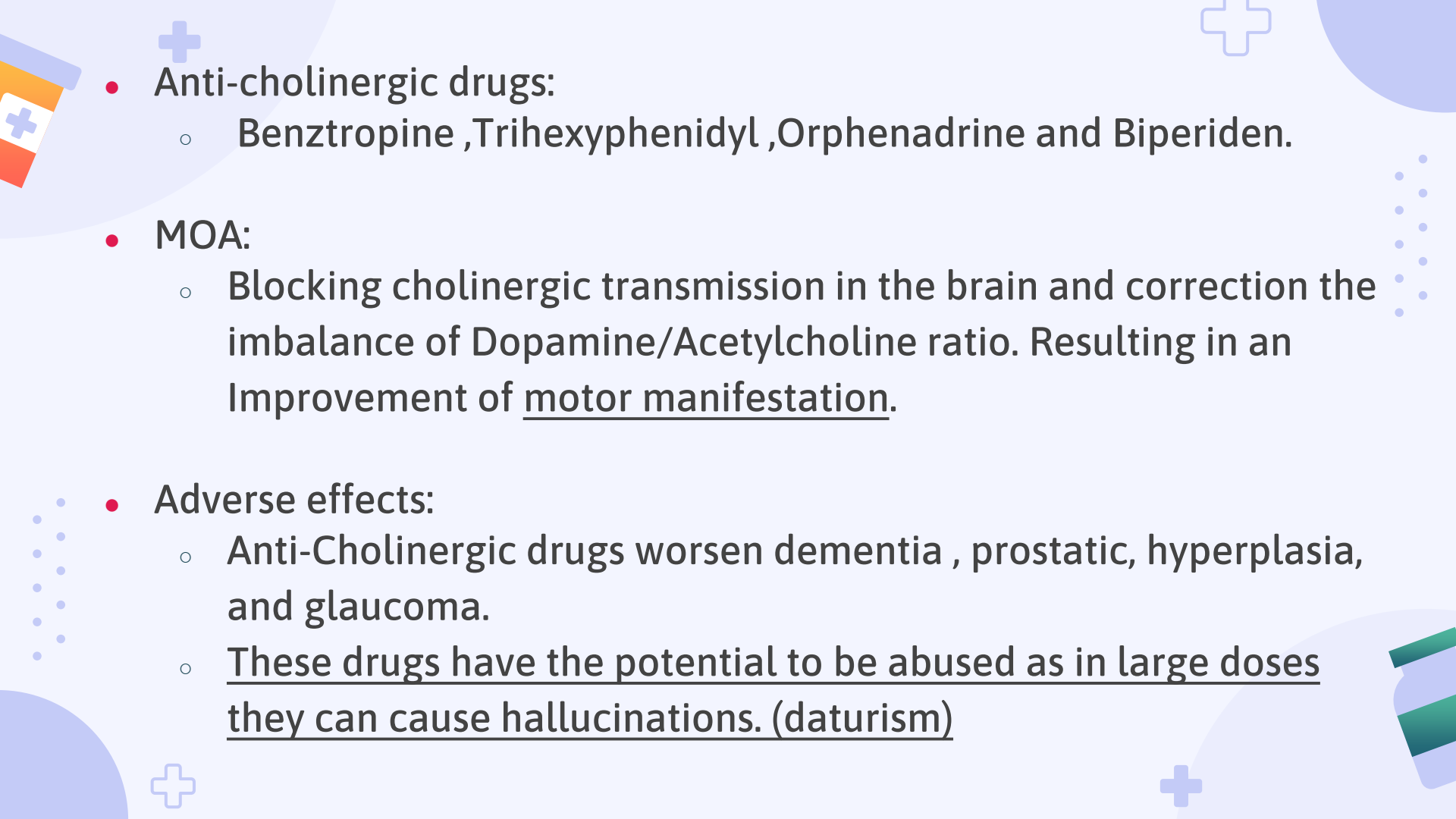
- MOA:

- is unclear , but it may potentiate dopamine synthesis ,release and reuptake.

- Side effects:

- Nausea , vomiting ,dizziness and postural hypotension.
- levido reticularis(mottled skin due to dilation of capillaries).



- 
- Anti-cholinergic drugs:
 - Benztropine ,Trihexyphenidyl ,Orphenadrine and Biperiden.
 - MOA:
 - Blocking cholinergic transmission in the brain and correction the imbalance of Dopamine/Acetylcholine ratio. Resulting in an Improvement of motor manifestation.
 - Adverse effects:
 - Anti-Cholinergic drugs worsen dementia , prostatic, hyperplasia, and glaucoma.
 - These drugs have the potential to be abused as in large doses they can cause hallucinations. (daturism)

Brands



Cases

- A 56-year-old man is diagnosed with Parkinson's disease.
- Carbidopa is prescribed. This agent reduces which of the following?
 - A. The activity of decarboxylase in the liver
 - B. The L -dopa dose necessary to achieve a therapeutic effect
 - C. The severity of L -dopa-associated dyskinesias
 - D. The time to onset of L -dopa's therapeutic effects

Answers

- Carbidopa, which does not penetrate the brain, reduces peripheral Dopa decarboxylase activity, and the metabolism of L -dopa. The therapeutic effect of L -dopa can be achieved at a lower dose than would be possible without carbidopa.

Cases

- Which one of the following combinations of antiparkinsonian drugs is an appropriate treatment plan?
 - A. Amantadine, carbidopa, and entacapone.
 - B. Levodopa, carbidopa, and entacapone.
 - C. Pramipexole, carbidopa, and entacapone.
 - D. Ropinirole, selegiline, and entacapone.
 - E. Ropinirole, carbidopa, and selegiline.

Answers

- Correct answer = B. To reduce the dose of levodopa and its peripheral side effects, the peripheral decarboxylase inhibitor, carbidopa, is coadministered. As a result of this combination, more levodopa is available for metabolism by catechol-O-methyltransferase (COMT) to 3-O-methyldopa, which competes with levodopa for the active transport processes into the CNS. By administering entacapone (an inhibitor of COMT), the competing product is not formed, and more levodopa enters the brain. The other choices are not appropriate, because neither peripheral decarboxylase nor COMT nor monoamine oxidase metabolizes amantadine or the direct-acting dopamine agonists, ropinirole and pramipexole.

Cases

- Peripheral adverse effects of levodopa, including nausea, hypotension, and cardiac arrhythmias, can be diminished by including which of the following drugs in the therapy?
 - A. Amantadine.
 - B. Ropinirole.
 - C. Carbidopa.
 - D. Tolcapone.
 - E. Pramipexole.

Answers

- Correct answer = C. Carbidopa inhibits the peripheral decarboxylation of levodopa to dopamine, thereby diminishing the gastrointestinal and cardiovascular side effects of levodopa. The other agents listed do not ameliorate adverse effects of levodopa.

CNS

By Dr.Riffat Ibrahim Riffat

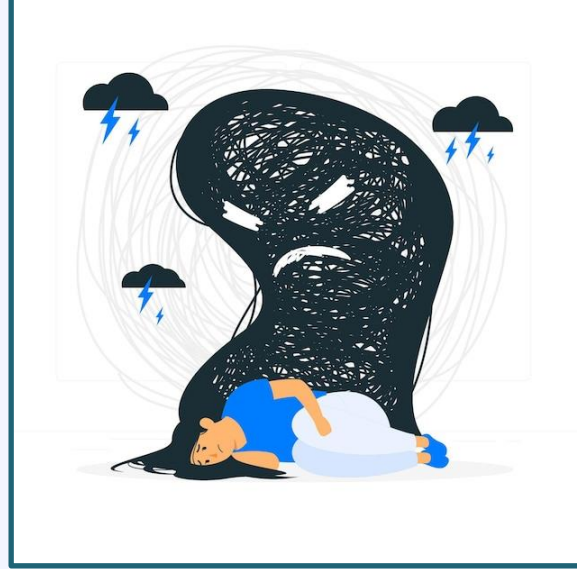
05 Depression





Introduction

- Depression is a disorder of mood.
- Globally it affects over 350 million people all over the world.
- It affects women more than men.
- Depression is considered a major cause of disability.
- Depression can lead to suicide.



Pathophysiology

Genetics:

- More than 4 genes were identified to cause depression.



- Biogenic amines and receptor theory:
 - Low levels of dopamine, serotonin, nor-epinephrine in brain.
 - Upregulation of certain receptors in the brain eg: 5HT2a and 5HT2c
 - This theory is the most accepted theory.
- Note: bio amine theory and genetics offer a short-handed explanation when it comes to trauma or experience caused depression.

Signs & Symptoms

- Symptoms of Depression:

- Emotional Symptoms :

- low levels of serotonin are the cause of emotional symptoms.
- loss of interest and enjoyment.
- Apathy. (lack of interest and emotional expression)
- loss of Self steam.
- Loss of Motivation.

- Biological Symptoms (Psycho-somatic disorder):

- low levels of nor-epinephrine are the cause of biological symptoms.
- Chronic pain.
- Loss of appetite.
- Sleep disorder.

BACKGROUND

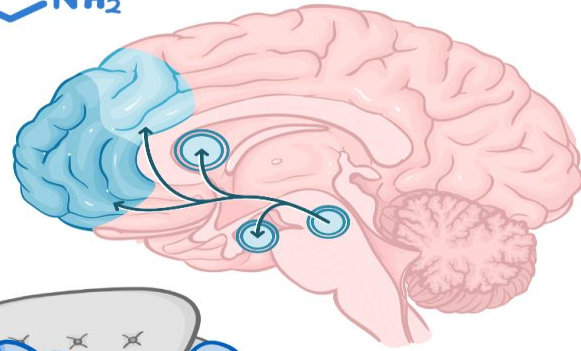
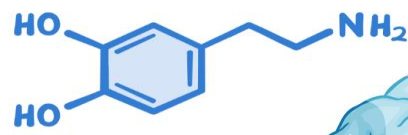
* LOSS of ABILITY to FEEL PLEASURE

~ COMMON SYMPTOM of DEPRESSIVE & SUBSTANCE USE DISORDERS



DIAGNOSIS

- * SNAITH-HAMILTON PLEASURE SCALE (SHAPS)
- * POSITIVE & NEGATIVE AFFECT SCHEDULE (PANAS)
- * ADDITIONAL TESTS
 - ~ SUBSTANCE USE
 - ~ VITAMIN DEFICIENCIES
 - ~ THYROID FUNCTION



CAUSES

- * DEPRESSIVE DISORDERS
- * TRAUMA or STRESS
- * DECREASED ACTIVITY in VENTRAL STRIATUM
- * DYSREGULATION of HORMONES
 - ~ GABA, DOPAMINE



TREATMENT

- * COGNITIVE BEHAVIORAL THERAPY
- * AEROBIC EXERCISE
- * ELECTROCONVULSIVE THERAPY
- * MEDICATION



Anti-depressant drugs

- Families are:
 - Monoamine oxidase inhibitors (MAO). 1960
 - Tricyclic anti-depressants (TCA). 1970
 - Selective serotonin reuptake inhibitors (SSRI). 1990
 - Atypical anti-depressants (heterocyclic). 2000
- SSRI and TCA are the most used drugs in treatment of depression.
- Anti-depressants need 2 weeks to produce a clinically significant effect, maximum benefit is observed at 12 weeks.



Medication and psychotherapy

- Despite their obvious clinical efficacy in treatment of mood disorder associated with depression and other diseases like general anxiety disorder (GAD), anti-depressants are not the first line treatment. **Psychotherapy with or without medication is the first line treatment to depression.**
- There is a great debate within the scientific community about when should anti-depressants be prescribed as only using anti-depressants without psychotherapy has caused cases of rebound depression once drug therapy ceases even when proper guidelines are followed.



Monoamine Oxidase inhibitors (MAO inhibitors)

Monoamine oxidase enzyme has to 2 sub types:

- Type A present in brain and GIT
- Type B in brain only and metabolizes dopamine only

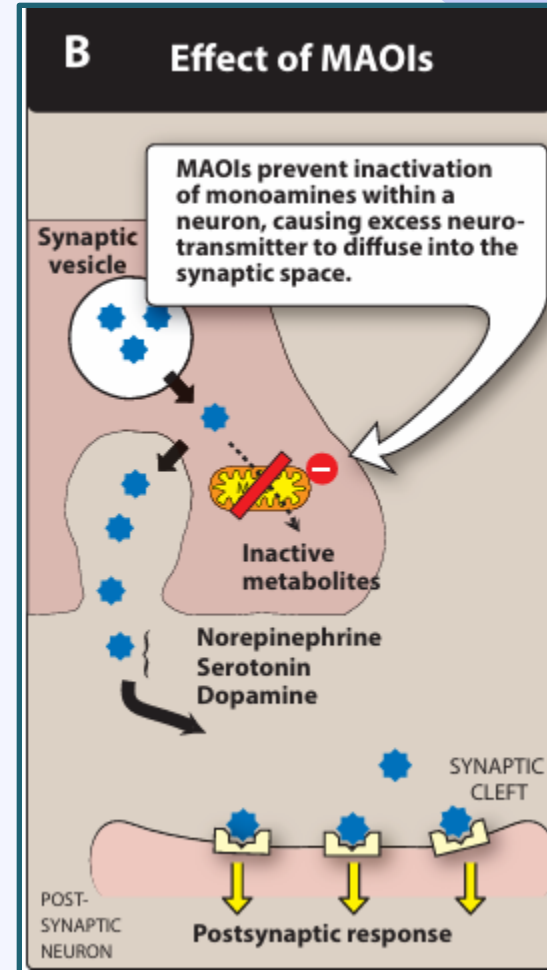
- Members:

- Phenelzine (A&B)
- tranylcypromine (A&b)
- selegiline. (B only)

- M.O.A:

- They prevent metabolism of reuptaken bio amines by MAO enzyme, thus increasing the amount available of bio amines to the receptors.

- MAO inhibitors are **not used as first line** anti-depressant drug due to their Severe side effects, but used for People who are unresponsive to other antidepressant drugs.



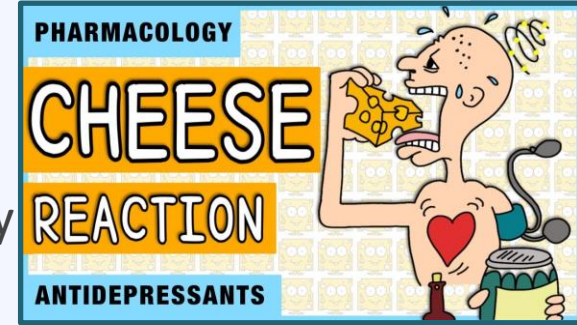
- Adverse effects:

- The following adverse effects are common with all anti-depressants:

- Orthostatic (Postural) hypotension, due to overall increase of bio amines in brain.
 - Sexual dysfunction in the form of delayed orgasm. This effect can treat pre-mature ejaculation, however, if the problem is not present this drug causes dysfunction.
 - Serotonin Syndrome (hyperthermia , arrhythmia and increased mortality rate) this effect increases significantly when combined with other serotonergic drugs.
 - CNS abnormalities depending on the drug
 - stimulation (Irritability and convulsions)
 - Sedation. (sleep and drowsiness)
- 

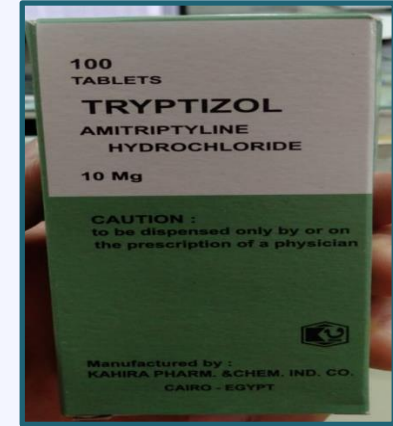
The following adverse effects are for MAO inhibitors only:

- Cheese Reaction (Hypertensive Crisis):
 - Certain foods like yogurt contain tyramine (has a sympathomimetic effect) which is metabolized by MAO.
 - Patients on MAO.I can not metabolize tyramine leading to its accumulation in the body
 - This accumulation results in:
 - Headache , Tachycardia ,hypertension , arrhythmia And seizures.
- MAO inhibitors are contraindicated in combination with other antidepressants to avoid Serotonin syndrome.
- CNS stimulation.



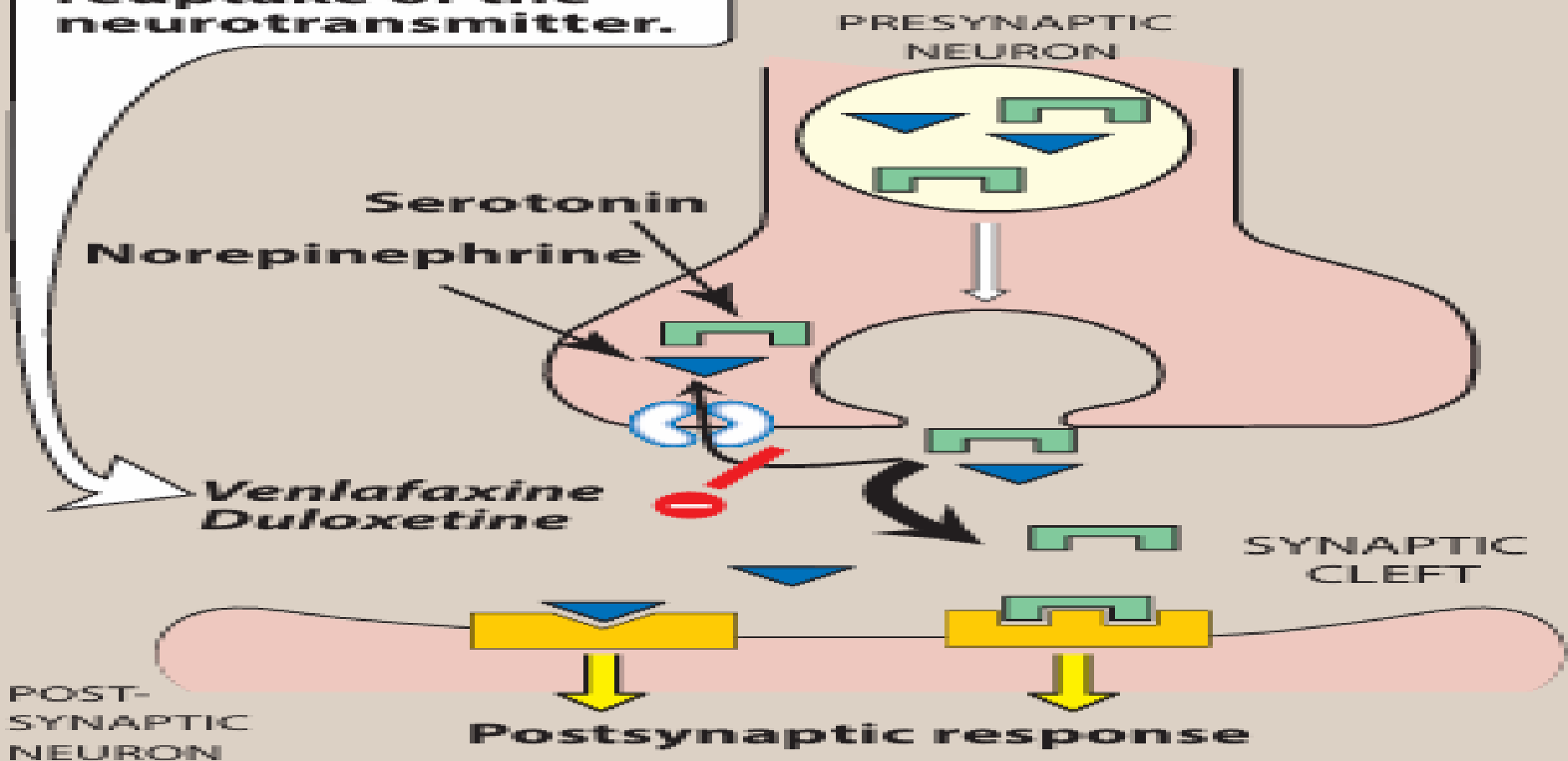
Tricyclic anti-depressants

- Members include :
 - Imipramine
 - Clomipramine
 - Amitriptyline
 - Nor-amitriptyline .



- MOA : Inhibition of Serotonin and Norepinephrine reuptake thus increasing the amount available in the synaptic space.
- Therapeutic Uses:
 - Major depression.
 - Chronic pain (Fibromyalgia)
 - Enuresis (Bed wetting problem) due to their blocking of muscarinic receptors.

Antidepressant drug blocks reuptake of the neurotransmitter.



- Adverse effects:

- CNS sedation (due to Atropine like action).
- Atropine like adverse effects.
- Hepatotoxicity in the form of cholestatic jaundice (very rare).

- (TCA toxicity):

- This toxicity is associated with large doses of the drug often suggesting a suicide attempt.
- Clinical manifestations are:
 - Atropine like toxicity.
 - Metabolic acidosis.
 - Cardiac arrhythmia.
- Treatment:
 - I.V NaHCO_3
 - I.V lidocaine if arrhythmia is not treated by NaHCO_3



+ Selective Serotonin Reuptake inhibitor

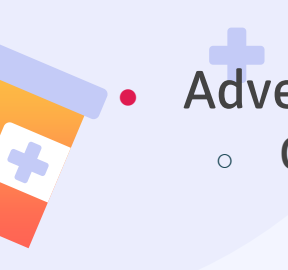
- Members include:

- Sertraline
- fluoxetine
- Citalopram
- escitalopram
- paroxetine.



- Therapeutic Uses:

- SSRI are first line treatment in major depression due to their safety even in large doses
- Obsessive compulsive disorder (OCD).
- Anxiety. (GAD)

- 
- 
- 
- 
- 
- Adverse effects:
 - CNS sedation or stimulation (usually disappears within 1-2 weeks).
 - GIT disturbance (prescribe PPI) and erectile dysfunctions are the most common side effects of SSRI and they disappear after 2 weeks of treatment.
 - SSRIs must be stopped gradually (over 4 weeks) to avoid Acute attack of Depression.
 - Weight gain or weight loss which is drug dependent.
 - Overdose with SSRIs does not usually cause cardiac arrhythmias, with the exception of citalopram, which may cause QT prolongation, however, QT prolongation with overdose is still a risk.

Atypical Antidepressants:

- Noradrenaline and Serotonin reuptake inhibitors :
 - Venlafaxine and duloxetine.
 - Duloxetine is commonly prescribed for neuropathic pain.
- Receptor Blocker: mirtazapine and trazodone
 - Trazodone blocks 5HT_{2A} and 5HT_{2C}
 - Mirtazapine blocks 5HT_{2C}
 - Mirtazapine causes Less GIT upset and sexual dysfunction
 - And thus, can used in case if the patient wants to stop SSRI, first by gradual withdrawal of the SSRI then using mirtazapine.
- SNRI & Serotonin receptor blockers are used similarly to SSRI

Brands



Bupropion is a weak dopamine and norepinephrine reuptake inhibitor that is used to alleviate the symptoms of depression.

- Bupropion is also useful for decreasing cravings and attenuating withdrawal of cigarette smoking.
- Side effects may include dry mouth, sweating, nervousness, tremor, and a dose dependent increased **risk for seizures.**



+ Patient specific considerations in antidepressant drug selection

- Given that many antidepressants are equally efficacious despite having different mechanisms of action and adverse effects. It becomes clinically significant to choose the best antidepressant for a specific patient in an individualized process.
- There are two aspects which come into consideration when choosing the best antidepressant for a patient:
 - Primary aspects for the patient.
 - Secondary considerations.

- 
- 
- 
- 
- Primary aspects for the patient:
 - Patient history with a specific antidepressant use.
 - Family history of antidepressant use
 - Adverse effect profile
 - Clinical presentation of depression
 - Comorbid medical/psychiatric conditions.
 - Secondary considerations include:
 - Age
 - Medication cost
 - Drug-food / drug-drug interactions
 - Genetic factors
 - Non-pharmacological intervention (psychotherapy)
 - History of non adherence to treatment.



Examples:

- Mirtazapine is known to cause weight gain and can cause poor compliance to treatment.
 - Bupropion is known to cause CNS stimulation and can be beneficial for patients experiencing hypersomnolence and detrimental to therapy when patient is experiencing insomnia.
 - The GIT and ED side-effects of SSRI often cause poor patient compliance and patient counseling is required.
- 

Sedating; may be useful for agitation

**SELECTIVE
SEROTONIN RE-UPTAKE
INHIBITORS**

Citalopram
Escitalopram
Fluoxetine
Fluvoxamine
Paroxetine
Sertraline

**SEROTONIN/
NOREPINEPHRINE
REUPTAKE INHIBITORS**

Duloxetine
Venlafaxine
Desvenlafaxine

**ATYPICAL
ANTIDEPRESSANTS**

Bupropion
Mirtazapine
Nefazodone
Trazodone

**TRICYCLIC/POLYCYCLIC
ANTIDEPRESSANTS**

Amitriptyline
Amoxapine
Clomipramine
Desipramine
Doxepin
Imipramine
Maprotiline
Nortriptyline
Protriptyline
Trimipramine

**MONOAMINE OXIDASE
INHIBITORS**

Phenelzine
Tranylcypromine

**High potential
for orthostatic
hypotension**

**Weight
gain**

**Gastro-
intestinal
distress**

Cases

- A 51-year-old woman with symptoms of major depression also has angle-closure glaucoma. Which of the following antidepressants should be avoided in this patient?
 - A. Amitriptyline.
 - B. Sertraline.
 - C. Bupropion.
 - D. Mirtazapine.
 - E. Fluvoxamine.

Answers

- Correct answer = A. Because of its potent antimuscarinic activity, amitriptyline should not be given to patients with glaucoma because of the risk of acute increases in intraocular pressure. The other antidepressants all lack antagonist activity at the muscarinic receptor.

Cases

- A 25-year-old woman has a long history of depressive symptoms accompanied by body aches and pain secondary to a car accident 2 years earlier. Physical and laboratory tests are unremarkable. Which of the following drugs might be useful in this patient?
 - A. Fluoxetine.
 - B. Sertraline.
 - C. Phenelzine.
 - D. Mirtazapine.
 - E. Duloxetine.

Answers

- Correct answer = E. Duloxetine is a SNRI that can be used for depression accompanied by symptoms of pain. SSRIs (fluoxetine and sertraline), MAOIs (phenelzine), and atypical antidepressants (mirtazapine) have little activity against pain syndromes.

Cases

- A 36-year-old man presents with symptoms of compulsive behavior. If anything is out of order, he feels that “work will not be accomplished effectively or efficiently.” He realizes that his behavior is interfering with his ability to accomplish his daily tasks but cannot seem to stop himself. Which of the following drugs would be most helpful to this patient?
 - A. Imipramine.
 - B. Fluvoxamine.
 - C. Amitriptyline.
 - D. Tranylcypromine.

Answers

- Correct answer = B. SSRIs are particularly effective in treating obsessive-compulsive disorder, and fluvoxamine is approved for this condition. The other drugs are less effective in the treatment of obsessive-compulsive disorder.

Cases

- A 55-year-old teacher began to experience changes in mood. He was losing interest in his work and lacked the desire to play his daily tennis match. He was preoccupied with feelings of guilt, worthlessness, and hopelessness. In addition to the psychiatric symptoms, the patient complained of muscle aches throughout his body. Physical and laboratory tests were unremarkable. After 6 weeks of therapy with fluoxetine, his symptoms resolved. However, the patient complains of stomach pains. Which of the following drugs might be useful in this patient?
 - A. Fluvoxamine.
 - B. Sertraline.
 - C. Citalopram.
 - D. Mirtazapine.

Answers

- D. Mirtazapine is largely free from gastric and sexual side effects, commonly associated with SSRIs (fluvoxamine, sertraline, and citalopram), as well as with TCAs, and SNRIs.

Discussions

- **1. The First-Line Controversy & The Pharmacist's Role**
- **Opening Question:** "The slides mention that psychotherapy, with or without medication, is the first-line treatment for depression, and that there's a debate about prescribing antidepressants alone. As the medication experts on the care team, what is our role when a prescriber wants to start an SSRI for a new patient?"
- **Potential follow-ups:**
 - "What key points should we include when counseling a patient starting their first antidepressant?"
 - "How do we manage patient expectations about the 2-week lag time for efficacy?"

Discussions

- **2. Adverse Effect Management & Patient Counseling**
- **Scenario:** "A patient on sertraline returns to the pharmacy after 10 days and complains of significant nausea and GIT upset. They are ready to stop the medication. What do we do?"
- **Potential follow-ups:**
 - How would your counseling for this side effects? (e.g., GIT upset)"
 - "The slides mention that mirtazapine can be a good alternative when SSRI side effects are intolerable. Why is that? What is its unique side effect profile we must warn about?"

Discussions

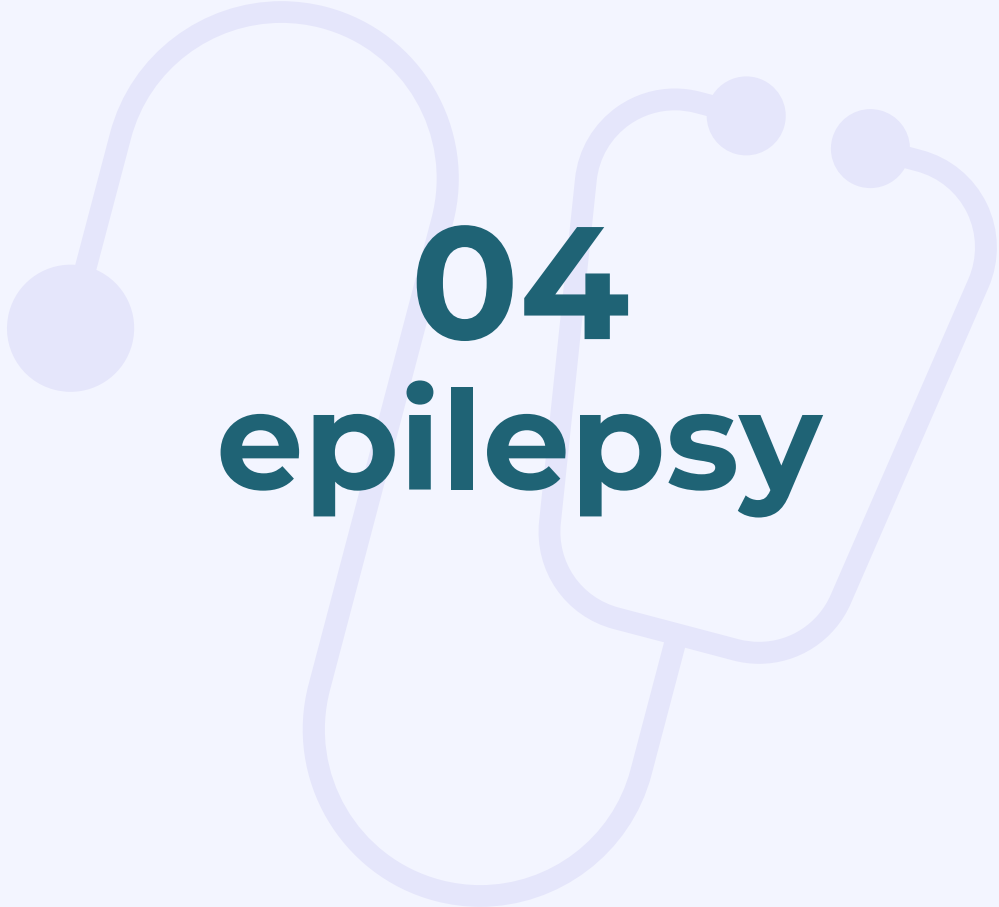
- **Case Study Rapid Fire & Rationale**
- **Use the cases from the lecture but ask for the clinical reasoning behind the answer.**
- **Case 1 (Glaucoma):** "Why exactly is amitriptyline contraindicated in angle-closure glaucoma? Let's talk about the receptor mechanism."
- **Case 2 (Chronic Pain):** "We chose duloxetine. Beyond 'it treats pain,' what is the pathophysiological reason an SNRI would be more effective for neuropathic pain than an SSRI?"

Discussions

- **Beyond the Pill: The Pharmacist's Holistic Role**
- **Discussion Starter:** "Let's look at the slide on 'Patient specific considerations.' A 22-year-old student and a 65-year-old retiree with the same diagnosis of major depression will likely have very different treatment plans. What factors from that list might be most important for each?"
- **Potential follow-ups:**
 - "For the student, how might medication cost and drug-drug interactions (e.g., with stimulants) influence our choice?"
 - "For the older adult, how do we factor in comorbidities and other medications (polypharmacy) when selecting an antidepressant?"
 - "How do we identify a patient who might be non-adherent and what strategies can we, as pharmacists, employ to help?"

CNS

By Dr.Riffat Ibrahim Riffat

A large, light purple outline of a human head in profile, facing right. A stethoscope is draped around the neck and over the head, with its chest piece resting on the forehead. The background is a light blue gradient with various medical icons: a clipboard with a checklist, a green pill bottle with a white cross, a red heart with a white cross, and a red and white pill bottle. There are also several small white plus signs and a trail of small blue dots.

04 epilepsy

Introduction

- epilepsy :
 - is a disease characterized by recurrent episodes of seizuers.
- Seizuers:
 - Are transient alterations of brain functions (motor, consciousness, visual, sensation, smell) due to abnormal neuronal discharge.
 - A single seizure does not qualify a patient to be diagnosed with epilepsy.
- Convulsions:
 - Involuntary often violent, spasms for skeletal muscles, could be related to seizures or not.



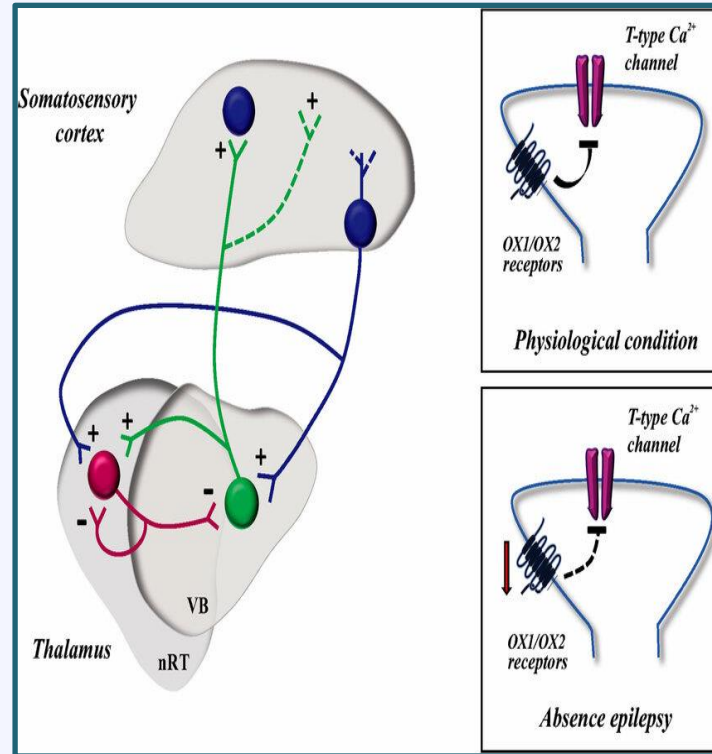
Patho physiology

All types of epilepsy can be explained by the following theory except absence seizures:

- An imbalance between glutamate and which activates NMDA receptors (excitatory) and the inhibitory GABA neurotransmitter leading to abnormal neuron activation.

In absence seizures :

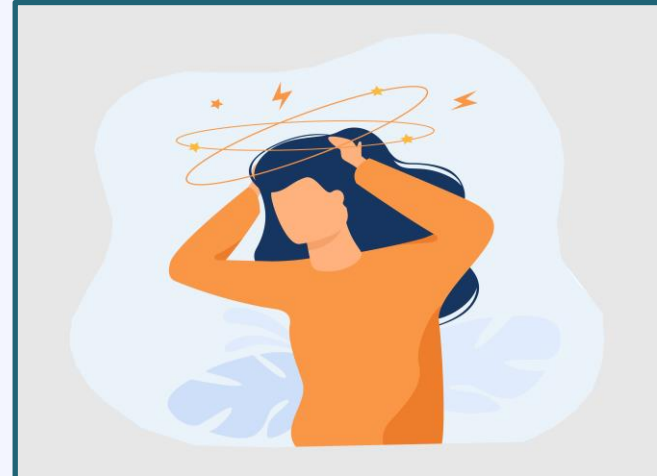
- Activation of t-type calcium channels in the relay neuron during the day causing a sleep like state.
- + This type is common in children



Causes

Causes of epilepsy maybe be:

- Genetic causes : affecting GABA production or receptors.
- Acquired:
 - Infections , eg: meningitis .
 - Head trauma .
 - Stroke (Brain bleeding) .



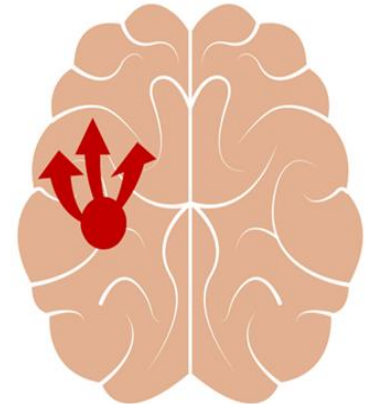
Classification of Epilepsy

Focal-onset epilepsy : localized in one area and in one hemisphere, the foci size will affect the symptoms (larger foci produces more noticeable symptoms)

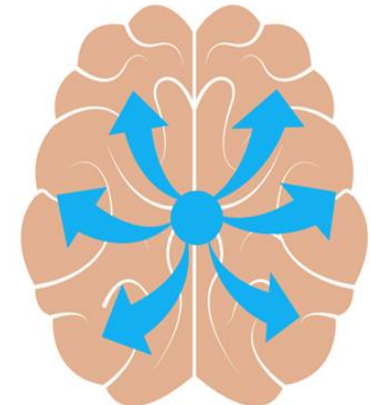
- Motor cortex: simple limb jerking or convulsions
- Visual area: visual abnormalities.
- Temporal cortex: auditory abnormalities.

Types :

- Simple focal seizures (With preserved awareness).
- Complex Focal seizures (impairment of awareness or loss of awareness)
- Focal seizures with secondary generalization .



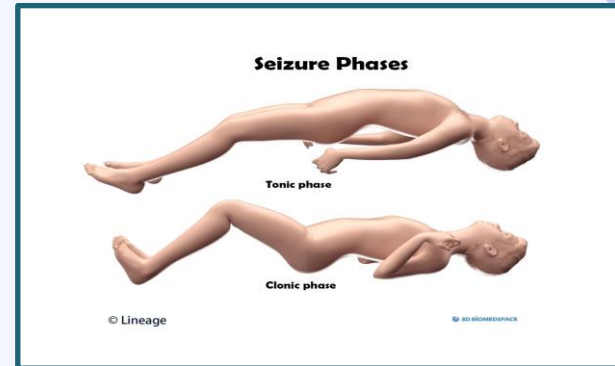
FOCAL



GENERALIZED

Generalized Onset seizures affecting both hemispheres of the brain:

- Absence seizures (Petit mal or little illness), a child can have over 100 mini episodes daily lasting 3-5 seconds.
- Tonic-Clonic seizures (Grand mal or Great illness).
- Myoclonic seizures with jerking movement.
- Loss of consciousness is constant with all three types of generalized seizures.



GENERALIZED SEIZURE: OTHER TYPES



Myoclonic Seizure

Non-rhythmic jerks resulting from involuntary muscle twitching that normally target the upper extremities, and after an episode, patients may describe the perceived sensation as momentary electrical shocks.

First generation drugs

- Members include:
 - Phenytoin
 - Carbamazepine
 - Valproic acid
 - Ethosuxiamide
 - phenobarbitone (very limited application)
 - Benzodiazepens. (limited to status epileptics I.V only and febrile seizures I.V or rectal suppositories)
- They mainly act on the inhibitory pathway.
- Adverse effect profile is very high with these agents, except ethosuxiamide.

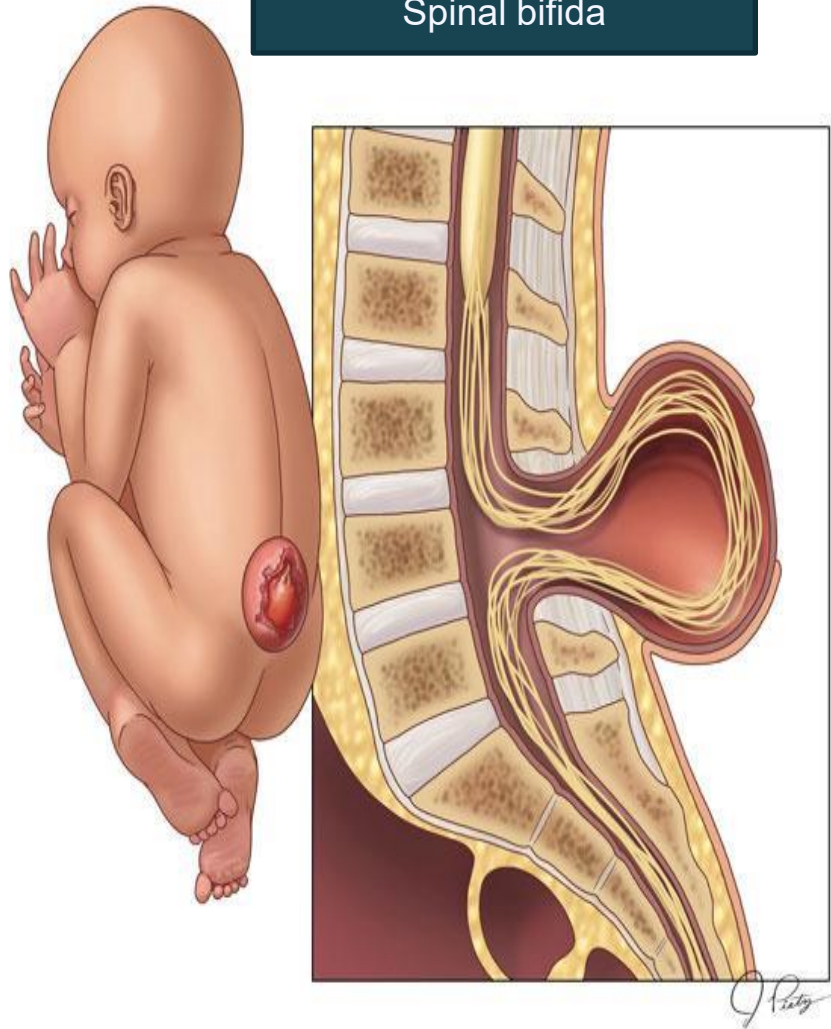
- Phenytoin or diphenyl hydantoin.
- MOA: Blocking sodium channels in Brain and heart.
- Therapeutic uses :
 - Treatment of Tonic-Clonic Seizures, other types of seiizures can be treated by phenytoin, however, newer agents are more superior and are safer.
 - Anti-arrhythmic drug .



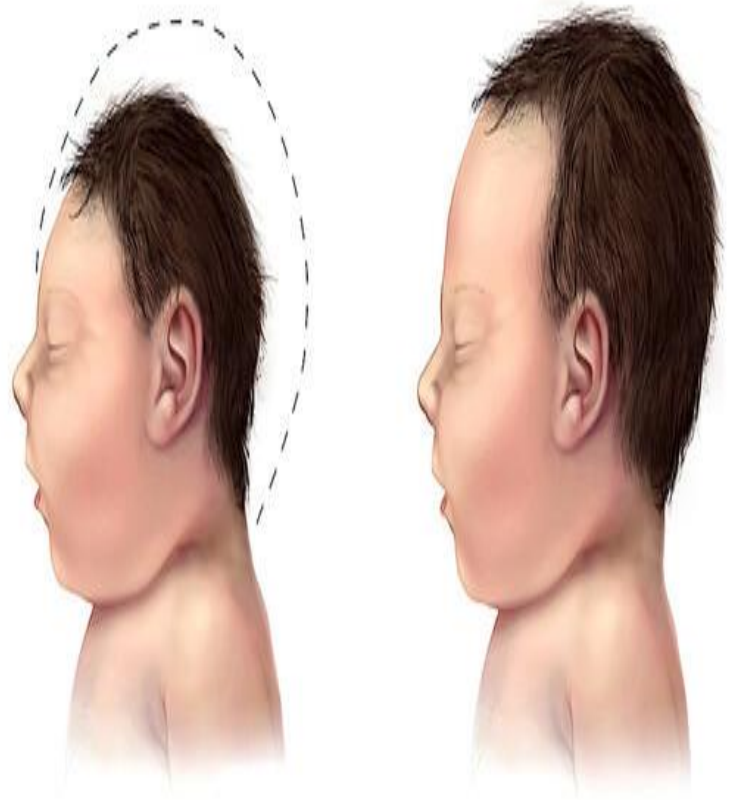
Adverse effects :

- The following adverse effects are common with all 1st generation epileptic drugs:
 - CNS (Nystagmus , Diplopia and Ataxia).
 - Liver (microsomal enzyme induction)
 - Blood (Megaloblastic anemia) due to decrease in folic acid caused by microsomal enzyme induction
 - Teratogenicity
 - It is about 3-4%, this effect is folic acid related, and supplementation is required with continuing the drug. Breast feeding is not an issue.
 - Craniofacial abnormality (Fetal hydantoin syndrome)
 - Neural tube defect (Spinal bifida)

Spinal bifida



Fetal hydantoin syndrome



Adverse effects :

- The following adverse effects are specific to phenytoin:

- Gingival hyperplasia.



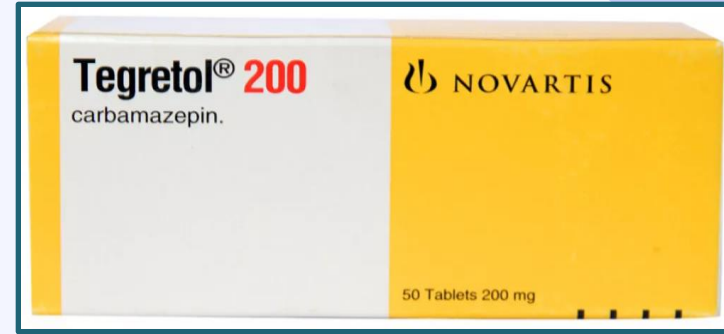
- Hirsutism.

- Osteomalacia, due to microsomal induction of vit. D metabolism.

- Lymphadenopathy.

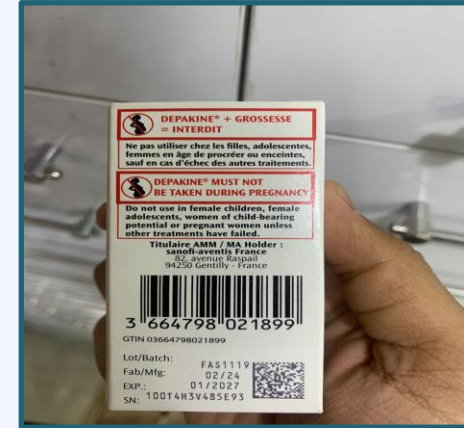


- Carbamazepine:
- MOA: Block Sodium channel in brain only
- Therapeutic Uses:
 - First choice in focal seizures.
 - First choice in trigeminal neuralgia.
 - Second choice in tonic-clonic seizures.
- Adverse effects:
 - Increase ADH secretion. (edema with hyponatremia)
 - Carbamazepine is characterized by auto-induction of its own metabolism.



- Valproic acid:
- MOA :
 - increase GABA inhibitory transmitter ,through inhibiting GABA transaminase that inhibits GABA metabolism.
 - Block Sodium channels in brain .
 - Block T-type calcium channels .

- Therapeutic Uses:
 - Treatment of all types of epilepsy.
- Adverse effects:
 - Liver : microsomal enzyme inhibitor.
 - Neutropenia .
 - Alopecia
 - Pancreatic and hepatic dysfunction (Rare but leads to death).
 - Valproic acid is strictly prohibited in pregnancy, as it has the ability to cause major abnormalities. (the drug comes with a black box warning)



- Ethosuximide:
- MOA: Selective T-type calcium channel blocker.
- Therapeutic uses: treatment of Absence seizures Only.
- Side effects:
 - Ethosuximide is the most safe anti-epileptic drug for absence seizures treatment.
 - GIT disturbance and Headache.





second generation drugs



- Members include:
 - Lamotrigine
 - Topiramate
 - Levetiracetam
 - Gabapentin and pregabalin
- They mainly act on the excitatory pathway.
- Adverse effect profile is much lower with these agents than 1st generation.



- Lamotrigine:
- MOA:
 - Blocking Sodium Channels
 - Inhibition of Glutamate.



- Therapeutic Uses:
 - Treats All types of Epilepsy Generalized, Focal and, even absence seizures.
 - Best choice in pregnancy
- Side effects:
 - Steven Johnson syndrome, Which increases with the combination of Lamotrigine and Valproic acid.

- Topiramate:
- MOA:
 - Block Sodium channels.
 - Potentiates GABA action.
 - Block AMBA receptors. (similar to NMDA receptor)
- Therapeutic uses:
 - Treatment of all types of epilepsy.
 - Migraine prophylaxis by an unknown mechanism
- Side effects:
 - Increase intraocular pressure (Glaucoma).
 - Renal stones.
 - Weight loss. (can be used for weight reduction, but it is still under clinical investigations)



- Levetiracetam:

- MOA:

- Binding Selectively to the vesicular protein SV2A and inhibition of Excitatory transmitters release.

- Therapeutic Uses:

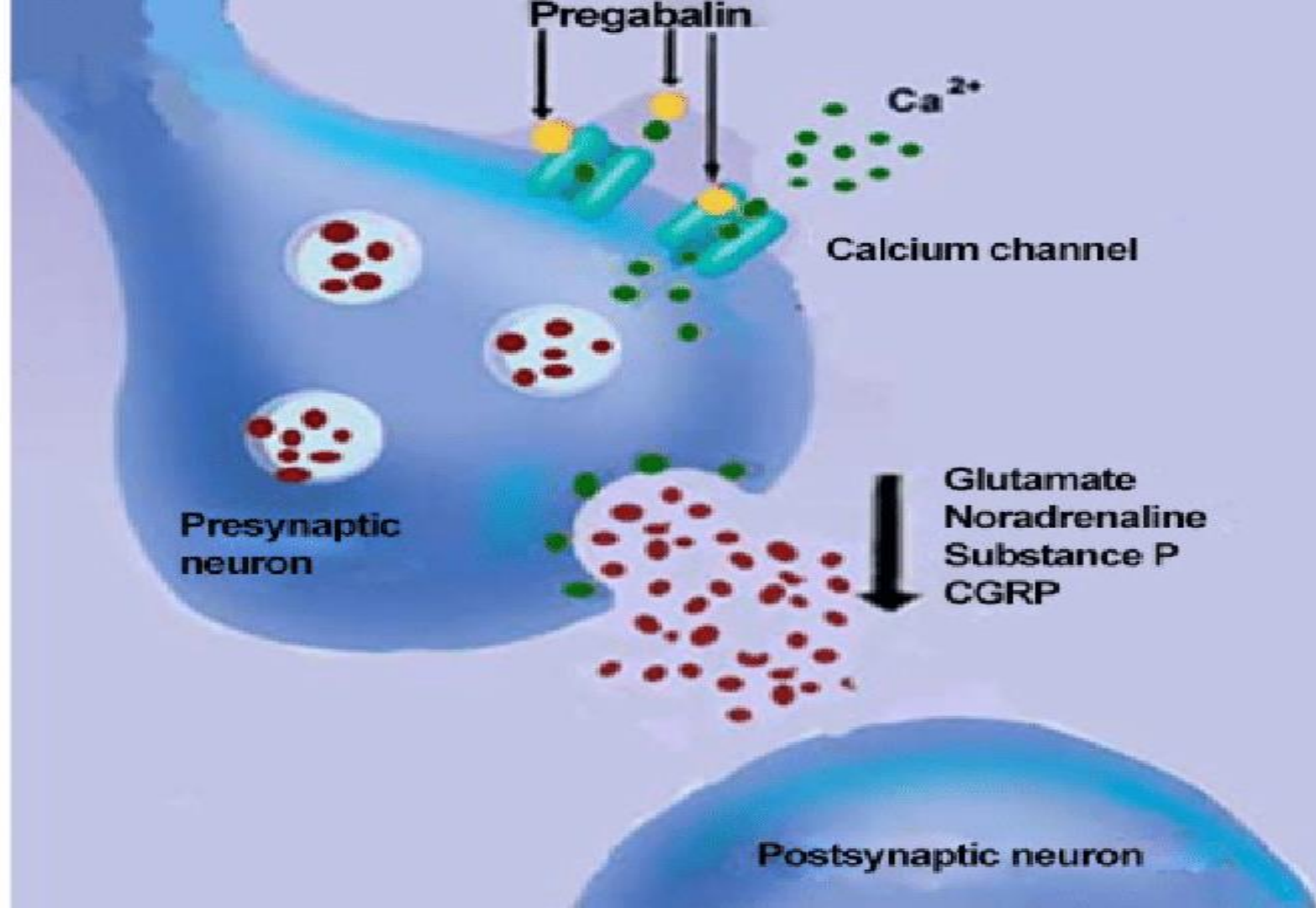
- Treatment of all types of epilepsy.

- Levetiracetam has very low side effects profile and is very safe in pregnancy.



- Gabapentin and pregabalin:
- MOA:
 - Binding to pre-synaptic P/Q calcium channels, preventing calcium influx and reducing glutamate release.
- Therapeutic Uses:
 - Adjuvant to other antiepileptic drugs.
 - Ideal drugs for neuropathy
- Adverse effects:
 - Sedation
 - Drug dependence and addiction, pregabalin is more likely to cause addiction than Gabapentin.





Cases

- A child is experiencing absence seizures that interrupt his ability to pay attention during school and activities.
- Which of the following therapies would be most appropriate for this patient?
 - Ethosuximide.
 - Carbamazepine.
 - Diazepam.
 - Carbamazepine plus primidone.
 - Watchful waiting.

Answers

- Correct answer = A. The patient has had many seizures that interrupt his ability to pay attention during school and activities, so therapy is justified. Monotherapy with primary agents is preferred for most patients. The advantages of monotherapy include reduced frequency of adverse effects, fewer interactions between antiepileptic drugs, lower cost, and improved compliance. Carbamazepine and diazepam are not indicated for absence seizures.

Cases

- A 25-year-old woman with myoclonic seizures is well controlled on valproate. She indicates that she is interested in becoming pregnant in the next year. With respect to her antiepilepsy medication, which of the following should be considered?

- A. Leave her on her current therapy.
- B. Consider switching to lamotrigine.
- C. Consider adding a second antiepilepsy medication.
- D. Decrease her valproate dose.

Answers

- Correct answer = B. Valproate is a poor choice in women of child-bearing age. A review of the medication history of this patient is warranted. If she has not tried any other antiepilepsy medication, then consideration of another anti-epilepsy medication may be beneficial. Studies show that valproate taken during pregnancy can have a detrimental effect on cognitive abilities in children.

Cases

- A 25-year-old woman with myoclonic seizures is well controlled on valproate. She indicates that she is interested in becoming pregnant in the next year. With respect to her antiepilepsy medication, which of the following should be considered?

- A. Leave her on her current therapy.
- B. Consider switching to lamotrigine.
- C. Consider adding a second antiepilepsy medication.
- D. Decrease her valproate dose.

Answers

- Correct answer = B. Valproate is a poor choice in women of child-bearing age. A review of the medication history of this patient is warranted. If she has not tried any other antiepilepsy medication, then consideration of another anti-epilepsy medication may be beneficial. Studies show that valproate taken during pregnancy can have a detrimental effect on cognitive abilities in children.

Skeletal muscle relaxants

By: Dr. Riffat Ibrahim Riffat

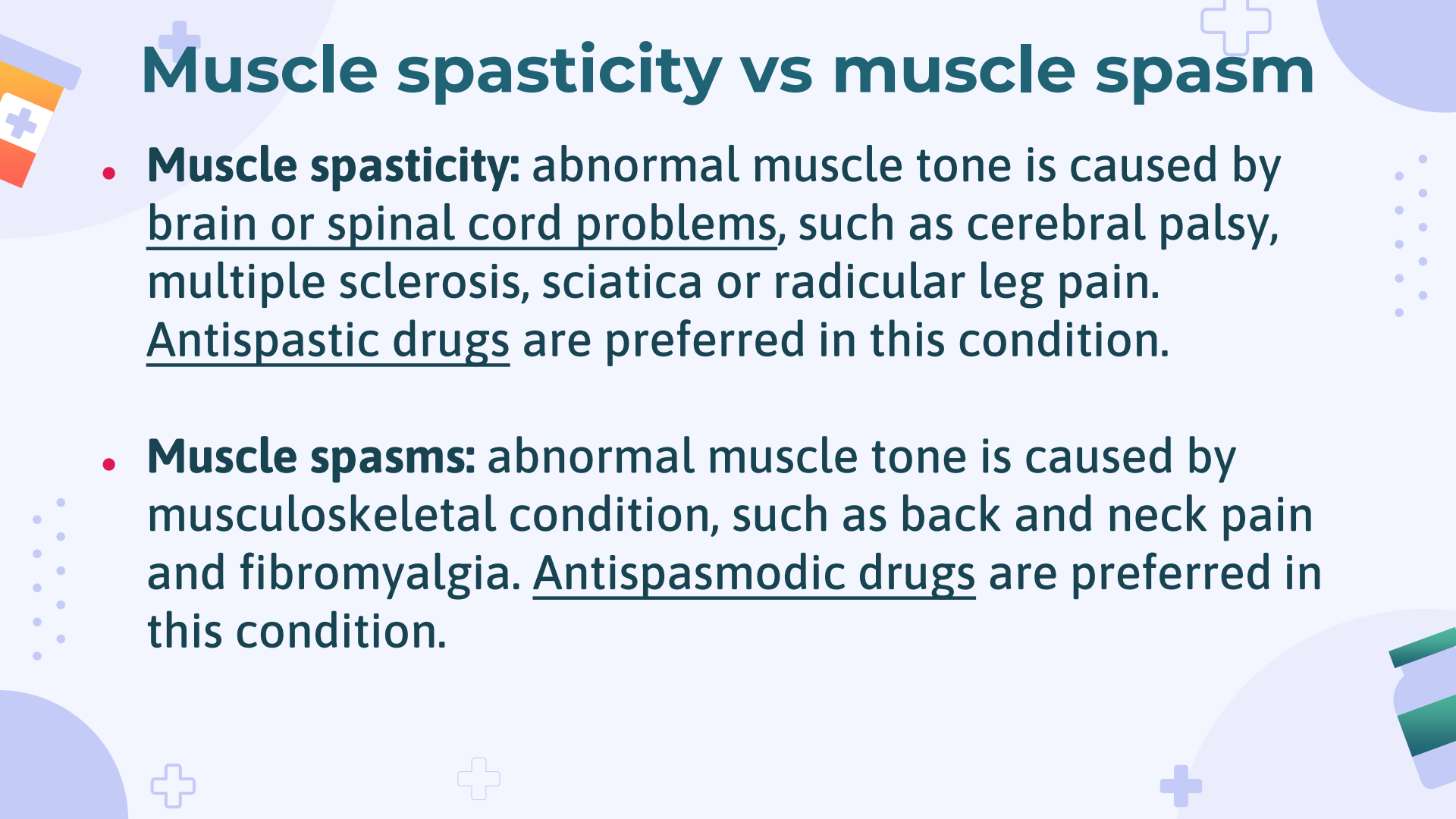


01 overview



Introduction

- Muscle relaxants are two major groups, neuromuscular blockers (NMBs) and spasmolytics.
 - Neuromuscular blockers act by blocking nicotinic acetylcholine receptors at the neuromuscular junction and have no CNS activity, they are often used during surgical procedures and in intensive care.
 - Spasmolytics: are centrally acting muscle relaxants, are used to relieve spasms (musculoskeletal pain) and spasticity (neurological conditions)



Muscle spasticity vs muscle spasm

- **Muscle spasticity:** abnormal muscle tone is caused by brain or spinal cord problems, such as cerebral palsy, multiple sclerosis, sciatica or radicular leg pain. Antispastic drugs are preferred in this condition.
- **Muscle spasms:** abnormal muscle tone is caused by musculoskeletal condition, such as back and neck pain and fibromyalgia. Antispasmodic drugs are preferred in this condition.

The background is a light blue gradient. A large, faint stethoscope is centered behind the text. In the top left, there is a clipboard with a checklist and a white plus sign. In the top right, there is a red and white pill bottle with a white plus sign. In the bottom left, there is a green pill bottle with a white plus sign. In the bottom right, there is a red heart with a white plus sign. There are also several small white plus signs scattered around the edges.

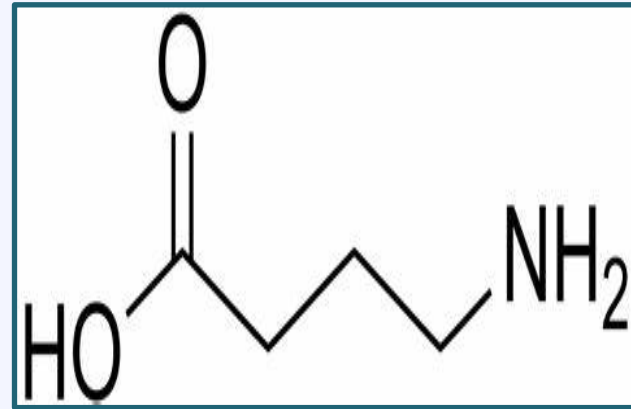
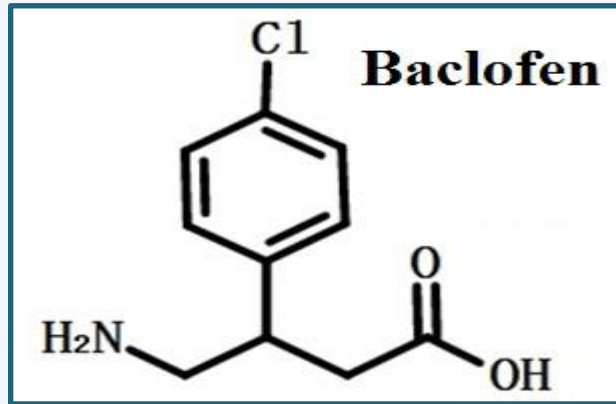
02

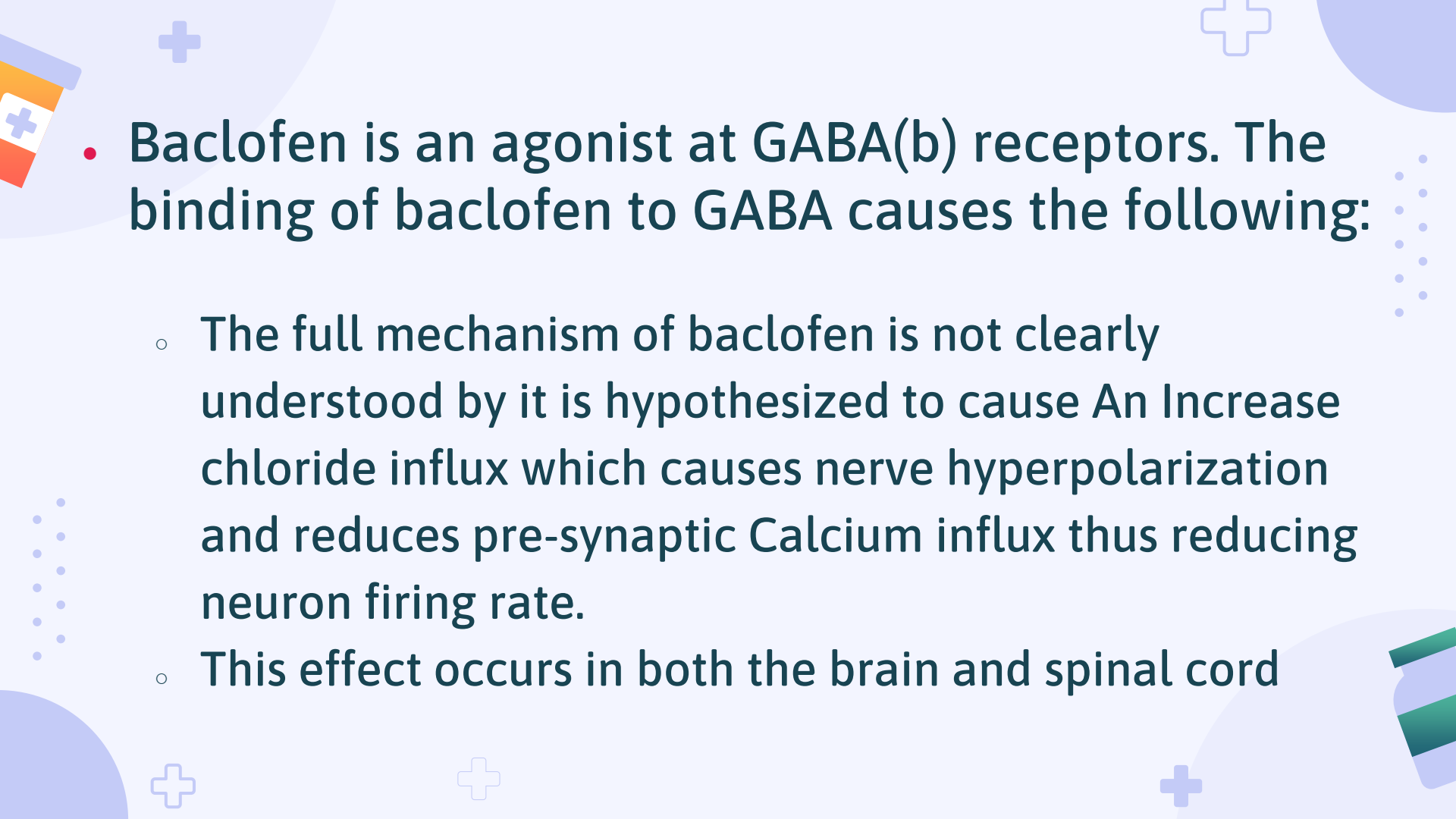
Central antispasticity agents

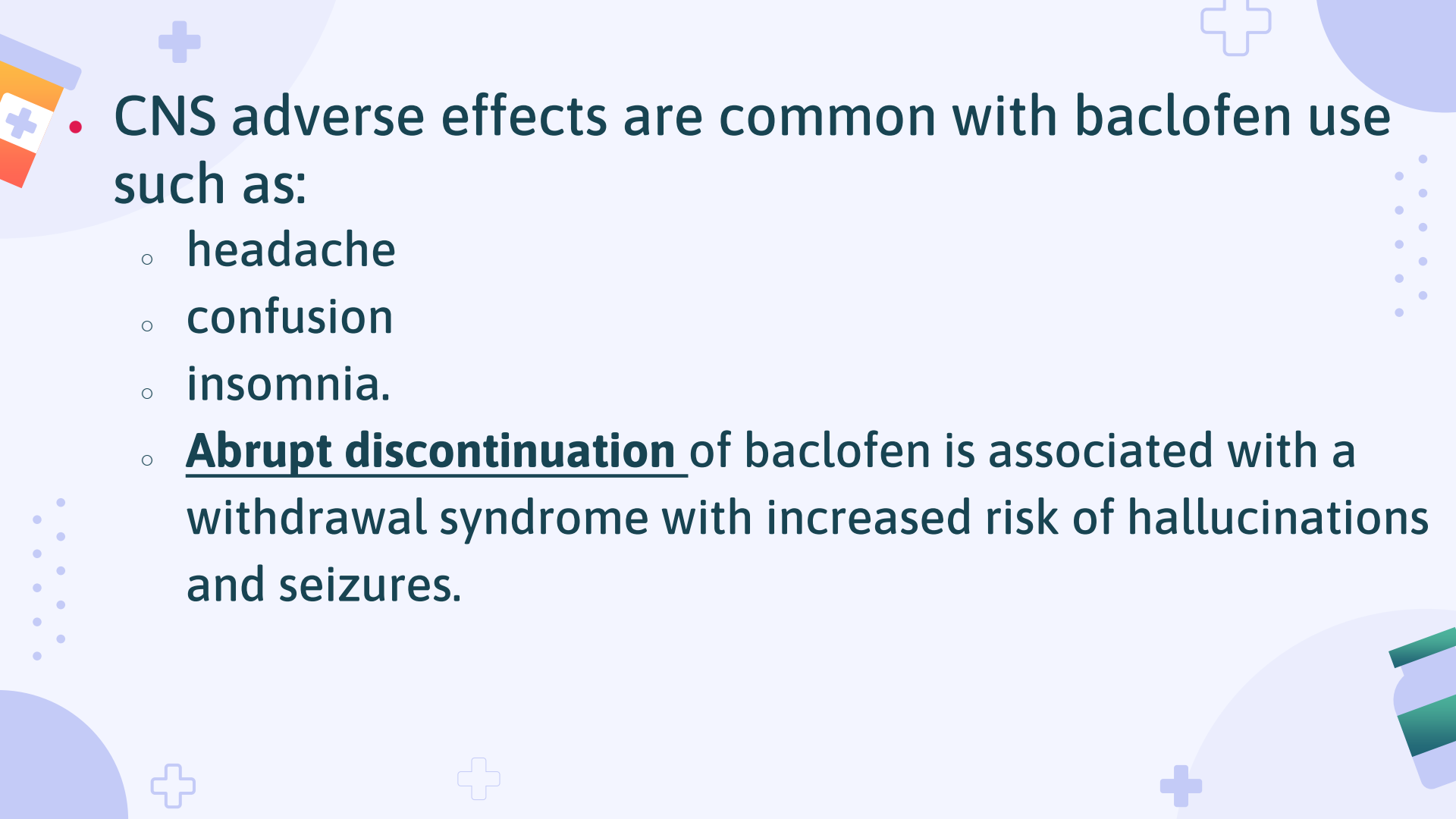
- 
- 
- 
- **Spasticity** is a disruption in muscle movement patterns that causes certain muscles to contract all at once when you try to move or even at rest. It's usually caused by damage to nerve pathways within your brain or spinal cord that control movement and stretch reflexes.
 - In general, antispastic medications act on your spinal cord or skeletal muscle directly to improve muscle tightness (hypertonicity) and involuntary spasms.
 - Certain conditions like: multiple sclerosis, spinal cord injury, cerebral palsy greatly benefit from antispastic drugs to alleviate muscle contraction and pain.
- 
- 
- 
- 

Baclofen

- Baclofen is a derivative of GABA (chlorophenyl-GABA) that was designed to be an orally active GABA- potentiating agent.



- 
- Baclofen is an agonist at GABA(b) receptors. The binding of baclofen to GABA causes the following:
 - The full mechanism of baclofen is not clearly understood by it is hypothesized to cause An Increase chloride influx which causes nerve hyperpolarization and reduces pre-synaptic Calcium influx thus reducing neuron firing rate.
 - This effect occurs in both the brain and spinal cord

- 
- CNS adverse effects are common with baclofen use such as:
 - headache
 - confusion
 - insomnia.
 - **Abrupt discontinuation** of baclofen is associated with a withdrawal syndrome with increased risk of hallucinations and seizures.

Brands



ميلوباك MYLOBAC
للتصلب اللويحي والم العضلات

Tizanidine

Tizanidine is centrally acting α_2 agonist that is used as an antispastic agent.

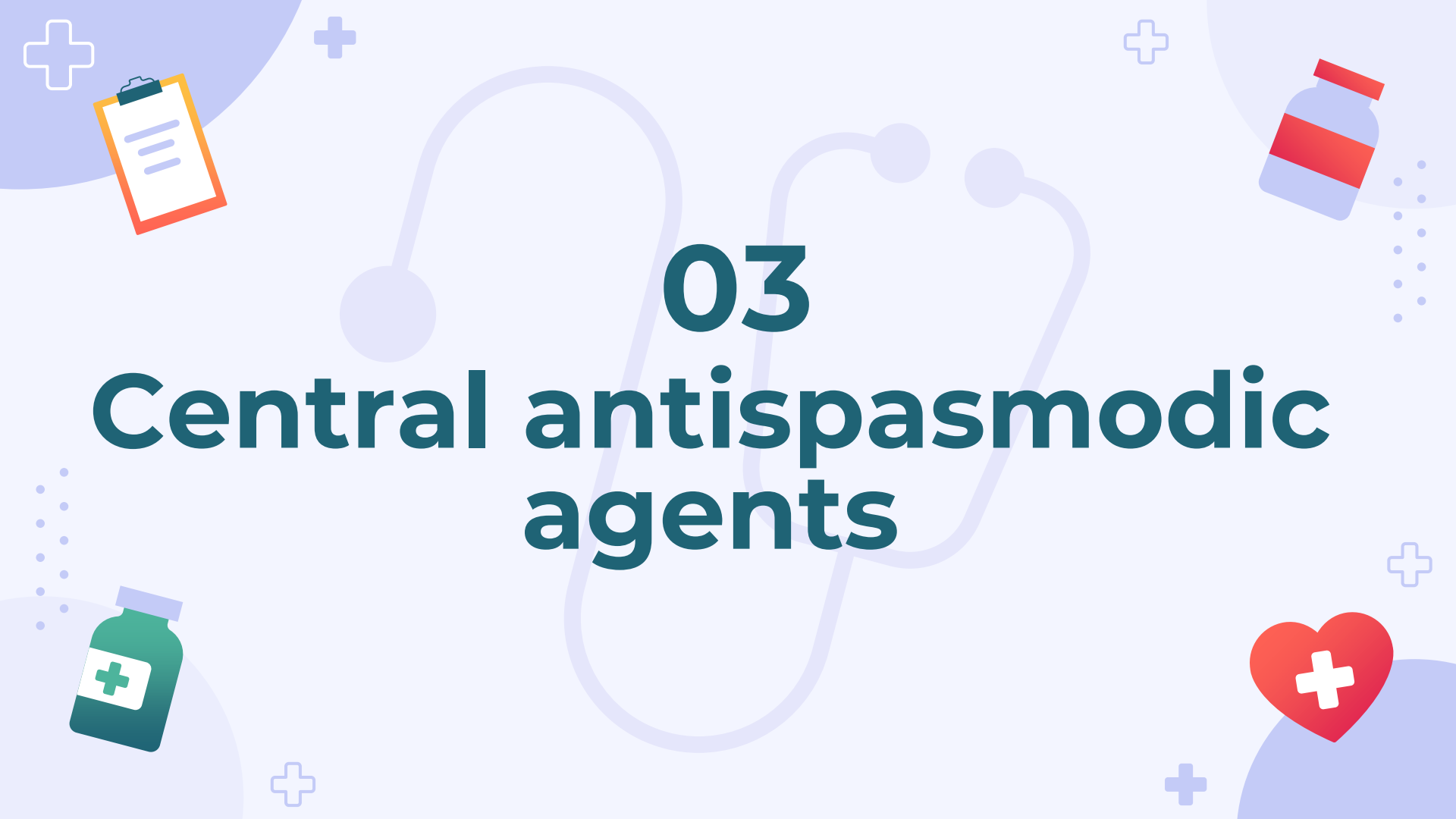
- Tizanidine is used to treat spasticity and spasm at doses much lower than the dose that would produce a cardiovascular effect.

Adverse effects are dose related such as:

- Hypotension, dry mouth, and sedation (common with α_2 agonist) .
- Abrupt withdrawal will cause rebound hypertension.
- Tizanidine is metabolized by CYP-1A2 and inhibitors of this enzyme (ciprofloxacin) will cause tizanidine toxicity.

Brands





03

Central antispasmodic agents

- 
- 
- 
- 
- 
- 
- 
- Providers mainly prescribe **antispasmodic skeletal muscle relaxants for musculoskeletal and myofascial pain, especially lower back pain, and muscle spasms.**
 - these medications aren't the first-line therapy (NSAIDs, physiotherapy, and lifestyle adjustment are used first) for the treatment of pain or spasms.
 - This is due to the adverse side effects of antispasmodics, including addiction potential for some types of the medication.

Cyclobenzaprine

- it is structurally related to TCA with a similar mechanism of action and adverse effects.
- It used for short term management of muscle spasm associated with acute musculoskeletal conditions and off-label for fibromyalgia.
- Important notes:
 - It should be used with caution in the elderly and hepatic dysfunction.
 - Discontinuation of cyclobenzaprine for 14 days is required before initiating MAOI therapy
 - Serotonin syndrome is a risk when combined with a serotonergic drug such as (SSRI, sumatriptan, ergot alkaloids)

Brands



Orphenadrine

Orphenadrine is an anticholinergic drug closely related to the 1st generation antihistamine [Diphenhydramine].

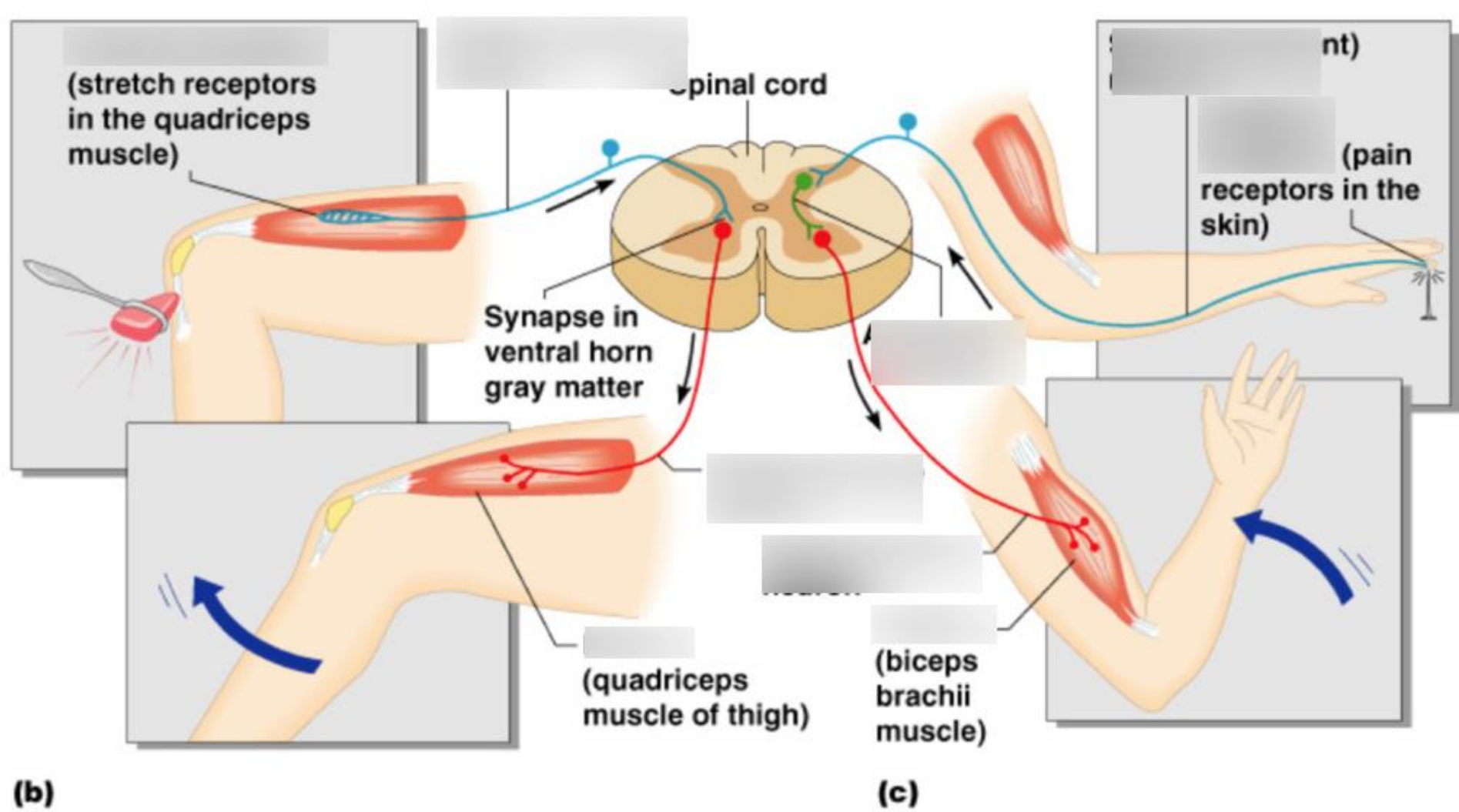
- It is used in combination with paracetamol for treatment of muscle spasm conditions. its mechanism of action in causing analgesia and skeletal muscle relaxation is not well defined.
- Its main adverse effects are anticholinergic side effects.
- orphenadrine can be abused in Large doses to cause hallucinations.

Brands

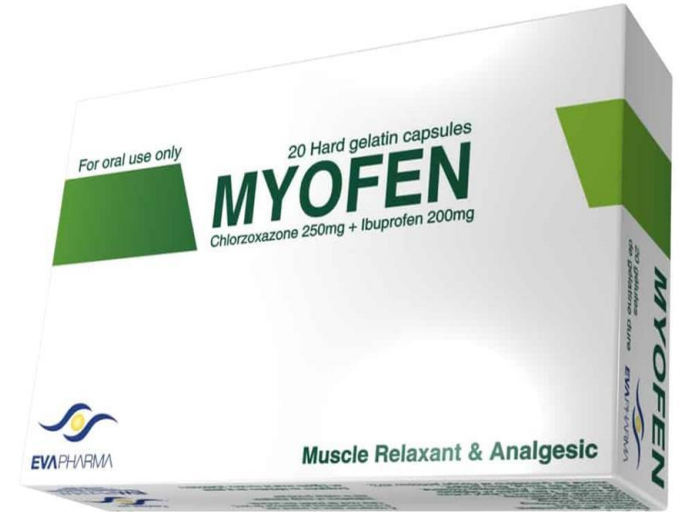


Chlorzoxazone

- The mechanism of action is not fully understood, but it works primarily in spinal cord by inhibiting multi-synaptic reflex arcs involved in producing and maintaining skeletal muscle spasm.
- Chlorzoxazone is available in combination with analgesics such as ibuprofen or paracetamol.
- Chlorzoxazone should be used only for short term management espically when combined with paracetamol due to a risk of fatal hepatocellular toxicity.



Brands



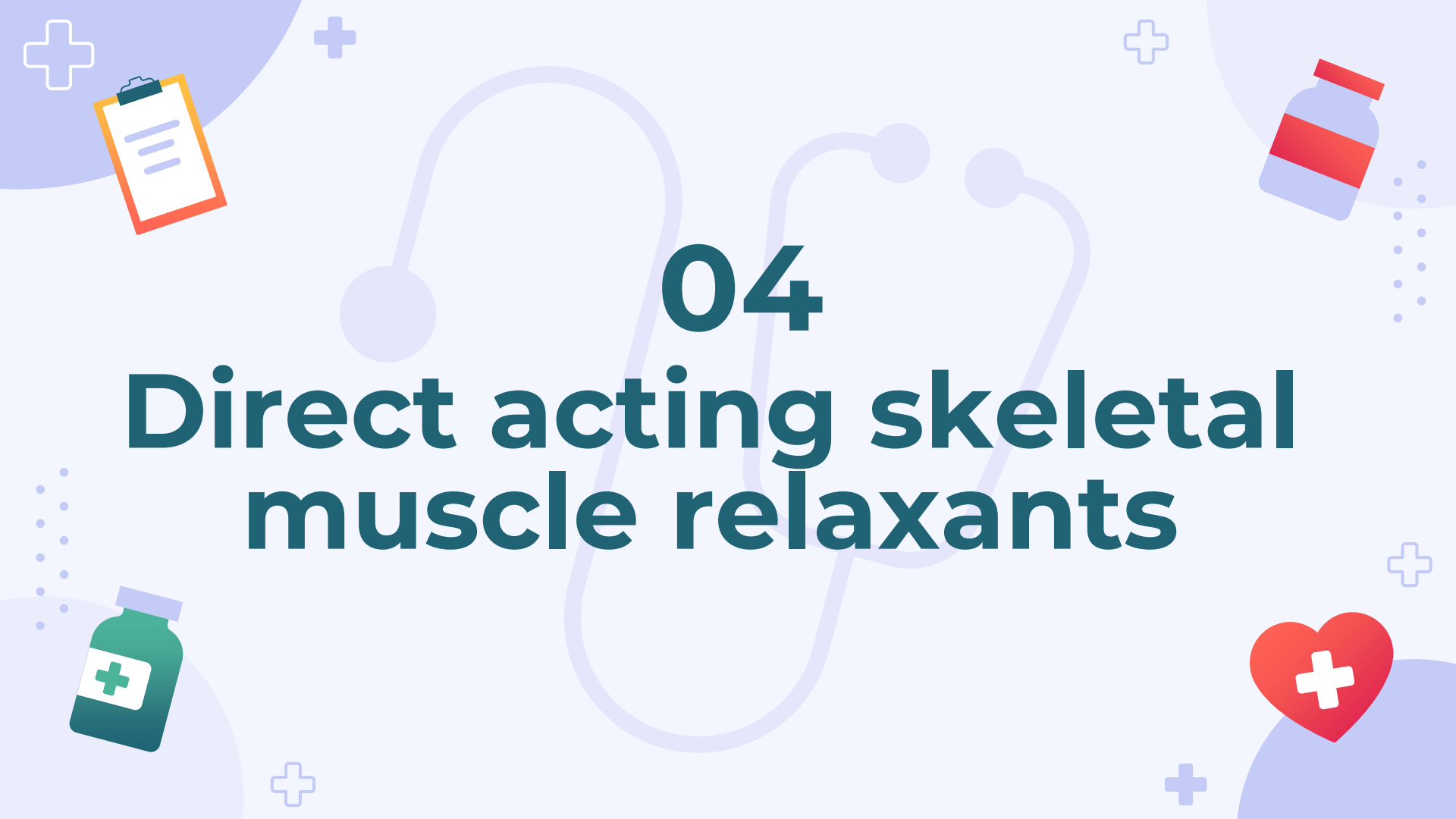
Methocarbamol

- Methocarbamol is carbamate derivative of guaifenesin. And is used in a manner similar to chlorzoxane.
- The mechanism of action of methocarbamol is not fully understood, however, it is used due to having the lowest abuse potential of all antispasmodics.
- Possible side effects are: dizziness, lightheadedness, and drowsiness. Methocarbamol is a well tolerated drug.

Brands



ميثوريلاكس Methorelax
دواء مرخي للعضلات



04

Direct acting skeletal muscle relaxants

Dantrolene

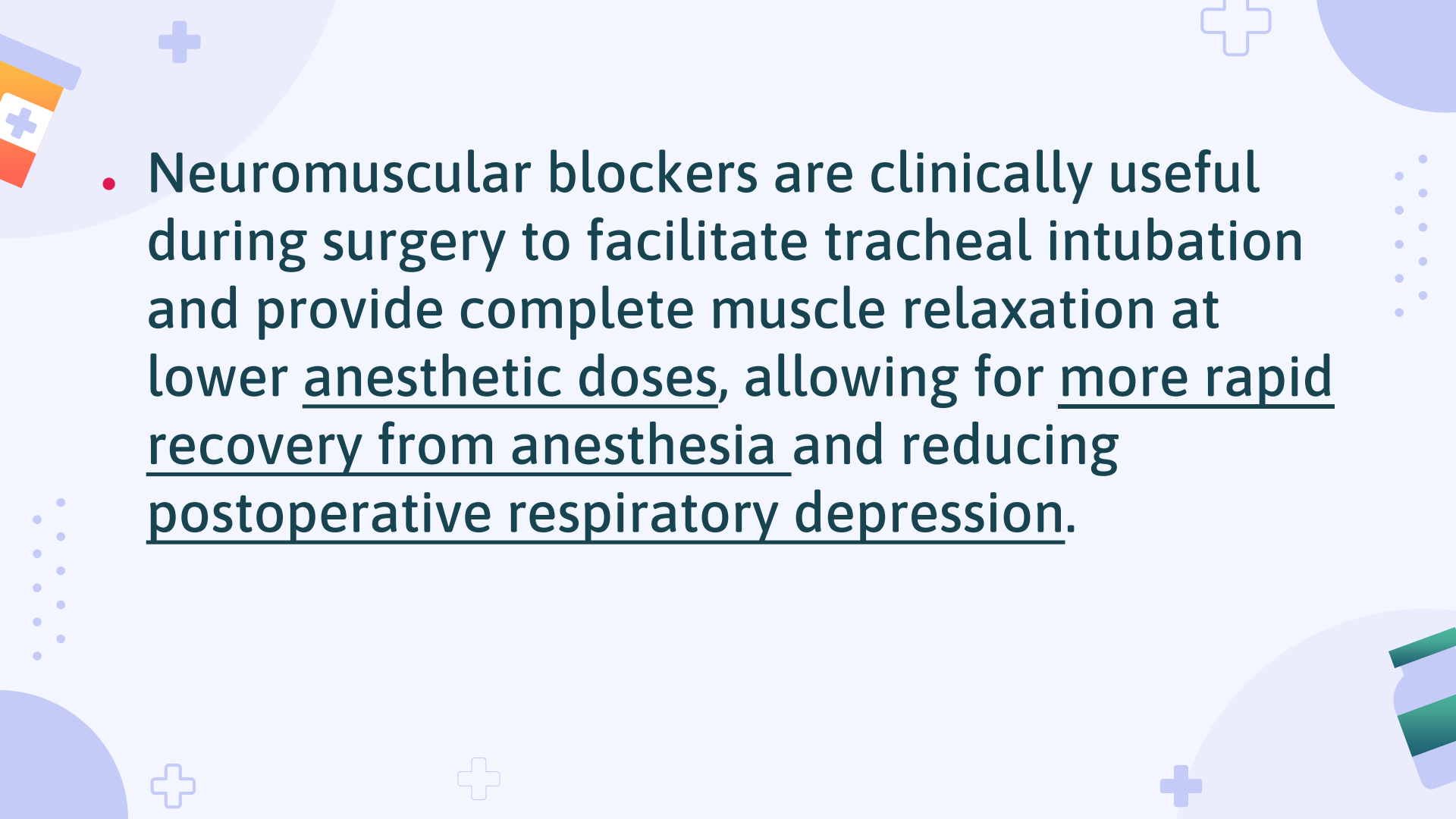
- Dantrolene is a hydration derivative related to phenytoin that has a unique peripheral mechanism of spasmolytic activity.
- The mechanism of action for dantrolene has a unique mechanism in which it inhibits Calcium ions release from sarcoplasmic reticulum.
- Dantrolene is used for treatment of various chronic muscle spasticity conditions such as MS, and spinal cord injury.
- Dantrolene (I.V) is also used for management of muscle rigidity associated with malignant hyperthermia (40 C) [a severe reaction to certain drugs used for anesthesia caused by genes].
- Drowsiness and dizziness are common with dantrolene use, the drugs comes with a black box warning for hepatotoxicity.

Brands

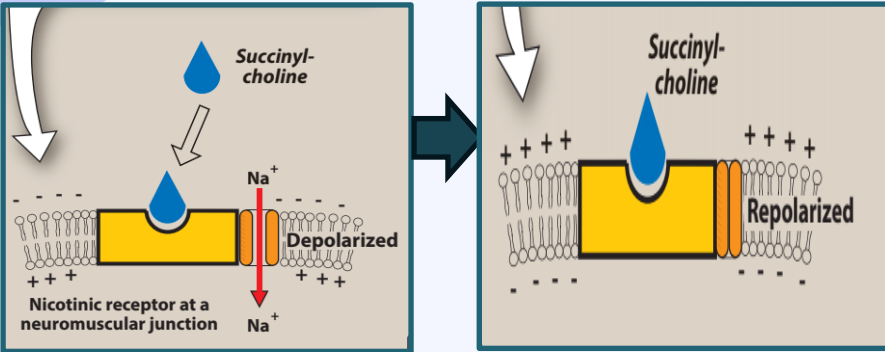


05 Neuromuscular blockers



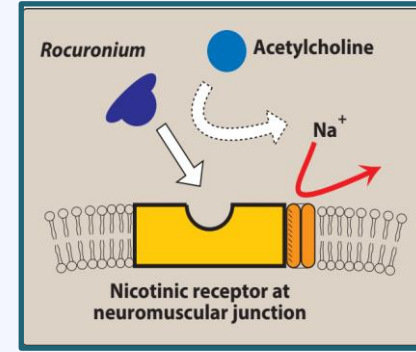
- 
- Neuromuscular blockers are clinically useful during surgery to facilitate tracheal intubation and provide complete muscle relaxation at lower anesthetic doses, allowing for more rapid recovery from anesthesia and reducing postoperative respiratory depression.

Classification



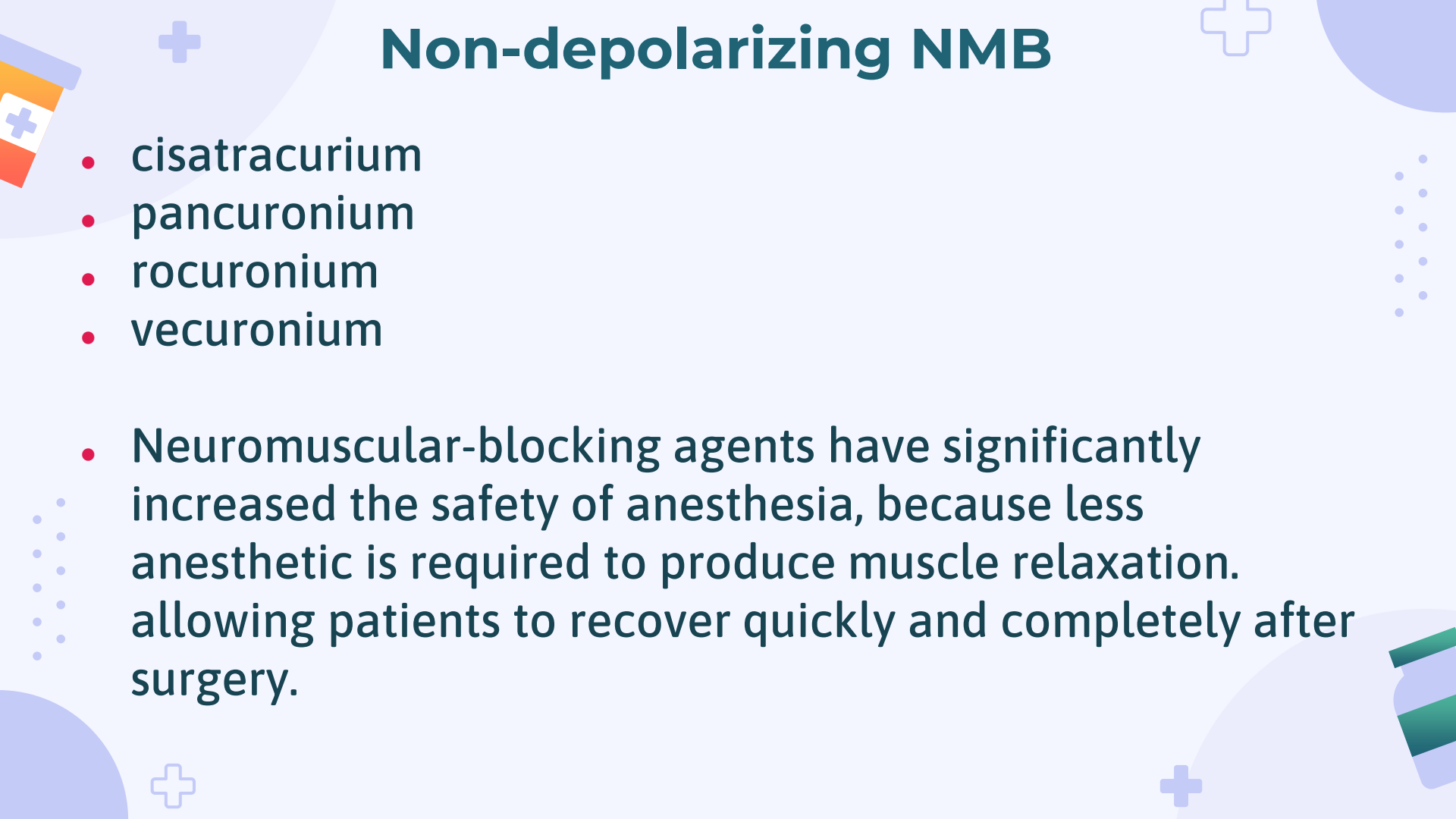
Depolarizing NMB

These drugs block cholinergic transmission between motor nerve endings and the nicotinic receptors on the skeletal muscle by constant activation of the receptor which causes a state of hyperpolarization. This state prevents muscle response to further stimulation.



Non-depolarizing NMB

These drugs block cholinergic transmission between motor nerve endings and the nicotinic receptors on the skeletal muscle by simply occupying the Nm receptor without affecting nerve polarization.

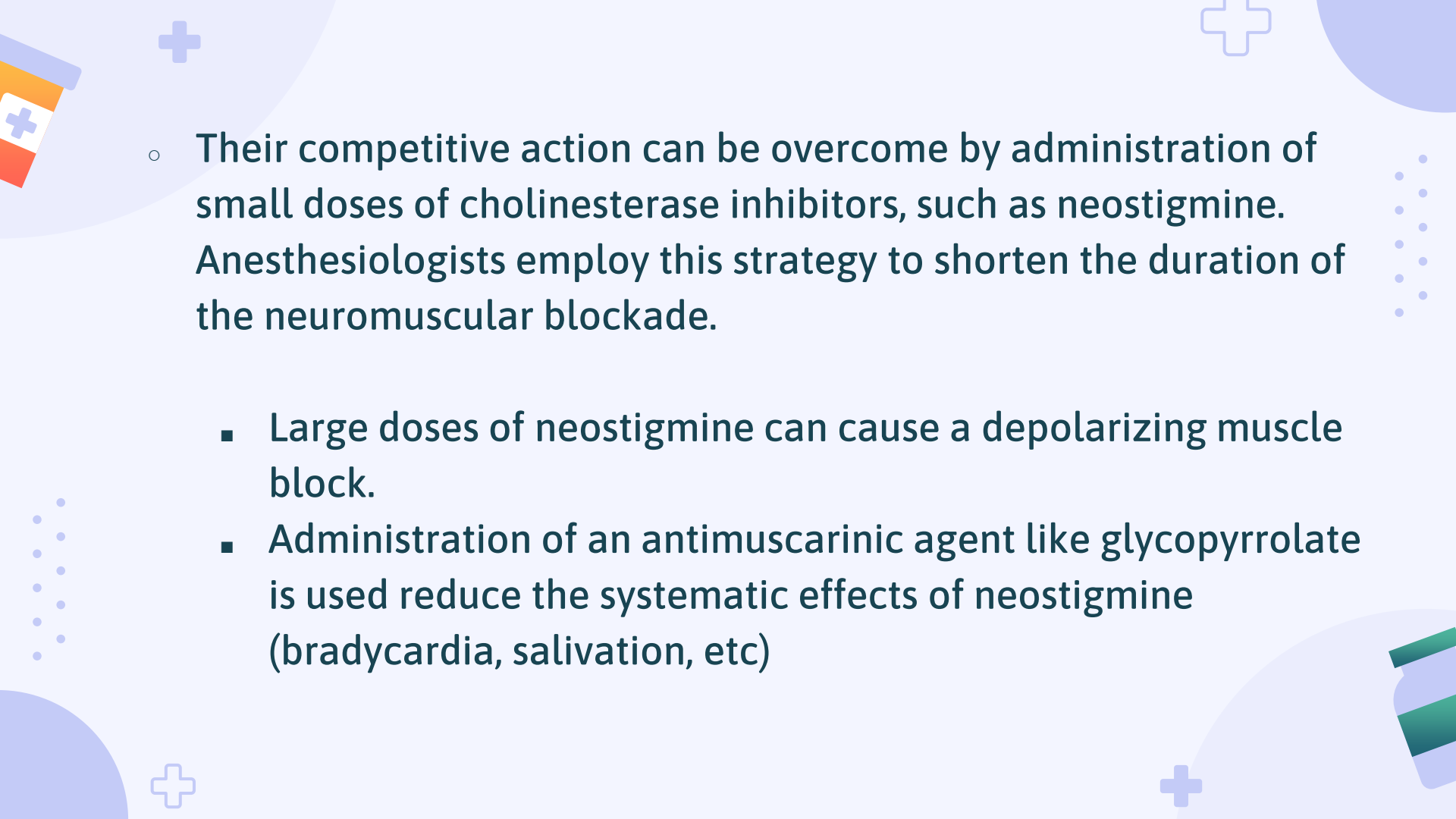


Non-depolarizing NMB

- cisatracurium
 - pancuronium
 - rocuronium
 - vecuronium
-
- Neuromuscular-blocking agents have significantly increased the safety of anesthesia, because less anesthetic is required to produce muscle relaxation. allowing patients to recover quickly and completely after surgery.

- Mechanism of action:

- Nondepolarizing agents competitively block ACh at the nicotinic receptors without stimulating it. Thus, these drugs prevent depolarization of the muscle cell membrane and inhibit muscular contraction.
- 

- 
- Their competitive action can be overcome by administration of small doses of cholinesterase inhibitors, such as neostigmine. Anesthesiologists employ this strategy to shorten the duration of the neuromuscular blockade.
 - Large doses of neostigmine can cause a depolarizing muscle block.
 - Administration of an antimuscarinic agent like glycopyrrolate is used reduce the systematic effects of neostigmine (bradycardia, salivation, etc)

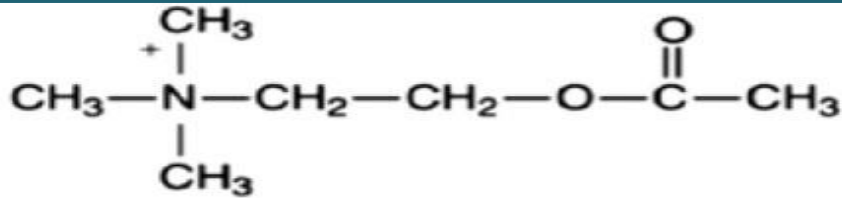


Pharmacokinetics:

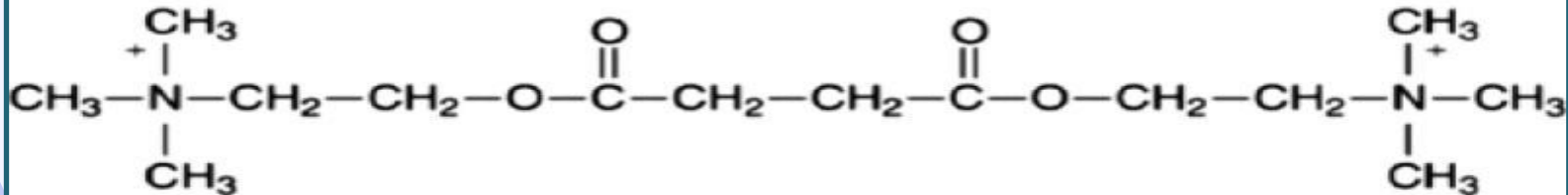
- These agents are administered I.V as they are very polar and can not be absorbed orally. They also can not cross B.B.B
- The choice of an agent depends on the desired onset and duration of the muscle relaxation and route of elimination.
 - vecuronium is hepatically metabolized to inactive metabolites. (not suitable for hepatic dysfunction)
 - pancuronium is eliminated renally unchanged (not suitable for kidney failure)
 - cisatracurium is metabolized in plasma and can be used in patients with multiple organ failure.
-  In general, these agents are safe with minimal side effects.

Depolarizing NMB

- Succinylcholine is the only member of this family used in clinical application, it is structurally similar to acetylcholine, however, it is resistant to synaptic acetylcholinesterase
- Succinylcholine is administered I.V where almost 90% of the administered dose is metabolized through hydrolysis by plasma pseudocholinesterase.



Acetylcholine



Succinylcholine

- 
- mechanism of action:
 - It's mode of action occurs in two phases:
 - Phase 1: activation of Nm receptors causes the opening of the sodium channel associated with the nicotinic receptors, which results in depolarization of the receptor. This leads to a transient twitching of the muscle.
 - Phase 2: Continued binding of the depolarizing agent renders the receptor incapable of transmitting further impulses. With time, continuous depolarization gives way to gradual repolarization as the sodium channel closes or is blocked. This causes a resistance to depolarization and flaccid paralysis.
 - N.B [phase 2 occurs as result of synaptic acetylcholine esterase not being able to breakdown succinylcholine.]
- 
- 
- 
- 



Adverse effects :

- **Malignant hyperthermia:** in a very small percentage of susceptible patients, exposure to halogenated hydrocarbon anesthetics or the neuromuscular blocker succinylcholine may induce malignant hyperthermia (MH)
- [a rare life-threatening condition. MH causes a drastic and uncontrolled increase in skeletal muscle oxidative metabolism, overwhelming the body's capacity to supply oxygen, remove carbon dioxide, and regulate temperature, eventually leading to circulatory collapse and death if not treated immediately.]
- H. Susceptibility to MH is often inherited

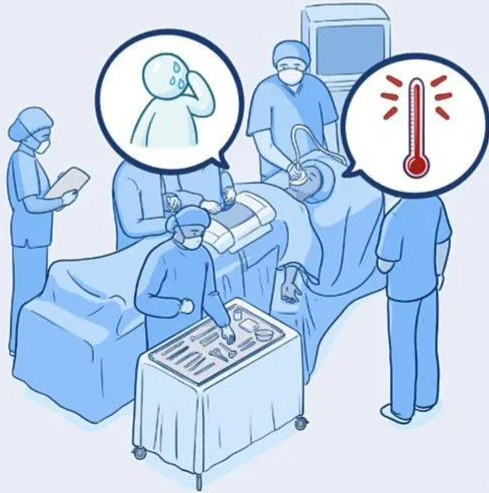


MALIGNANT HYPERTHERMIA

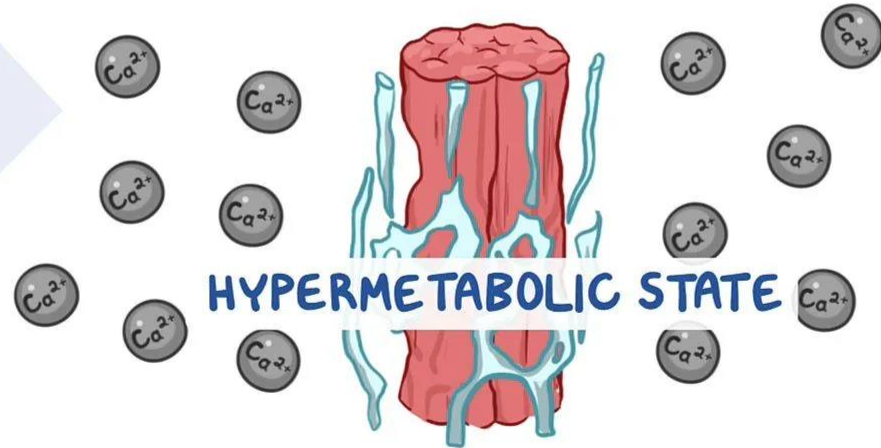
SERIOUS ANESTHETIC COMPLICATION

GENETIC MUTATION

* DIHYDROPYRIDINE
or RYANODINE
RECEPTORS



CALCIUM OVERLOAD causes
SUSTAINED MUSCLE CONTRACTION
& BREAKDOWN

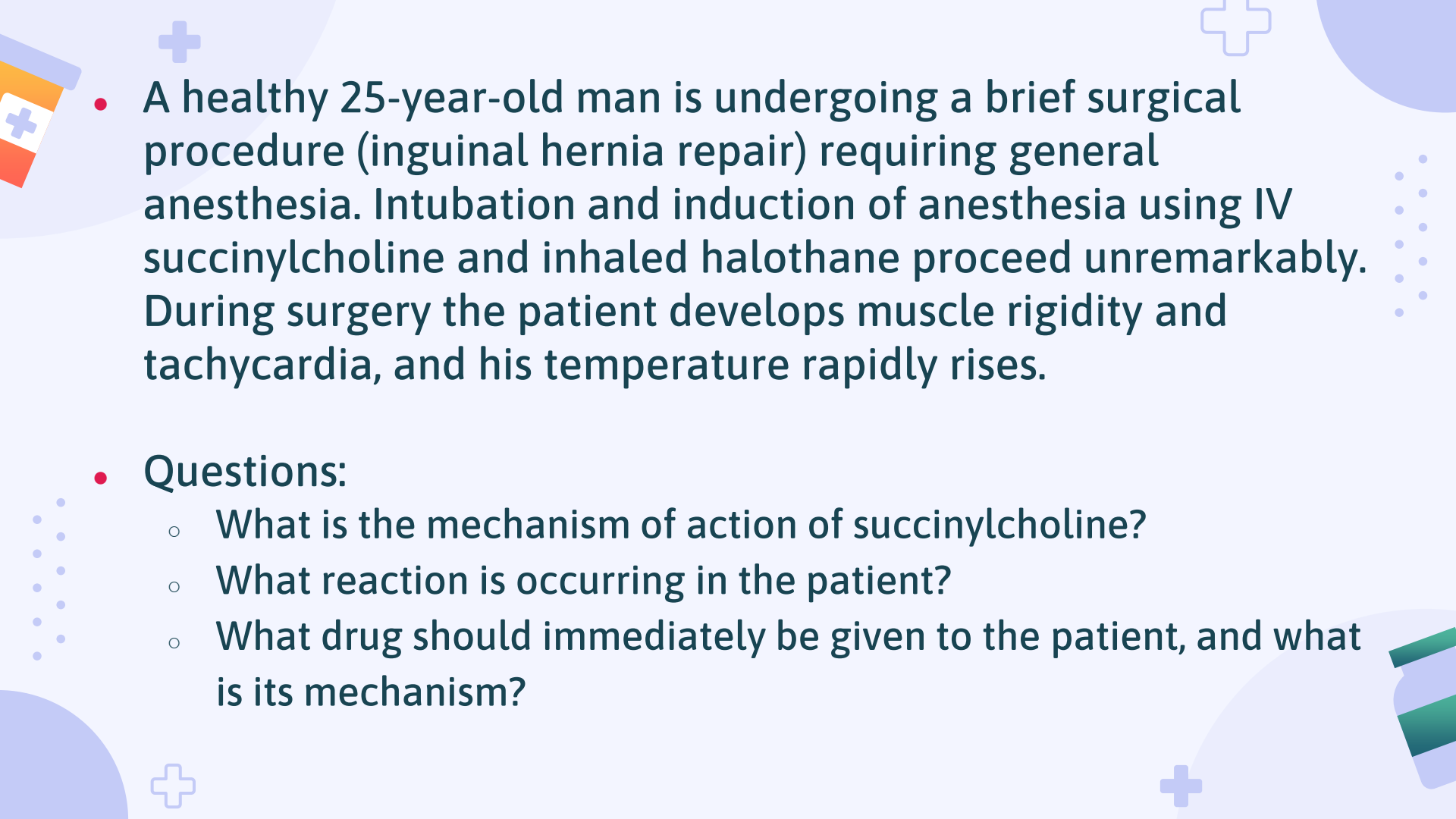


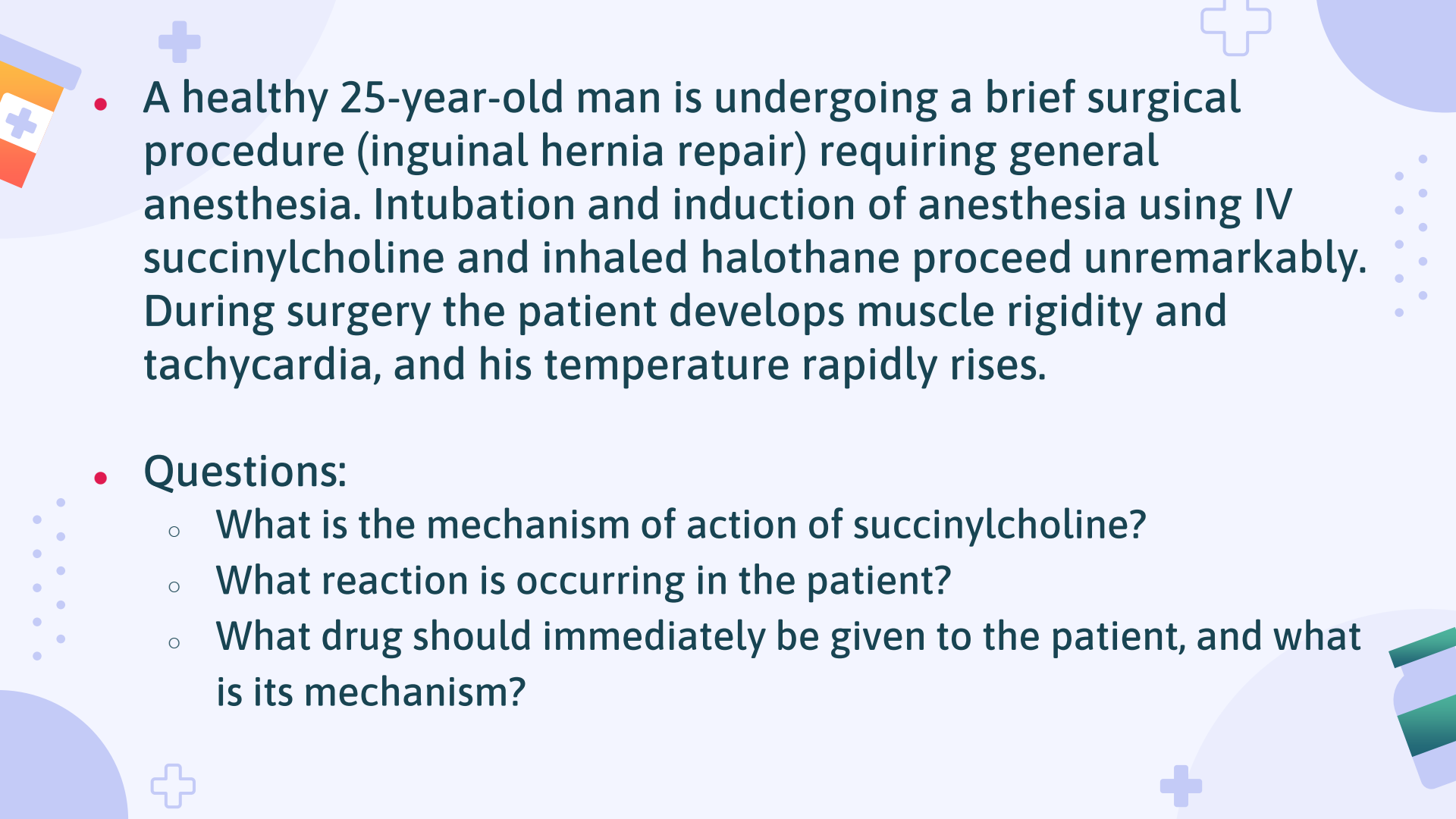
- **Apnea:** Administration of succinylcholine to a patient who is deficient in plasma cholinesterase or who has an atypical form of the enzyme can lead to prolonged apnea due to paralysis of the diaphragm.

A large, light purple stethoscope graphic is centered in the background, with its chest piece at the top and the tubing looping around the text.

**06
Cases**



- 
- A healthy 25-year-old man is undergoing a brief surgical procedure (inguinal hernia repair) requiring general anesthesia. Intubation and induction of anesthesia using IV succinylcholine and inhaled halothane proceed unremarkably. During surgery the patient develops muscle rigidity and tachycardia, and his temperature rapidly rises.

- Questions:
 - What is the mechanism of action of succinylcholine?
 - What reaction is occurring in the patient?
 - What drug should immediately be given to the patient, and what is its mechanism?
- 



• Answers:

- Nicotinic receptor agonist at the motor endplate of the neuromuscular junction, which causes persistent stimulation and depolarization of muscle cells.
 - Malignant hyperthermia.
 - Dantrolene, which acts by interfering with calcium release from the sarcoplasmic reticulum.
- 



Anxiolytic and Hypnotic drugs

By: Dr. Riffat Ibrahim Riffat



Table of contents

01

Overview

02

Benzodiazepines

03

Other anxiolytic
drugs

04

Barbituates

05

Other hypnotic
drugs

06

Cases



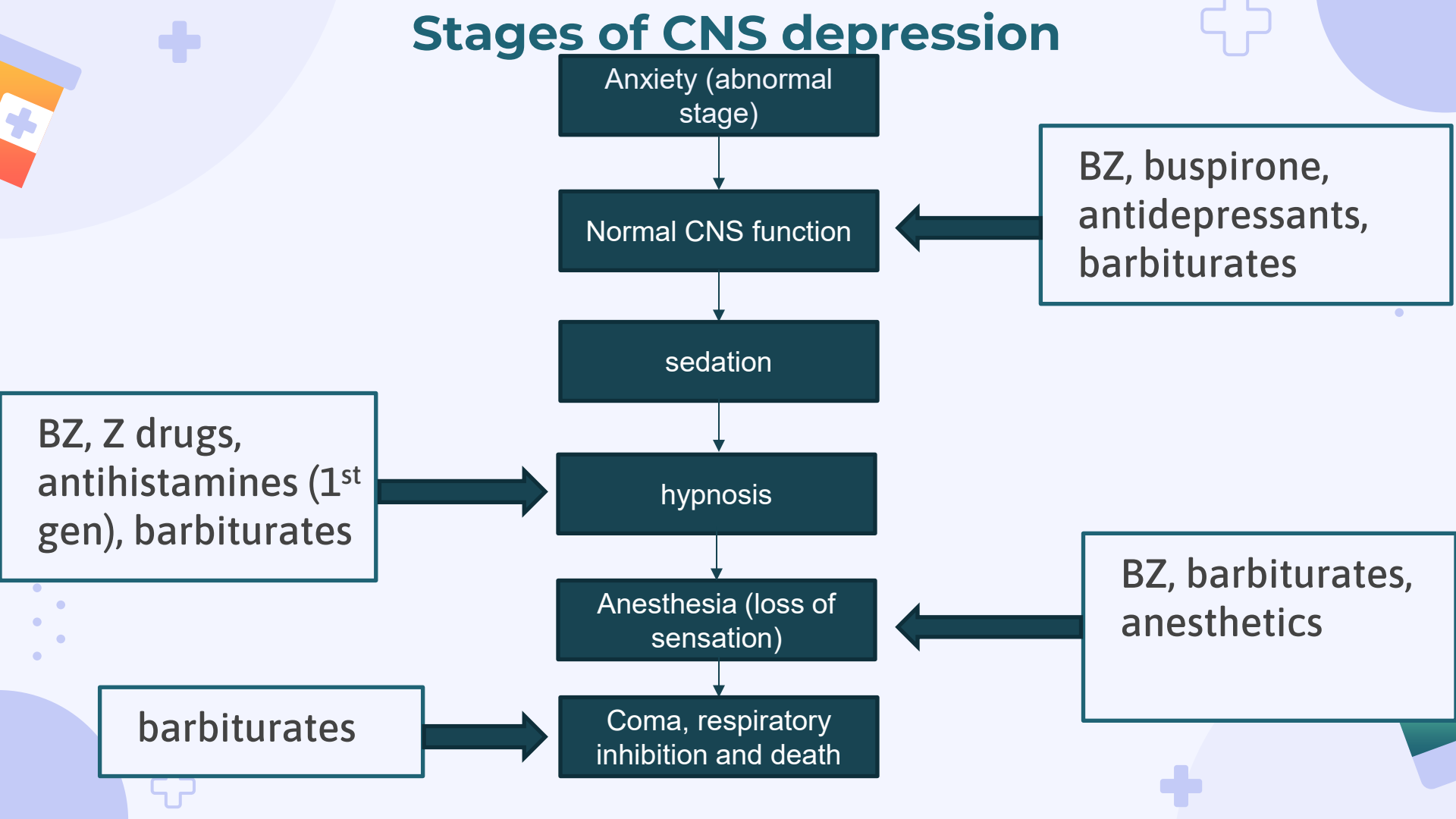
01 Overview



Introduction

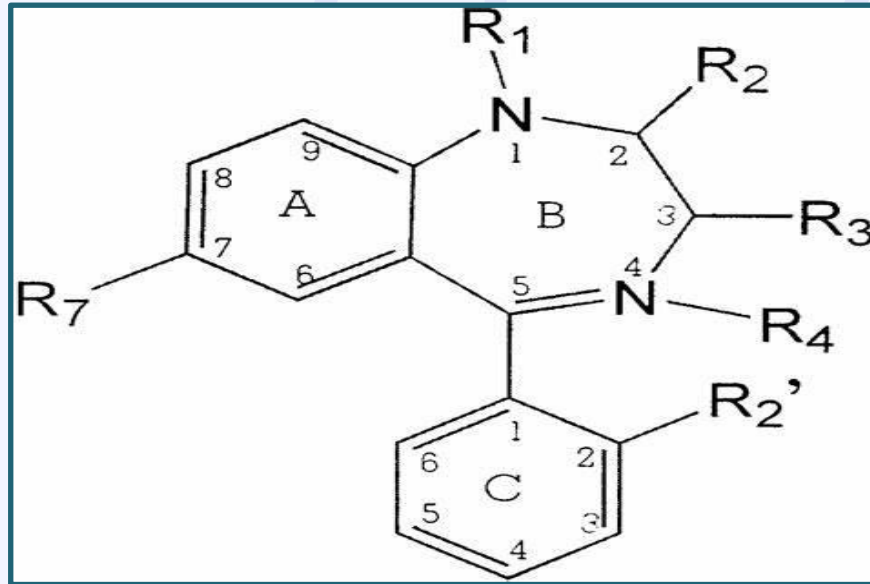
- Disorders involving anxiety are among the most common mental disorders. Anxiety is an unpleasant state of tension, apprehension, or uneasiness (a fear that arises from either a known or an unknown source).
- The physical symptoms of severe anxiety are similar to those of fear (such as tachycardia, sweating, trembling, and palpitations) and involve sympathetic activation. Episodes of mild anxiety are common life experiences and do not warrant treatment.
- severe, chronic, debilitating anxiety (most days for at least 6 months) may be treated with antianxiety drugs (sometimes called anxiolytics) and/or some form of psychotherapy.

Stages of CNS depression



02

Benzodiazepines



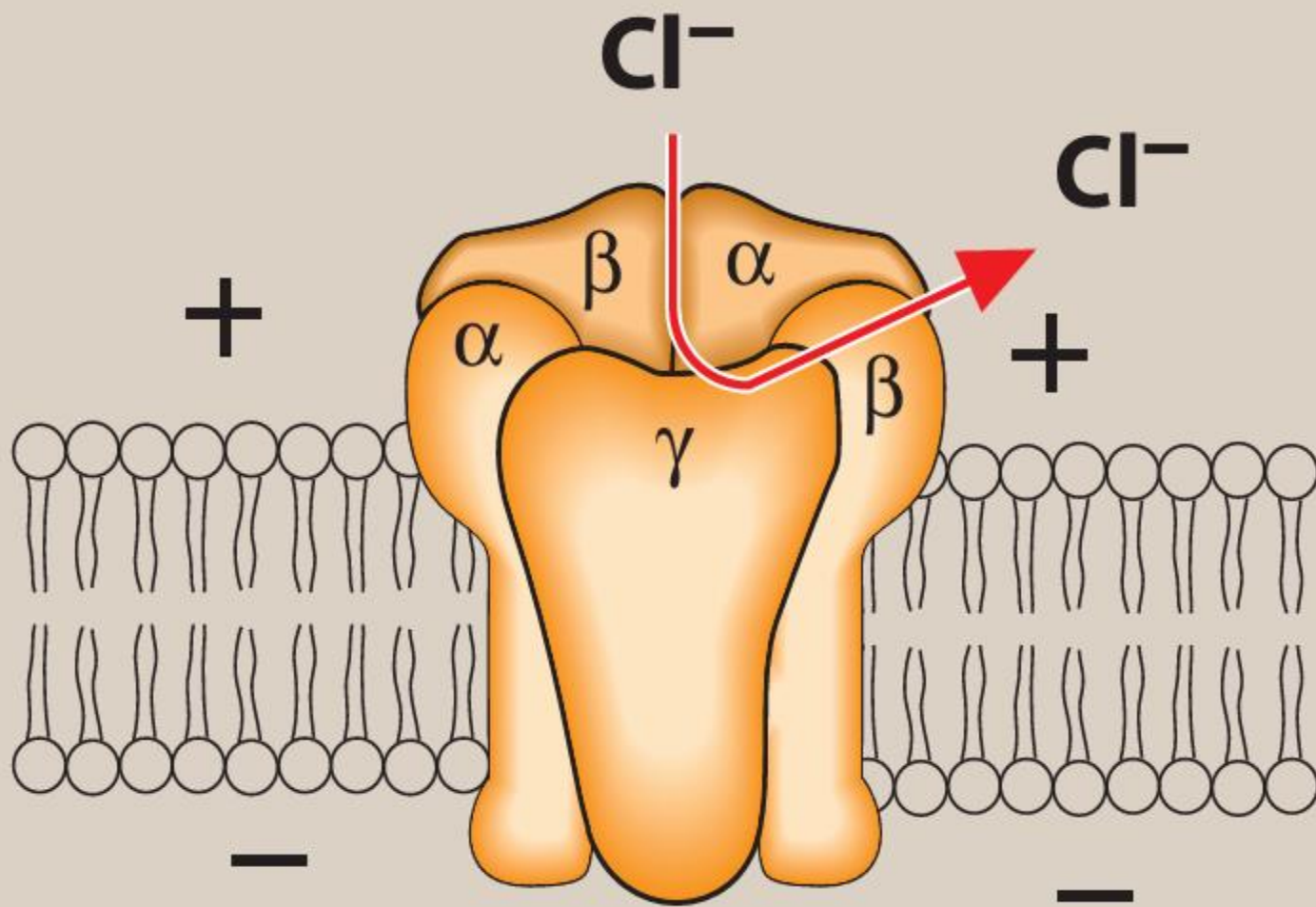
Introduction

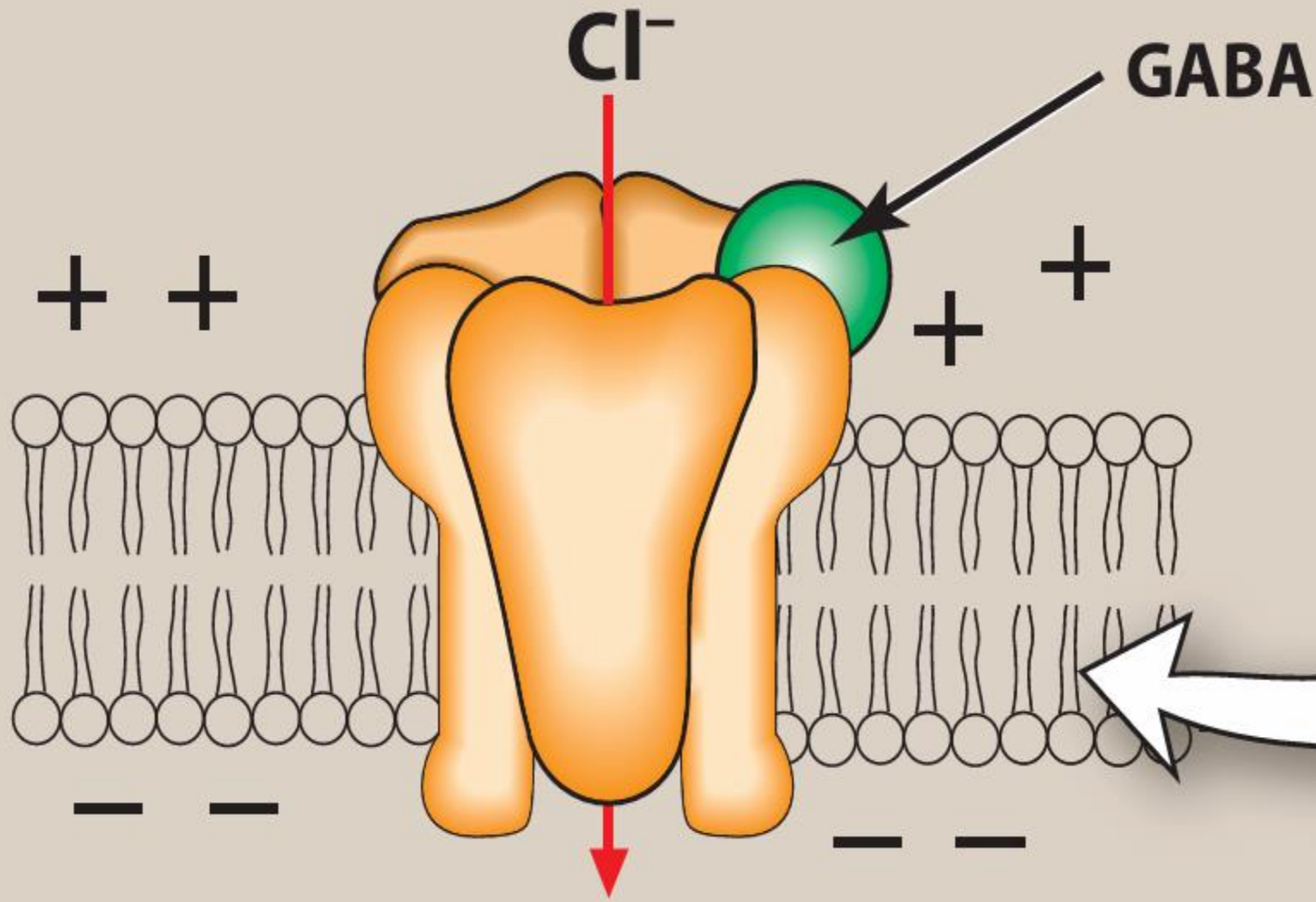
- Benzodiazepines are widely used anxiolytic drugs. They have largely replaced barbiturates in the treatment of anxiety and insomnia, because benzodiazepines are generally considered to be safer and more effective.
- However, they are not the 1st choice in treatment of chronic anxiety or chronic insomnia, they are used for short term management or as adjuvant to standard treatment. This limitation is due to risk of abuse and dependence often associated with benzodiazepines.

Mechanism of action

- GABA is the major inhibitory neurotransmitter in the central nervous system (CNS). The GABA receptors are composed of a combination of five α , β , and γ subunits that span the postsynaptic membrane
- Binding of GABA to its receptor triggers an opening of the central ion channel, allowing chloride through the pore. The influx of chloride ions causes hyperpolarization of the neuron and decreases neurotransmission by inhibiting the formation of action potentials.

- 
- 
- Benzodiazepines modulate GABA effects by binding to a specific, high-affinity site (distinct from the GABA-binding site) located at the interface of the α subunit and the γ subunit on the GABA receptor.
 - Note: These binding sites are sometimes labeled “benzodiazepine (BZ) receptors.”
 - Benzodiazepines increase the frequency of channel openings produced by GABA.
 - This mechanism is dubbed allosteric modulation, since benzodiazepines do not affect the opening of the ion channel but cause configurational changes in GABA receptor which enhances GABA binding.
- 
- 
- 







Benzodiazepine

Cl⁻

GABA

+++

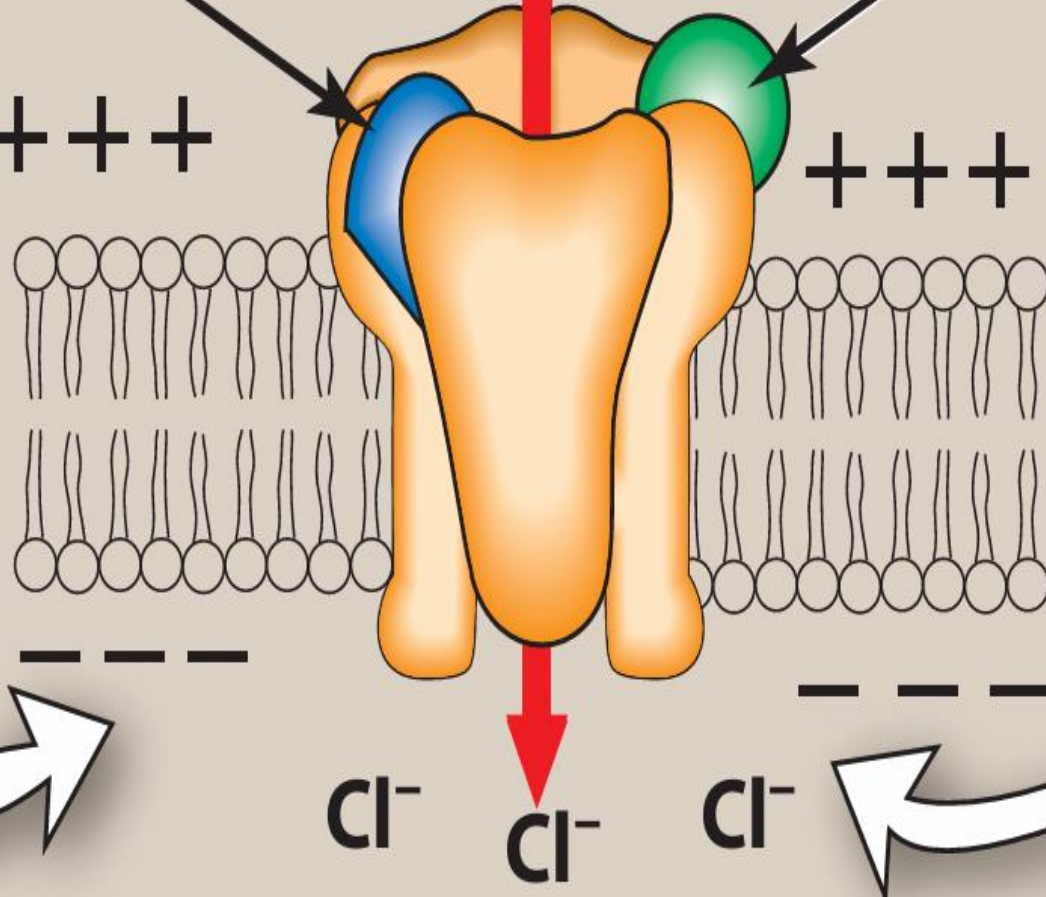
+++

**ing
fore**

Cl⁻

Cl⁻

Cl⁻



Types of BZ

- The individual benzodiazepines show small differences in their relative anxiolytic, anticonvulsant, and sedative properties. However, the duration of action varies widely among this group, and pharmacokinetic considerations are often important in choosing one benzodiazepine over another.
- Short acting BZ ($t_{1/2}$ 3-8 hours)
 - Alprazolam (tab)
 - Midazolam (inj)
 - Triazolam
- intermediate acting BZ ($t_{1/2}$ 10-20 hour)
 - Diazepam (tab) (supp) (inj)
 - Clonazepam
 - Oxazepam (a metabolite of Diazepam)
- Long acting BZ ($t_{1/2}$ 1-3 days): chlordiazepoxide

Pharmacokinetics

- Absorption and distribution:

- The benzodiazepines are lipophilic. They are rapidly and completely absorbed after oral administration, distribute throughout the body and penetrate into the CNS.

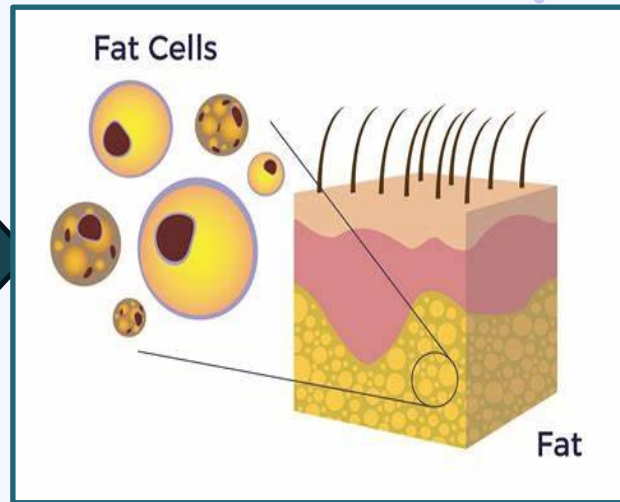
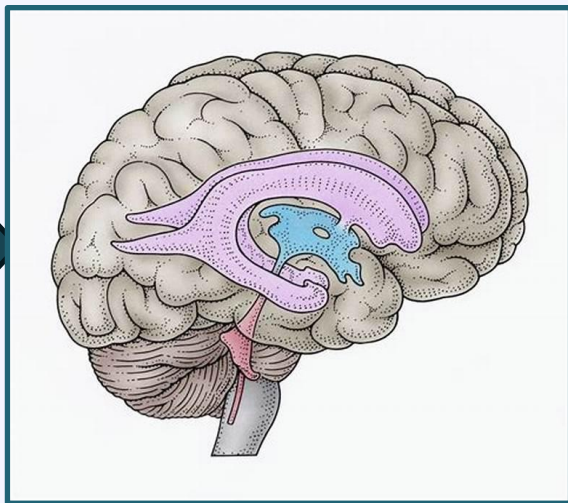
- Duration of action:

- The half-lives of the benzodiazepines are important clinically, because the duration of action may determine the therapeutic usefulness
- However, the clinical duration of action does not correlate with the actual half-life (otherwise, a dose of diazepam could conceivably be given only every other day, given its active metabolites). This may be due to receptor dissociation rates in the CNS and subsequent redistribution to fatty tissues and other areas.



• Metabolism and excretion:

- Most benzodiazepines are metabolized by the hepatic microsomal system to compounds that are also active.
 - For these benzodiazepines, the apparent half-life of the drug represents the combined actions of the parent drug and its metabolites.
 - Termination of action is through distribution and further metabolism by conjugation into glucuronide conjugates to make them more hydrophilic.
- 
- 
- 
- 



- Dependence:

- Psychological and physical dependence on benzodiazepines can develop if high doses of the drugs are given for a prolonged period.
- Abrupt discontinuation of the benzodiazepines results in withdrawal symptoms, including confusion, anxiety, agitation, restlessness, insomnia, tension, and (rarely) seizures.
- BZ with shorter half life cause more severe withdrawal symptoms.

- Pregnancy and lactation:

- BZ cross the placenta and are teratogenic, they also are secreted in breast milk and thus a nursing mother can expose the infant to BZ.

Therapeutic uses

- Anxiety disorders (low dose):
 - Benzodiazepines are effective for the treatment of the anxiety symptoms secondary to panic disorder, generalized anxiety disorder (GAD). obsessive-compulsive disorder, and extreme anxiety associated with phobias, such as fear of flying.
 - The benzodiazepines are also useful in treating anxiety related to depression and schizophrenia.



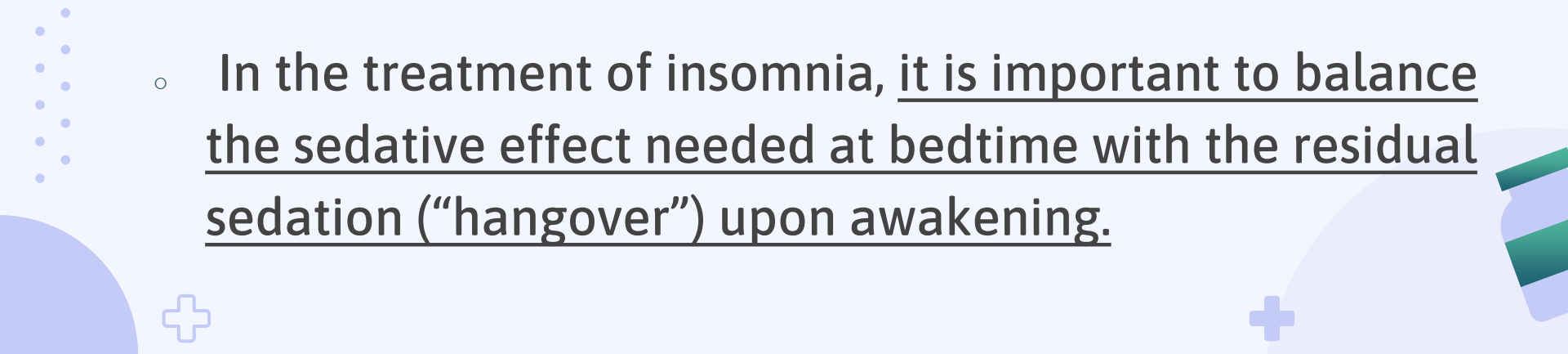
• Anxiety disorders:

- The longer-acting agents are often preferred in those patients with anxiety that may require prolonged treatment. The antianxiety effects of the benzodiazepines are less subject to tolerance than the sedative and hypnotic effects.
- For panic disorders, alprazolam is effective for short and long-term treatment, although it may cause withdrawal reactions in about 30% of patients.





• Sleep disorders (high dose):

- These agents decrease the latency to sleep onset and increase stage II of non-rapid eye movement (REM) sleep. Both REM sleep and slow-wave sleep are decreased.
 - In the treatment of insomnia, it is important to balance the sedative effect needed at bedtime with the residual sedation (“hangover”) upon awakening.
- 



Sleep disorders:

- Short acting agents with rapid onset of action are effective in treating individuals who have problems falling asleep but carry the risk of withdrawal and rebound insomnia after discontinuation.
 - Long and intermediate acting agents are useful for patients who experience frequent awakenings or have difficulty staying asleep, however, drug effect can extend to day time which will cause day time drowsiness (hang-over effect)
- 



Amnesia:

- Anterograde amnesia: Temporary impairment of memory with use of the benzodiazepines. In which The ability to learn and form new memories is also impaired.
 - The shorter-acting agents are often employed as premedication for anxiety-provoking and unpleasant procedures (endoscopy) which makes the patient more compliant and has little to no memory of the experience.
 - Midazolam is the most commonly used BZ to provide  this level of sedation. 
- 



Seizures:

- The use of BZ in treatment of epilepsy is no longer accepted due to development of better and much safer anti-epileptic drugs.
 - Diazepam is the drug of choice in terminating status epilepticus.
 - Diazepam supp are used to treat seizures associated with febrile seizures.
- 
- 
- 



• Muscular disorders:

- Diazepam is useful in the treatment of skeletal muscle spasms, such as occur in muscle strain, and in treating spasticity from degenerative disorders, such as multiple sclerosis and cerebral palsy.
 - However, the experience with BZ use in muscular disorders is limited.
- 

Adverse effects

- Physical dependence and tolerance
- Rebound insomnia after discontinuation.
- Hangover effect: a state of psychomotor disturbance that occurs the following day after use of long acting drugs (I.E residual effect)
- Anterograde amnesia and memory disturbance.
- CYP inhibitors can prolong the action of these drugs.
- Benzodiazepines are very safe as toxic dose is 1000 fold the therapeutic dose.

BZ antagonist

• Flumazenil

- is a GABA receptor antagonist that can rapidly reverse the effects of benzodiazepines. The drug is available for intravenous (IV) administration only.
- Administration of flumazenil may precipitate withdrawal in dependent patients or cause seizures if a benzodiazepine is used to control seizure activity.

Brands



03 Other anxiolytic drugs



Anti-depressants

- SSRI and SNRI are considered the first line treatment for chronic anxiety, GAD, and OCD due to absence of risk of dependence and tolerance.
- Escitalopram, and venlafaxine are approved for these conditions which may be used alone or in combination with a benzodiazepine.
- This combination is beneficial for sever cases as SSRI and SNRI require at least 4-6 weeks to produce a clinically significant effect.

Buspirone

- Buspirone is useful for the chronic treatment of GAD and has an efficacy comparable to that of the benzodiazepines. It has a slow onset of action and is not effective for short-term or “as-needed” treatment of acute anxiety states.
- The actions of buspirone appear to be mediated by serotonin (5-HT_{1A}) and (5-HT_{2A}) receptors
- buspirone lacks the anticonvulsant and muscle-relaxant properties of the benzodiazepines.
- The frequency of adverse effects is low, with the most common effects being headaches, dizziness, nervousness, nausea, and light-headedness. Sedation and psychomotor and cognitive dysfunction are minimal, and dependence is unlikely.

Brands





Clinical application: Management of GAD

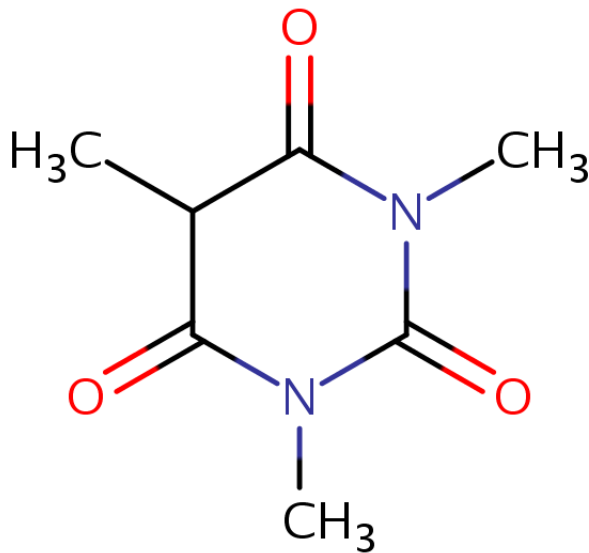
- Psychotherapy is the 1st step in treatment of GAD and OCD (acute or chronic).
- Acute anxiety is relieved by the use of BZ (preferably a short acting BZ like alprazolam for a period of 1-2 week)
- Chronic anxiety:
 - SSRI or SNRI, clinical effect is observed within 4-6 weeks.
 - If response is not adequate:
 - Switch to another SSRI or SNRI
 - Add BZ or buspirone or antipsychotic agent



Clinical application: Management of GAD

- The combination is used for severe GAD with anxiety symptom
- Week 1: SSRI (Escitalopram) + intermediate acting BZ (lorazepam).
- Week 4-6: SSRI (Escitalopram) + tapering (dose reduction of) lorazepam.
- Short acting BZ are not preferred in chronic treatment due to severe withdrawal symptoms and difficulty in drug tapering.

03 Barbitutes



Introduction

- The barbiturates were formerly the mainstay of treatment to sedate patients or to induce and maintain sleep.
- Today, they have been largely replaced by the benzodiazepines, primarily because:
 - Barbiturates induce rapid tolerance and physical dependence.
 - They are associated with very severe withdrawal symptoms.
 - Lethal in overdose.

Mechanism of action

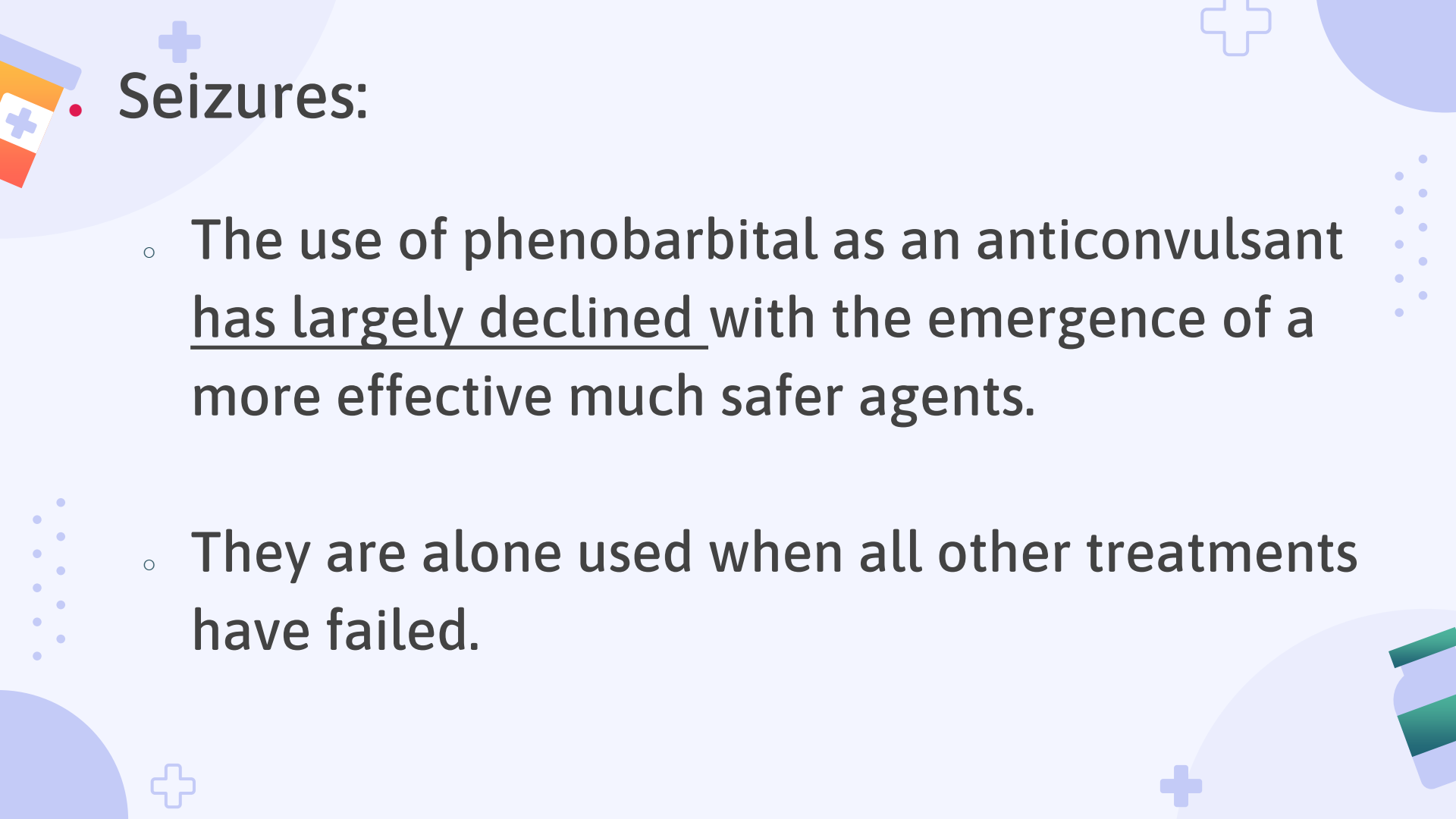
- Allosteric modulation of GABA receptor by binding to a site that is distinct from both GABA site and BZ site. This produces an action similar to BZ action.
- They also block the release of the excitatory glutamate causing further CNS inhibition and decreased neuronal activity.
- At low doses barbiturates cause sedation (calming, relaxing effect), at high doses they cause hypnosis, anesthesia.
- Overdoses can cause coma and death with inhibition of the respiration process.

Types of Barbiturates

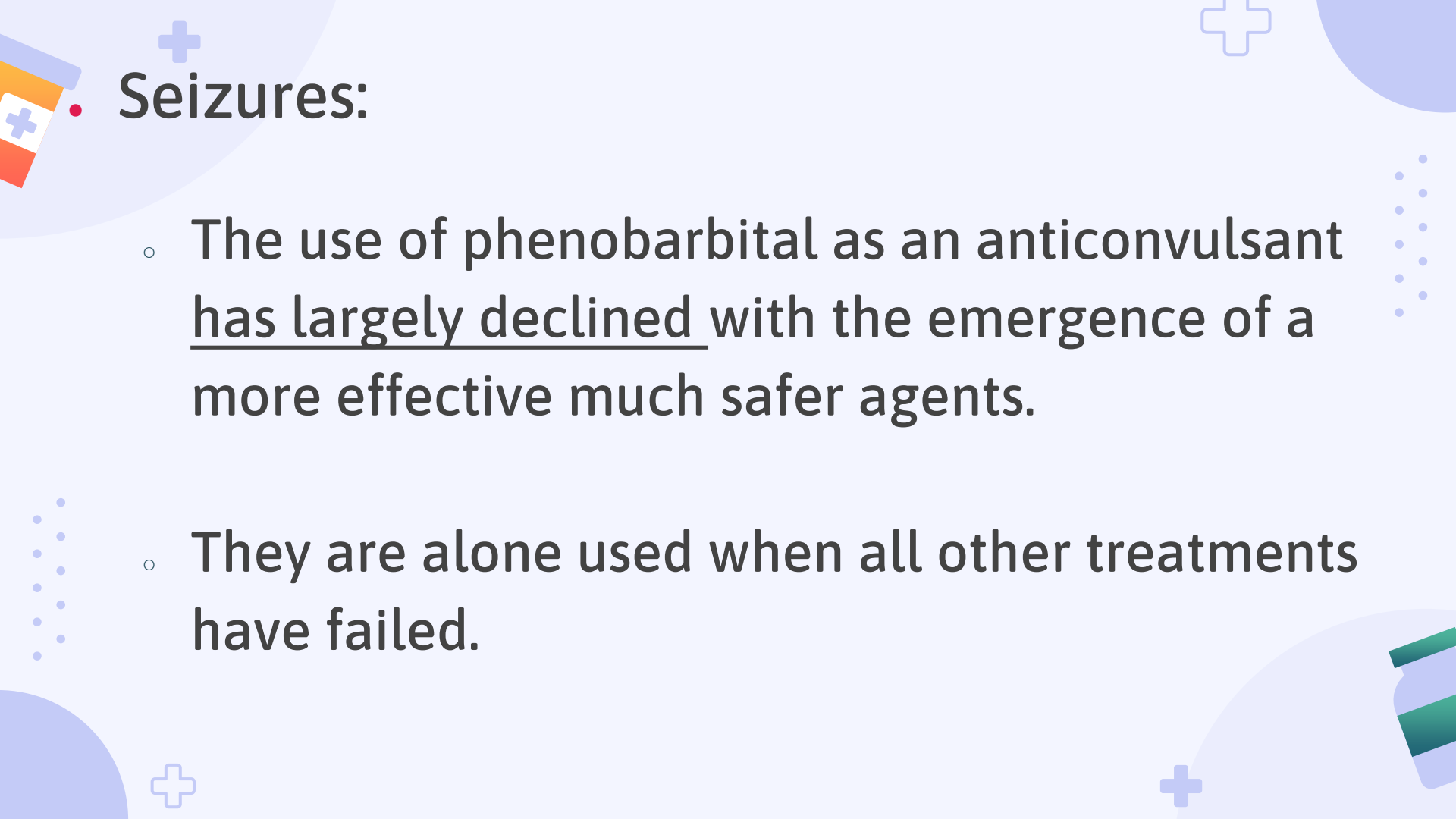
- Ultrashort acting Barbiturates (onset is within a few seconds)
 - Thiopental
- Short acting Barbiturates ($t_{1/2}$ 3-8 hours)
 - Pentobarbital
 - Secobarbital
- Long acting Barbiturates ($t_{1/2}$ 1-2 days):
 - Phenobarbital

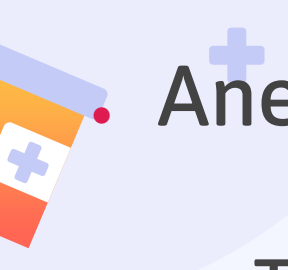
Therapeutic uses

- Anxiety disorders, sedative/hypnotic:
 - the use of barbiturates for insomnia is no longer generally accepted, given their adverse effects and potential for tolerance.



Seizures:

- The use of phenobarbital as an anticonvulsant has largely declined with the emergence of a more effective much safer agents.
 - They are alone used when all other treatments have failed.
- 



Anesthesia:

- Thiopental was used to induce anesthesia, however, it has been replaced by better and safer agents.
 - I.V pentobarbital is used to induce a “barbiturate coma” in patients with traumatic brain injury or those awaiting skin grafts in severe burns.
- 

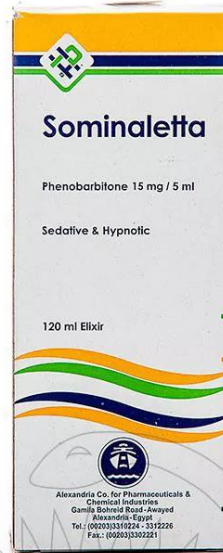


Adverse effects and pharmacokinetics



- The kinetics and adverse effects of barbiturates and BZ are very similar, the difference is that:
 - Barbiturates are associated with very severe withdrawal effects, and can lower the seizure threshold.
 - Barbiturates are CYP-450 inducers and can reduce the activity of many drugs.
 - Because they are CYP-450 inducers they can be used to breakdown bilirubin in neonates with hyperbilirubinemia, however, U.V light therapy is the 1st choice in this condition.
- 
- 
- 

Brands



04 Other hypnotic drugs



Z drugs

- Zolpidem, zaleplon, and eszopiclone are collectively called the Z drugs.
- Non of these drugs is structurally related to BZ, but They do not significantly alter the various sleep stages and, hence, are often the preferred hypnotics. This may be due to their relative selectivity for the BZ receptor.
- they exhibit no anticonvulsant or muscle-relaxing properties or anxiolytic effect.
- They show few withdrawal effects, exhibits minimal rebound insomnia, and little tolerance occurs with prolonged use.

Zolpidem

- Zolpidem is rapidly absorbed from the gastrointestinal (GI) tract, and it has a rapid onset of action and short elimination half-life (about 2 to 3 hours). It provides a hypnotic effect for approximately 5 hours.
- Zolpidem undergoes hepatic oxidation by the CYP450 system to inactive products , and so it is subject to CYP450 related adverse effects.
- Adverse effects of zolpidem include:
 - Nightmares
 - Agitation
 - Daytime drowsiness

Zaleplon

- Zaleplon is an oral nonbenzodiazepine hypnotic similar to zolpidem;
- zaleplon causes fewer residual effects on psychomotor and cognitive function compared to zolpidem or the benzodiazepines.
- This may be due to its rapid elimination, with a half-life of approximately 1 hour. The drug is metabolized by CYP3A4.

Eszopiclone

- Eszopiclone is an oral nonbenzodiazepine hypnotic that also acts on the BZ1 receptor.
- It has been shown to be effective for insomnia for up to 6 months.
- Eszopiclone is rapidly absorbed (time to peak, 1 hour), extensively metabolized the CYP450 system

Brands



Antihistamines

- Some antihistamines with sedating properties such as:
 - Diphenhydramine
 - Hydroxyzine
 - Doxylamine
- they are effective in treating mild types of situational insomnia.

Antidepressants

- The use of sedating antidepressants with strong antihistamine profiles has been ongoing for decades.
- Doxepin an older tricyclic agent with SNRI mechanisms of antidepressant and anxiolytic action, was recently approved at low doses for the management of insomnia.

Clinical application: Management insomnia

- Short term insomnia (less than 3 months) can be caused by an acute life stressor (such as a death in the family) and chronic insomnia both benefit from employing sleep hygiene and stress management.
- Short term insomnia can be managed by: BZ, Z drugs, and antihistamines.
- Long term and chronic insomnia will require psychic and medical evaluation. Certain medical conditions likes: sleep apnea, pulmonary disease, HF, chronic pain can be the cause of insomnia.
- The use of Z drugs, and anti-depressants are employed in chronic insomnia as they carry a lower risk of dependence and tolerance than BZ, however, the risk with Z drugs is still present since they are controlled substance.

Cases

- An 18-year-old male is having difficulty sleeping because of the death of his grandfather. He is given a benzodiazepine that does which of the following?
 - A. Binds to serotonin 5-HT 1 receptors
 - B. Binds to GABA A receptors
 - C. Is an antagonist at α -adrenoceptors
 - D. Is an antagonist at dopamine D 2 receptors

Answers

- B. Benzodiazepines enhance the binding of GABAA to its receptor, which increases the permeability of chloride.
- The benzodiazepines do not relieve pain but may reduce the anxiety associated with pain.

Cases

- Which one of the following is a short-acting hypnotic?
 - A. Phenobarbital.
 - B. Diazepam.
 - C. Chlordiazepoxide.
 - D. Triazolam.

Answers

- D. Triazolam is a short-acting drug. It has little daytime sedation. The other drugs listed are longer acting

Cases

- A 45-year-old man who has been injured in a car accident is brought into the emergency room. His blood alcohol level on admission is 275 mg/dL. Hospital records show a prior hospitalization for alcohol-related seizures.
- His wife confirms that he has been drinking heavily for 3 weeks. What treatment should be provided to the patient if he goes into withdrawal?
- A. None. B. Lorazepam. C. Pentobarbital. D. Phenytoin. E. Buspirone.

Answers

- B. It is important to treat the seizures associated with alcohol withdrawal. Benzodiazepines, such as chlordiazepoxide, diazepam, or the shorter-acting lorazepam, are effective in controlling this problem.
- They are less sedating than pentobarbital or phenytoin

CNS

By Dr.Riffat Ibrahim Riffat

General Anesthetics



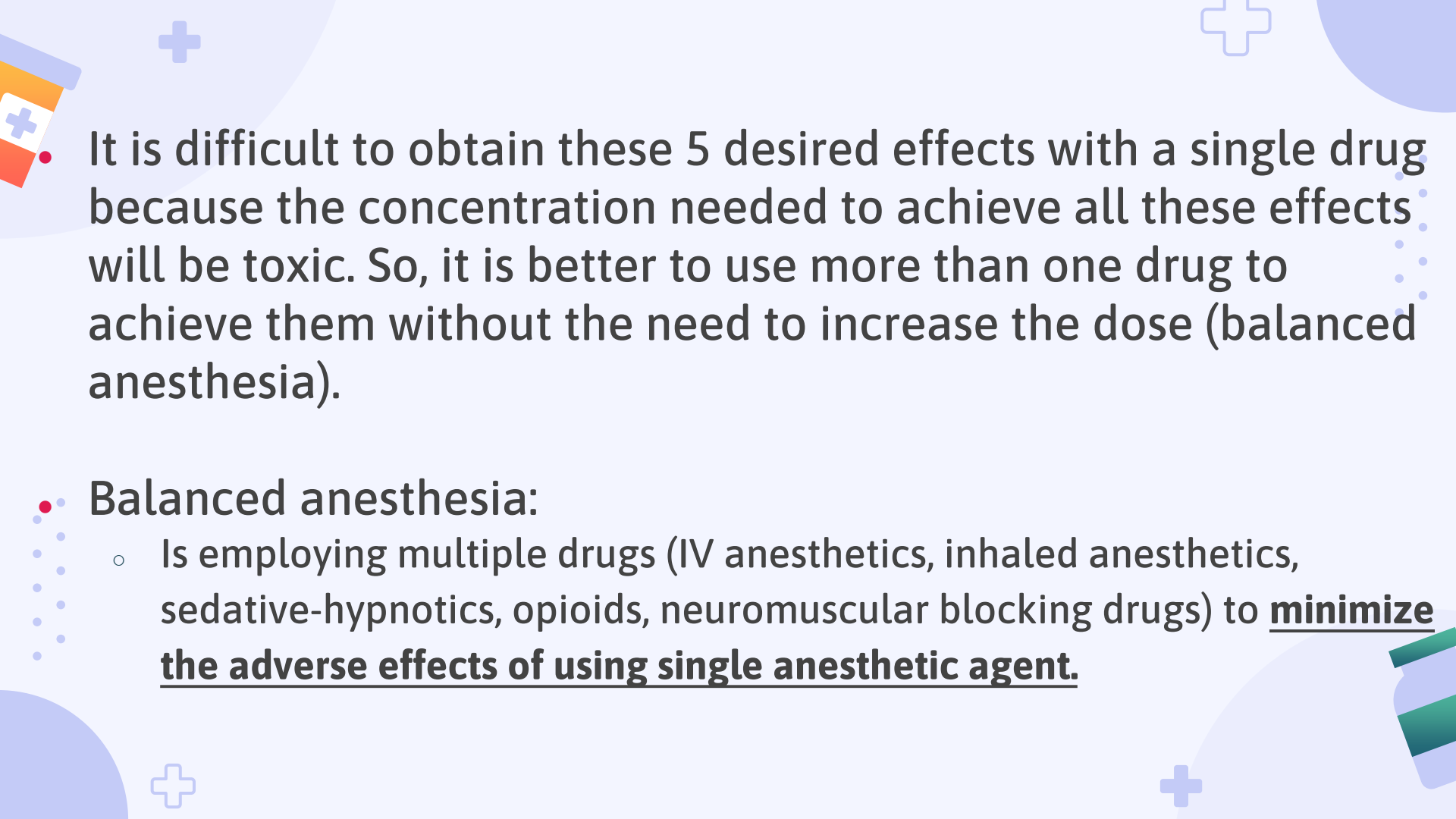
Introduction

General anesthesia is intended to produce 5 desired effects:

- Unconsciousness
- amnesia
- Analgesia
- skeletal muscle relaxation
- and inhibition of autonomic reflexes



- The purpose of these effects is loss of response to surgical incision.



• It is difficult to obtain these 5 desired effects with a single drug because the concentration needed to achieve all these effects will be toxic. So, it is better to use more than one drug to achieve them without the need to increase the dose (balanced anesthesia).

• **Balanced anesthesia:**

- Is employing multiple drugs (IV anesthetics, inhaled anesthetics, sedative-hypnotics, opioids, neuromuscular blocking drugs) to **minimize the adverse effects of using single anesthetic agent.**



• Phases of anesthesia:

• 1. Before surgery (preanesthetic medications):

- One hour before operation, the patient is given anxiolytic to relieve anxiety and may be given metoclopramide to hasten gastric evacuation and prevent vomiting and aspiration pneumonia.

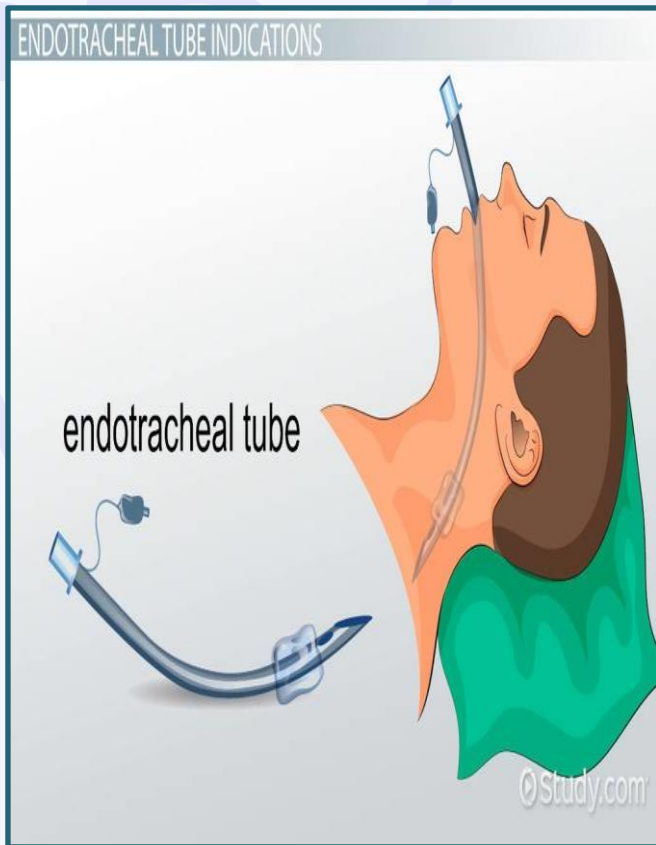
• 2. During surgery:

- Induction of anesthesia usually by an IV anesthetic
- Maintenance is achieved by inhalation anesthetic
- The patient is also given NMBs

• 3. After surgery:

- The patient is not left until respiration is normal
- Relieve of pain by an opioid





Classification of general anesthetics:

- I. Inhalation agents:
 - Volatile liquids: isoflurane – sevoflurane – desflurane – enflurane – halothane (halogenated anesthetics)
 - Gases: nitrous oxide – xenon
- II. IV agents:
 - Propofol – thiopental – etomidate – ketamine – midazolam



Pharmacodynamics

- 1. Activation of GABAA receptor:
- Most anesthetic agents directly and indirectly facilitate a GABA-mediated increase in chloride conductance to hyperpolarize and inhibit neuronal membrane activity.
- Examples:
 - Propofol, etomidate, isoflurane, benzodiazepines (midazolam, lorazepam, diazepam), and barbiturates (e.g. sodium thiopental)

Pharmacodynamics

- 2. Blocking of NMDA receptor: Nitrous oxide and ketamine inhibit excitatory glutamate-gated ion channels (NMDA receptors).
- 3. Opening of two-pore potassium channels (K2P): Opening of these channels causes neuronal hyperpolarization which reduces neuronal excitability.
 - Examples: halogenated inhalation anesthetics

Adverse effects of inhaled anesthetics:

Malignant hyperthermia:

Is a rare (1:20000), but life-threatening condition that may occur when anesthetics are used with NMBs like succinylcholine.

It is due to genetic abnormality in Ca^{2+} release channels (ryanodine receptor) in sarcoplasmic reticulum (SR) of muscle cells, leading to massive release of Ca^{2+} from SR following exposure to inhaled anesthetics or succinylcholine.

- **Symptoms include:** muscle spasm, hyperthermia, hypertension, tachycardia, and hyperkalemia.
- **Treatment includes:** dantrolene, a muscle relaxant that blocks Ca^{2+} release channels + supportive measures.

Adverse effects of inhaled anesthetics:

2. CNS: Rise of IC pressure due to VD of cerebral blood vessels.

3. CVS:

- Hypotension: due to myocardial depression and peripheral VD.
- Cardiac arrhythmia: especially with halothane.

4. Respiratory:

- Desflurane is a pulmonary irritant and can cause bronchospasm.
- Inhibition of mucociliary function.

5. Liver: Halothane can cause acute liver injury (halothane hepatitis).

6. Uterus: Inhaled anesthetics cause uterine relaxation which can be helpful when uterine relaxation is required for intrauterine fetal manipulation or manual extraction of a retained placenta. However, it can also lead to increased uterine bleeding after delivery when uterine contraction is desired.



+ INTRAVENOUS ANESTHETICS



- Propofol – thiopental – etomidate – ketamine – midazolam
 - They are commonly used for rapid induction of anesthesia.
 - IV anesthetics are highly lipophilic drugs, upon administration, they rapidly distribute in the highly vascular tissues including the brain causing rapid induction of anesthesia.
 - Termination of anesthesia is due to redistribution of the drug from nervous tissue to other tissues such as muscles, viscera, and adipose tissue.
 - Prolonged administration (cont. IV infusion) can lead to saturation of adipose tissue and slowing of the redistribution phase → prolongation of anesthesia, however, this can lead to undesired side effects.
- 

Propofol (Diprivan 1%)

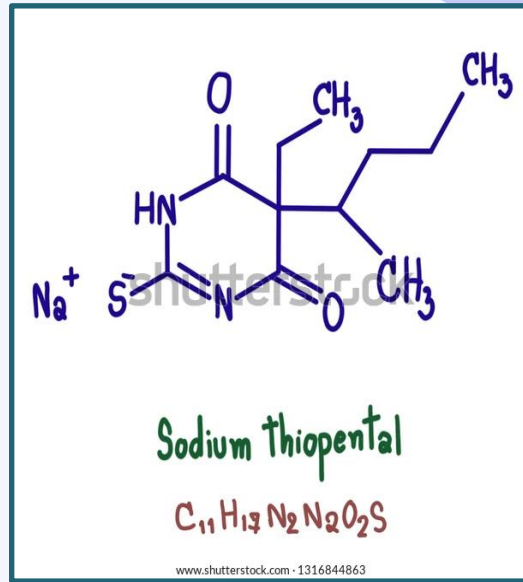
The most commonly used IV agent for induction of anesthesia.

- It is administered as an oil-in-water emulsion which can cause pain at the injection site. **Fospropofol** is a water-soluble derivative that is less painful and rapidly converted to propofol in the body.
- Adverse effects:
 - Pain at the site of injection
 - Propofol infusion syndrome occurs with prolonged infusion and consists of severe metabolic acidosis, skeletal muscle necrosis (rhabdomyolysis), hyperkalaemia, lipaemia, hepatomegaly, renal failure, arrhythmia and cardiovascular collapse.



Thiopental

- Thiopental is the only remaining barbiturate in common use.
- Used only for induction of anesthesia (not for maintenance).
- It is highly lipophilic drug. On IV injection, thiopental causes unconsciousness within about 20s, lasting for 5–15 min before it redistributes to fatty tissues.
- Adverse effects:
 - High dose can cause hypotension, RC depression, and myocardial depression.
 - The solution of thiopental is highly alkaline, if injected SC or IM, it causes local tissue necrosis and ulceration that can result in gangrene.
 - Like other barbiturates, it is hepatic microsomal enzyme inducer.



Ketamine (Ketalar)

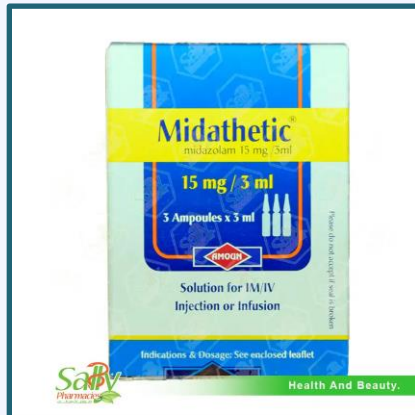
The characteristic state observed after an induction dose of ketamine is known as “dissociative anesthesia,” wherein the patient’s eyes remain open with a slow nystagmic gaze.

- Ketamine’s mechanism of action includes inhibition of the NMDA receptor.
- It differs from most other IV anesthetics in:
 - It produces significant analgesia
 - It raises the blood pressure
 - Does not affect respiration.
- The adverse effects are less marked in children making it preferred in pediatrics.
 - Increase IC tension
 - Hallucinations and psychiatric disturbances on recovery.



Midazolam (Dormicum)

- It is a very short acting BZ. It has rapid onset and short duration.
- May be used preoperatively for sedation and to reduce anxiety.
- It is used as a sole agent for surgical and diagnostic procedures that do not require analgesia (endoscopy, cardiac catheterization).
- The actions of the BZs can be reversed with flumazenil.



Cases

- Which of the following is correct regarding malignant hyperthermia?
 - A. It is triggered by dantrolene.
 - B. It is triggered by local anesthetics.
 - C. It is generally mild and clinically insignificant. D. It has no familial component.
 - E. It involves increased skeletal muscle oxidative metabolism.

Answers

- Correct answer = E. Malignant hyperthermia involves increased skeletal muscle oxidative metabolism and is a life-threatening condition. Dantrolene is the specific pharmacologic treatment. Local anesthetics have been shown to be safe. Triggering agents include succinylcholine and halogenated hydrocarbon volatile anesthetic agents in susceptible individuals. Susceptibility to malignant hyperthermia is inherited in an autosomal dominant fashion.

Cases

- Which of the following agents is preferred I.V agent for children
 - A. Ketamine
 - B. Propofol
 - C. isoflurane
 - D. NO

Answers

- Correct answer = A, The adverse effects are less marked in children making it preferred in pediatrics.

Cases

- A 36-year-old man I said to have an endoscopy, which of the following agents can be used as the sole anesthetic for this procedure
 - A. Ketamine
 - B. Propofol
 - C. Midazolam
 - D. sevofulrane

Answers

- Correct answer = D. Midazolam is used as a sole agent for surgical and diagnostic procedures that do not require analgesia (endoscopy, cardiac catheterization).

CNS

By Dr.Riffat Ibrahim Riffat

Local Anesthetics

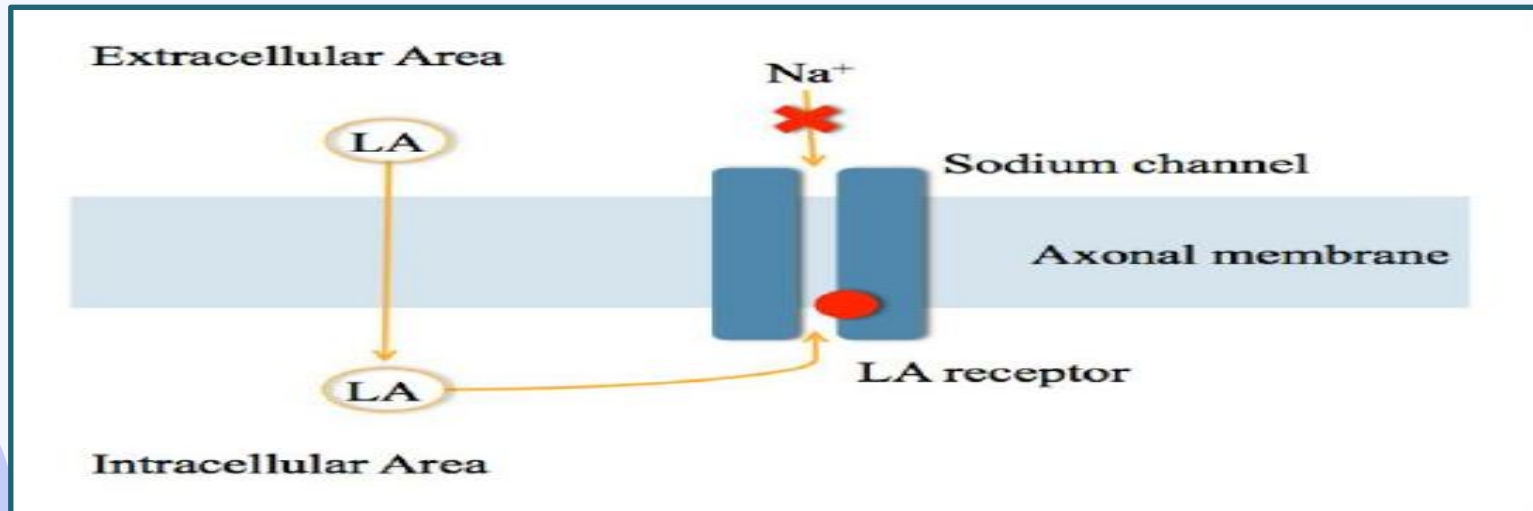


Introduction

local anesthesia is intended to produce Reversible loss of sensation in small part of the body as these used in dentistry and dermatology. Not including the brain.

- Notes: Small “ Type A delta “ and unmyelinated nerve fibers “ Type C” for pain are most sensitive for local anesthetics than the large fibers (motor).

- **MOA:**
- 1- Block voltage gate sodium channel within the nerve fibers so decrease nerve conduction.
- 2- The non ionized form (lipid soluble) cross the lipid membrane.
- 3- Once inside, It is converted into ionized form (active) which bind to the inner vestibule of the sodium channel



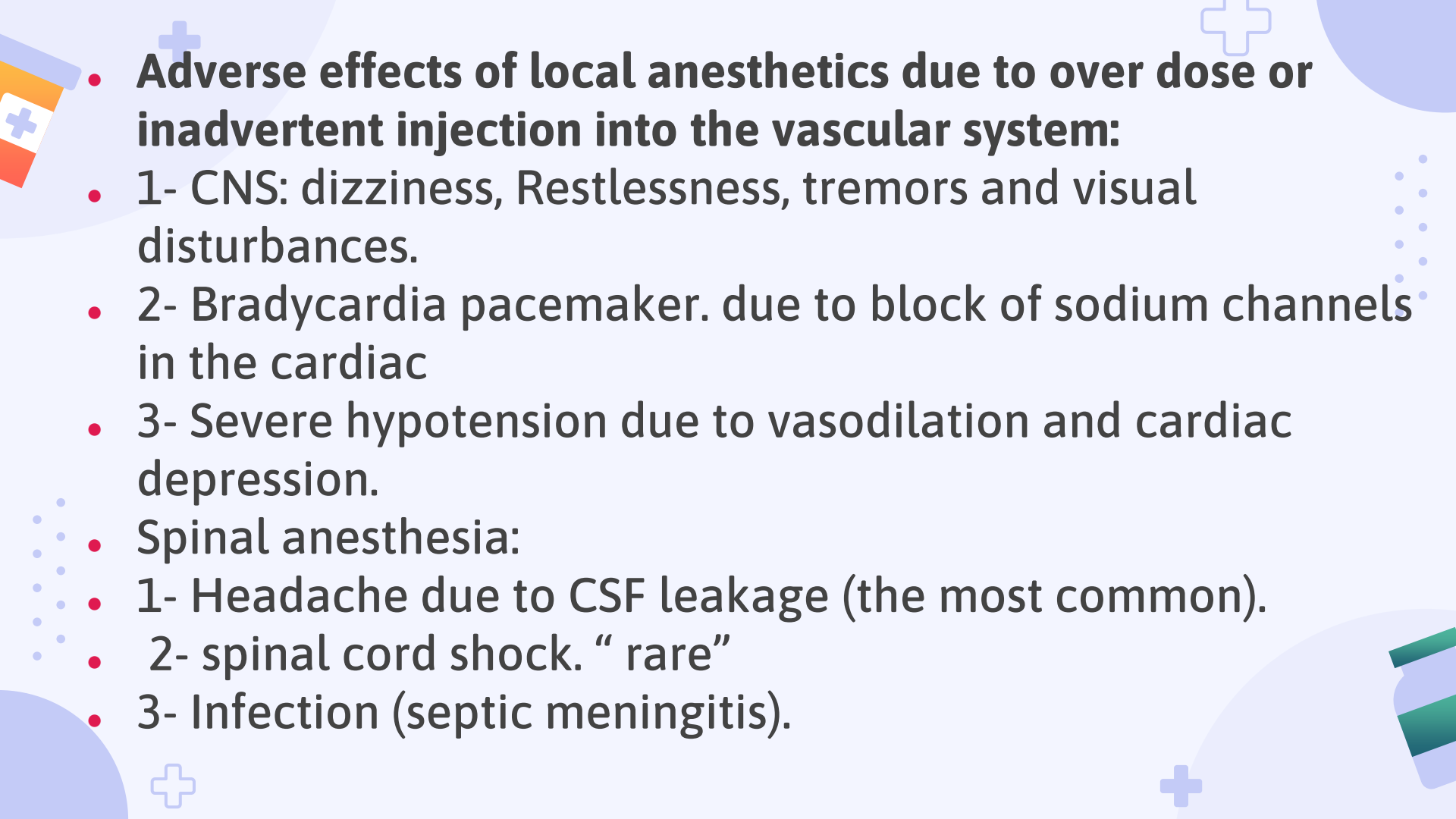


• Local anesthetic drugs:

- Esters: Cocaine, Procaine, Benzocaine.
 - Have short duration and metabolized by plasma and tissues esterase.
 - 1 Procaine: Prototype (The first synthetic agent). No Longer used.
 - 2 Cocaine: Natural source and rarely used.
 - 3 Benzocaine: - It is used for surface (topical) anesthesia only as creams and oral gel for oral ulcers.
 - The topical used of higher concentration of benzocaine (teething gel) may cause methemoglobinemia, this side effect is most common in children less than 2 years.
- 
- 
- 

- **Local anesthetic drugs:**

- **Amides:** (topical local anesthetics for the eyes). **Lidocaine, Mepivacaine, Bupivacaine and Ropivacaine.**
- Have long duration and metabolized by the liver.
- Lidocaine (lignocaine): Prototype.
- Widely used local anesthetics.
- Used also by IV for ventricular arrhythmia. “ Class Ib”.
- Used topically as eye drops.
- All LAs are Vasodilators except cocaine so added to as epinephrine a vasoconstrictor to reduce bleeding , increase duration of anesthesia and decrease amount of the drug from reaching systemic circulation.

- 
- **Adverse effects of local anesthetics due to over dose or inadvertent injection into the vascular system:**
 - 1- CNS: dizziness, Restlessness, tremors and visual disturbances.
 - 2- Bradycardia pacemaker. due to block of sodium channels in the cardiac
 - 3- Severe hypotension due to vasodilation and cardiac depression.
 - Spinal anesthesia:
 - 1- Headache due to CSF leakage (the most common).
 - 2- spinal cord shock. “rare”
 - 3- Infection (septic meningitis).

Cases

- A 32-year-old woman requests an epidural to ease labor pains. She reports that she has an allergic reaction to Novocain(procaine) at the dentist's office. Which of the following local anesthetics would be appropriate for use in an epidural for this patient?
- A. Chloroprocaine.
- B. Mepivacaine.
- C. Ropivacaine.
- D. Tetracaine.

Answers

- Correct answer= C.- Procaine is an ester local anesthetic. Since this patient has an allergy to procaine, other ester anesthetics (Chloroprocaine and tetracaine) should not be used. - Mepivacaine an amide local anesthetics, should not be used due to potential for increased toxicity to the neonate.- Ropivacaine is an amide anesthetic

Cases

- Which of the following is correct regarding local anesthetics?
- A. They affect only small , unmyelinated nerve fibers.
- B. They have either a lipophilic or a hydrophilic group.
- C. They have either an amide or an ester linkage.
- D. They are unaffected by PH of the tissue and pKa of the drug.
- E. In their ionized form , they interact with the protein receptor of calcium channels.

Answers

- Correct answer= C. - The small , unmyelinated nerve fibers that conduct impulses for pain, temperature and autonomic activity are most sensitive to the action of local anesthetics, but other nerve fibers are affected also.- Local anesthetics have a lipophilic group (aromatic ring) , joined by either an amide or ester linkage to a carbon chain that in turn is joined to amine hydrophilic group. - Onset and duration of local anesthetics are affected by both PH of the tissue and PKa of the drug. - Local anesthetics work by blocking sodium ion channels.

Cases

- A 36-year-old man I said to have an endoscopy, which of the following agents can be used as the sole anesthetic for this procedure
 - A. Ketamine
 - B. Propofol
 - C. Midazolam
 - D. sevofulrane

Answers

- Correct answer = D. Midazolam is used as a sole agent for surgical and diagnostic procedures that do not require analgesia (endoscopy, cardiac catheterization).